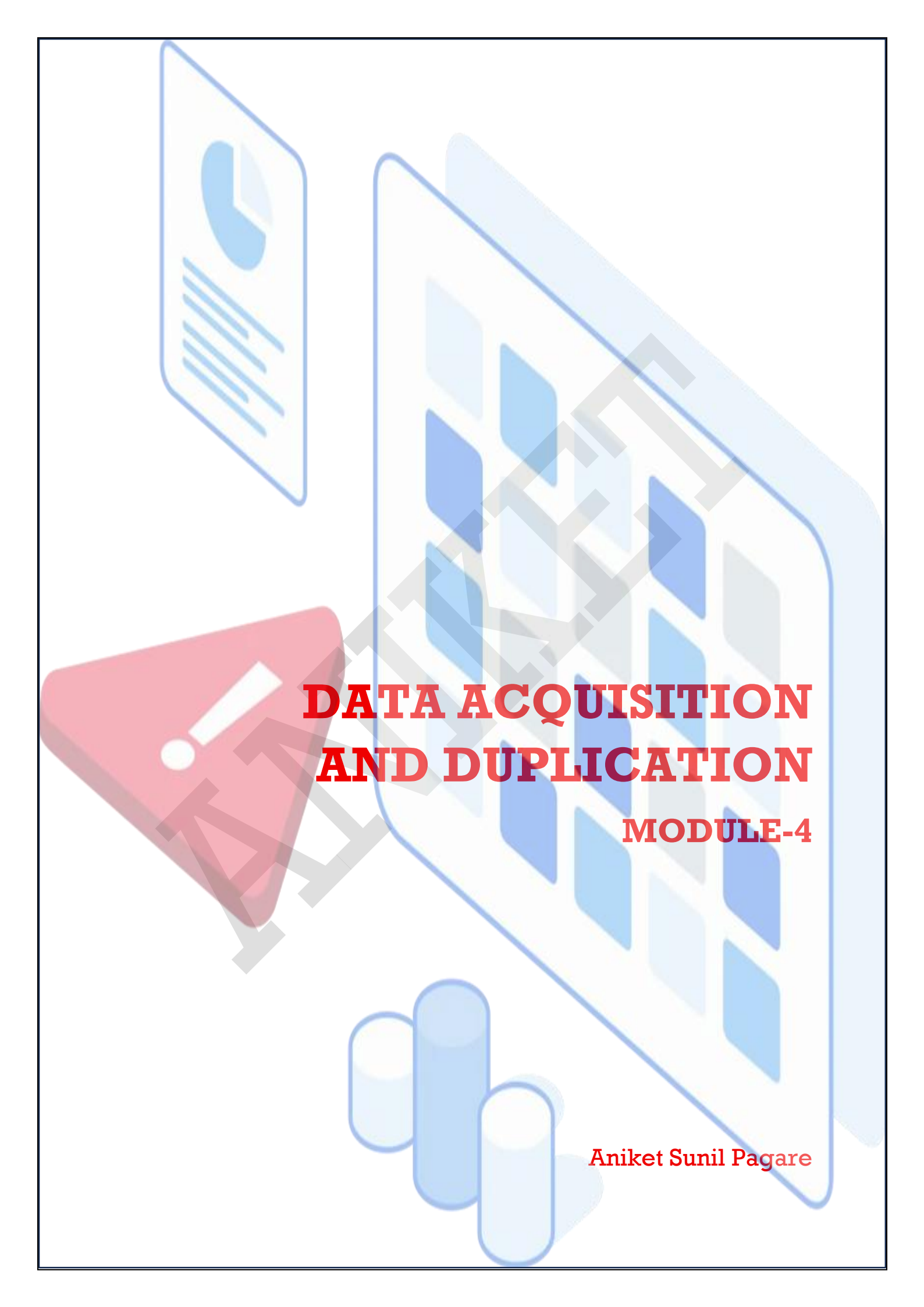




DATA ACQUISITION AND DUPLICATION

MODULE-4

Aniket Sunil Pagare

What is Data Acquisition in Cyber Forensics?

Definition:

Data acquisition is the process of collecting or capturing digital evidence from electronic devices (e.g., hard drives, phones, memory) **without altering or damaging** the original data.

✓ Purpose:

To **preserve the integrity** of digital evidence for analysis, legal investigation, or court use.

🔄 Types of Acquisition:

Type	Description	Use Case
Physical Acquisition	Bit-by-bit copy of the entire storage (including deleted/hidden files)	Deep forensic investigation
Logical Acquisition	Only collects active (visible) files	Quick data preview or triage
Targeted Acquisition	Acquires specific files, folders, or logs	Faster and focused investigation
Live Acquisition	Captures volatile data (RAM, network) from a running system	For active incidents or malware analysis

What is Data Duplication in Cyber Forensics?

Definition:

Data duplication refers to creating an **exact forensic copy (clone)** of the original digital media (hard disk, phone, SD card, etc.).

This ensures:

- The **original is untouched**
- Investigators work on the **duplicate (image)**
- The copy is **forensically sound and verifiable** via hash values (MD5, SHA1)

✓ Why it's important:

- Maintains **evidence integrity**
 - Supports **chain of custody**
 - Allows multiple analysts to work on the same data without altering the original
-

🔗 How They Relate:

- **Acquisition** is the overall process of collecting evidence.
 - **Duplication** is the step where a **forensic copy** of that evidence is made to ensure the original remains unaltered.
-

💡 Example:

A forensic investigator uses FTK Imager to:

- **Acquire** a hard drive image (capture all data including deleted files)
 - Then creates a **duplicate copy** of the image to work on
 - Verifies both original and duplicate using **hash matching**
-

Lab-1 Create a dd image of a system drive and compute MD5 hash for future validation Using DD Tool

The **dd.exe** tool is the **Windows version** of the Unix/Linux dd command-line utility, widely used in **digital forensics and system administration** to perform **low-level copying and conversion of data**.

What Does dd.exe Do?

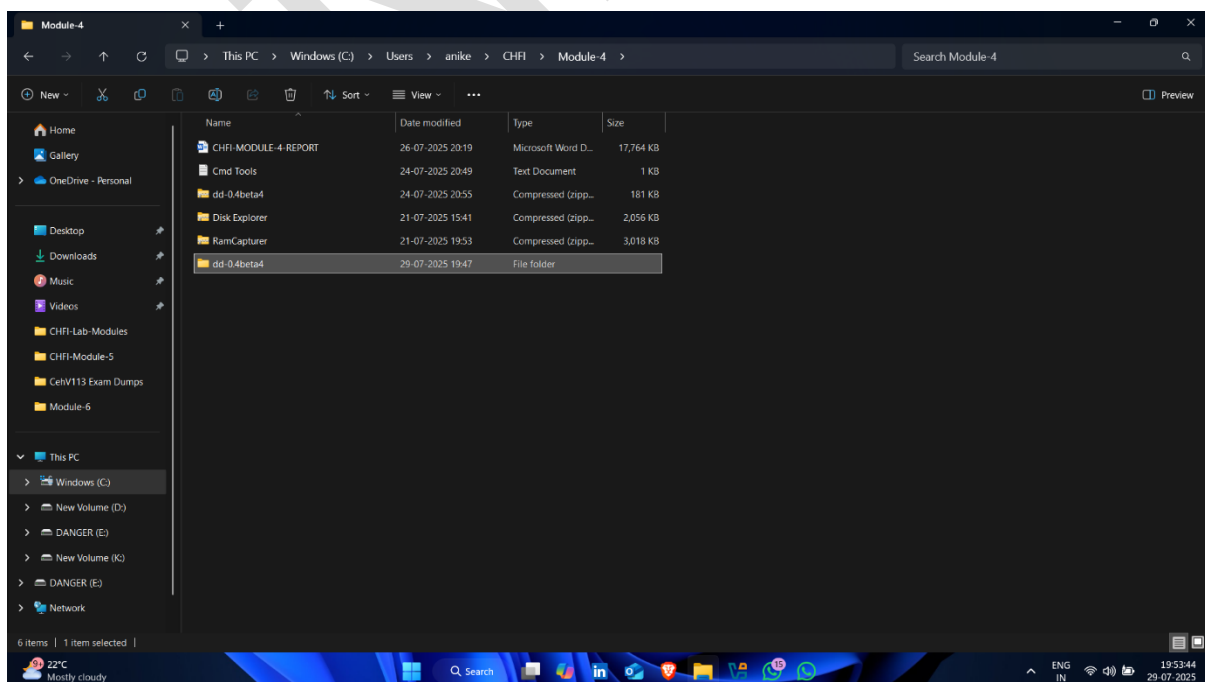
It is used to:

- **Create bit-by-bit copies of drives** (disk imaging)
- **Clone drives or partitions**
- **Create forensic images** (exact copies) of hard disks, USBs, etc.
- **Copy damaged or corrupted data** for recovery purposes

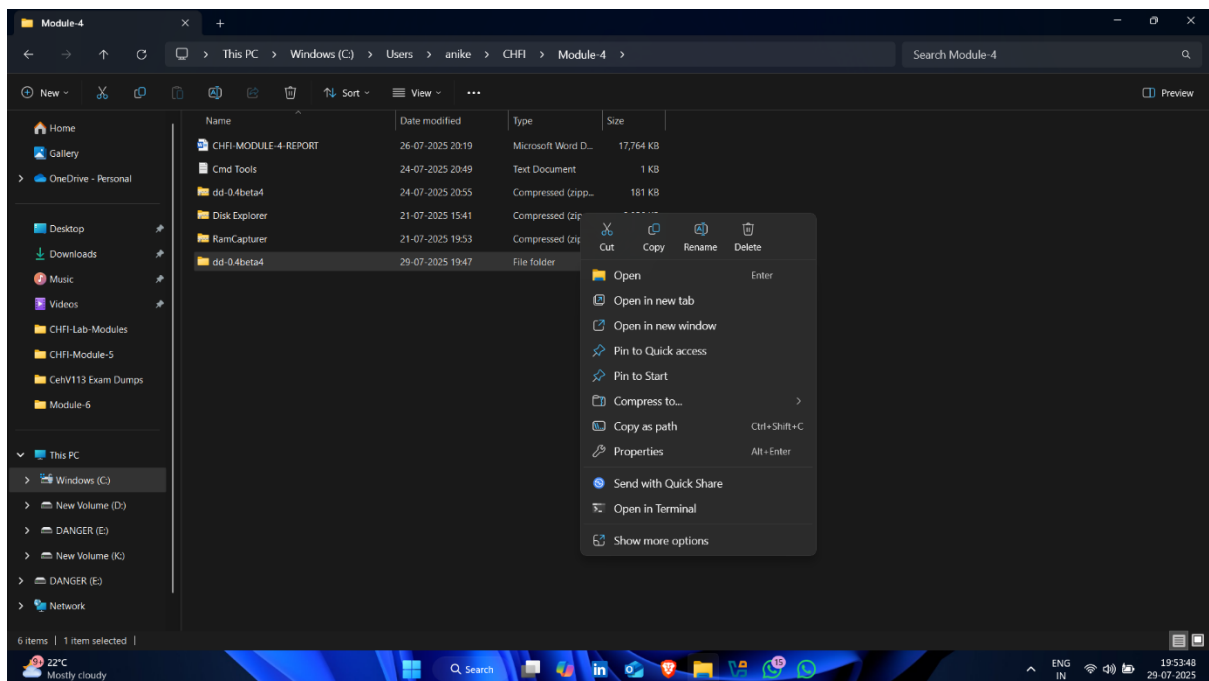
In Windows forensics, dd.exe is often used to **preserve evidence** by making an exact replica of a suspect drive.

How to use it -:

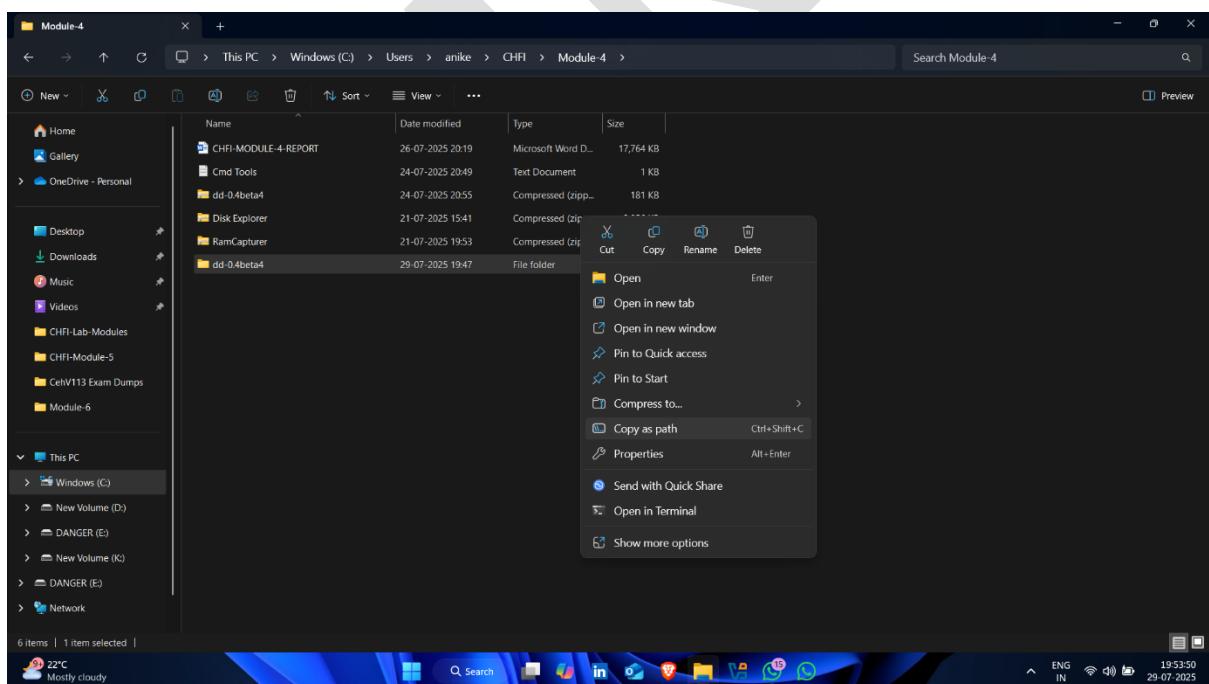
- DD tool Folder



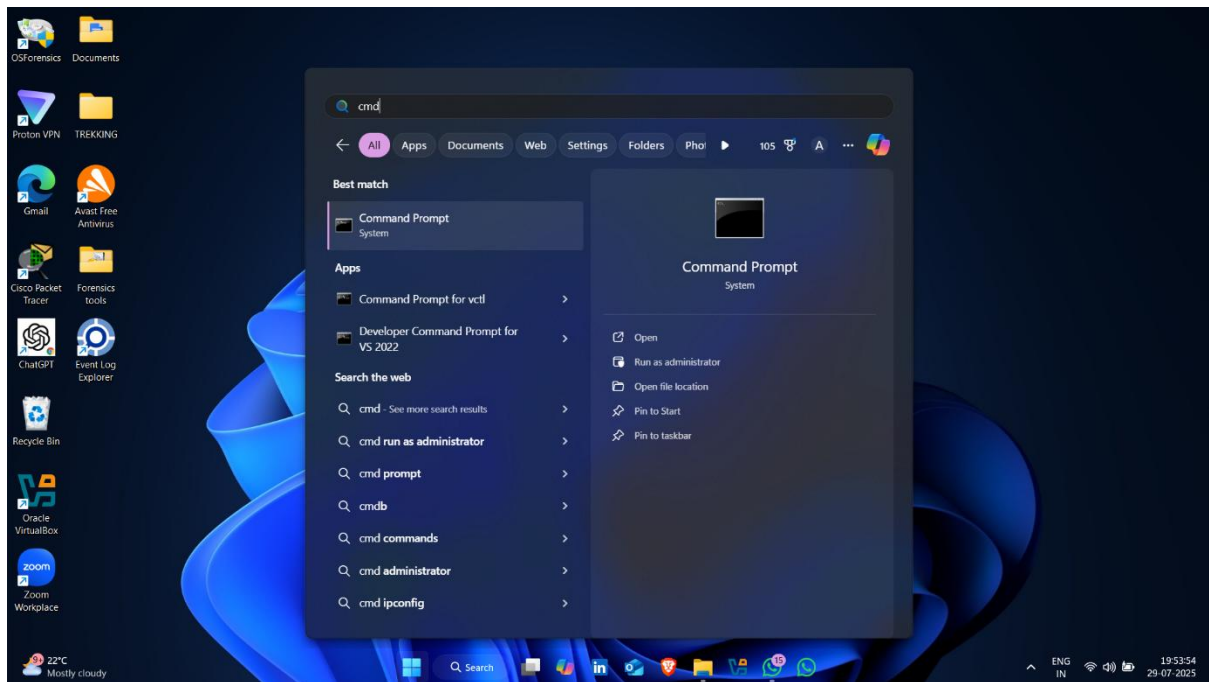
- **Right click on folder**



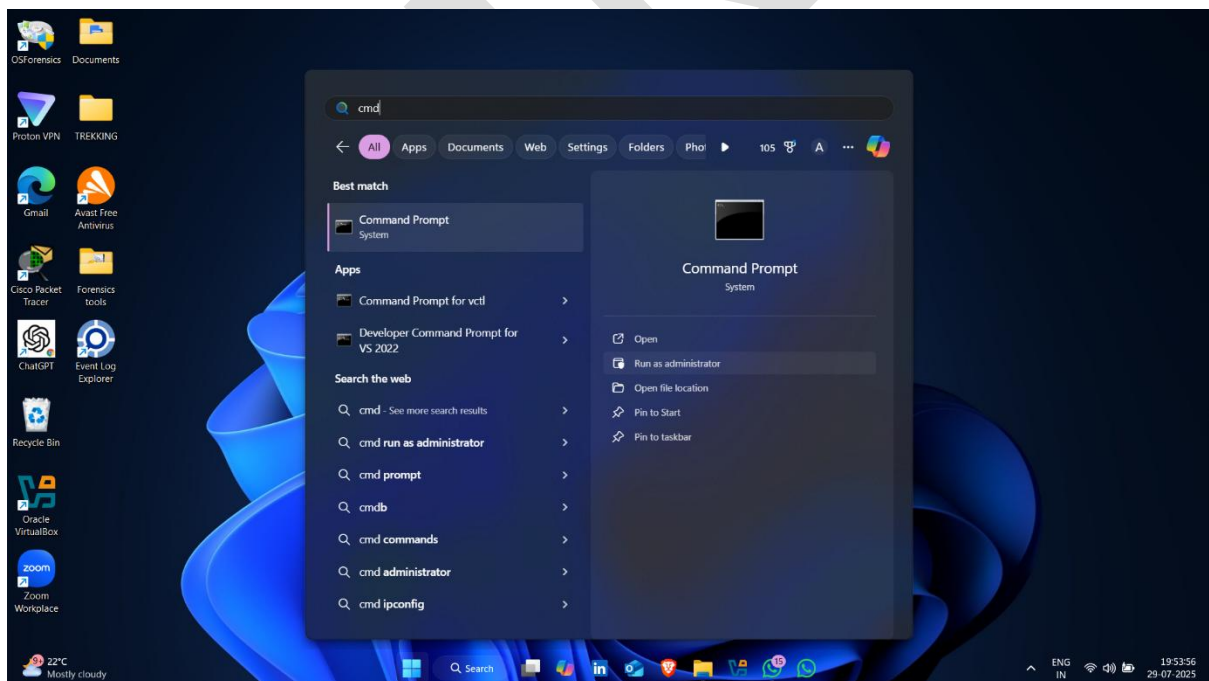
- **click on copy as path**



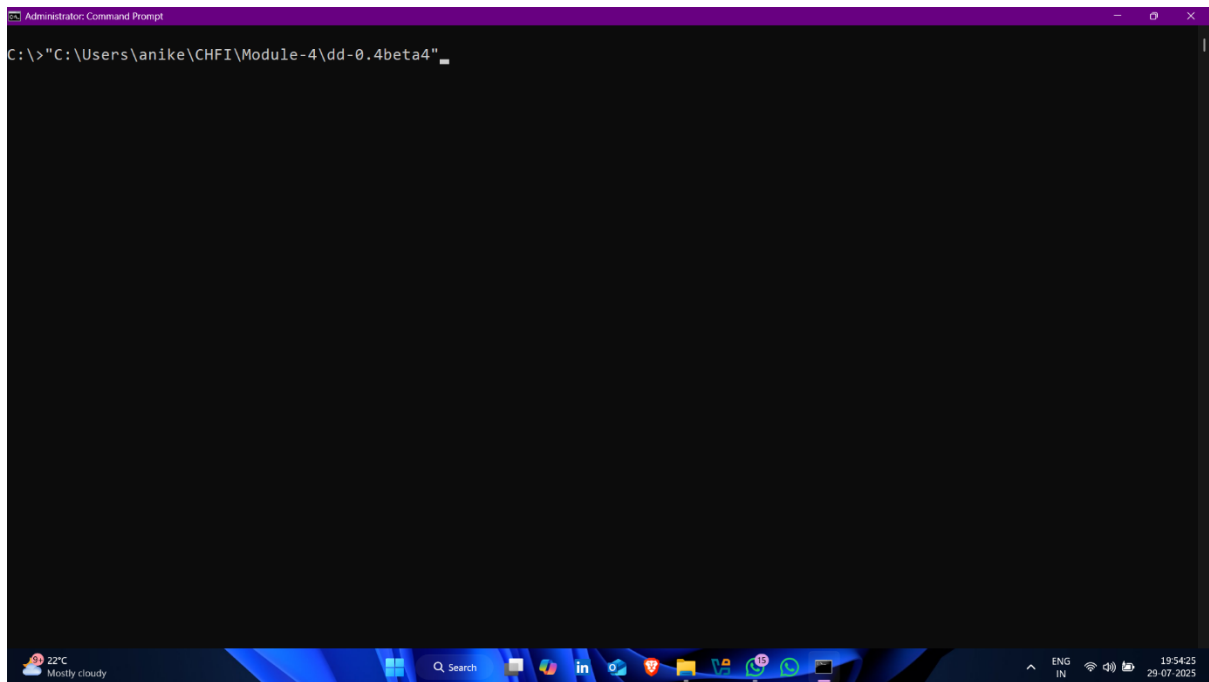
- open command prompt



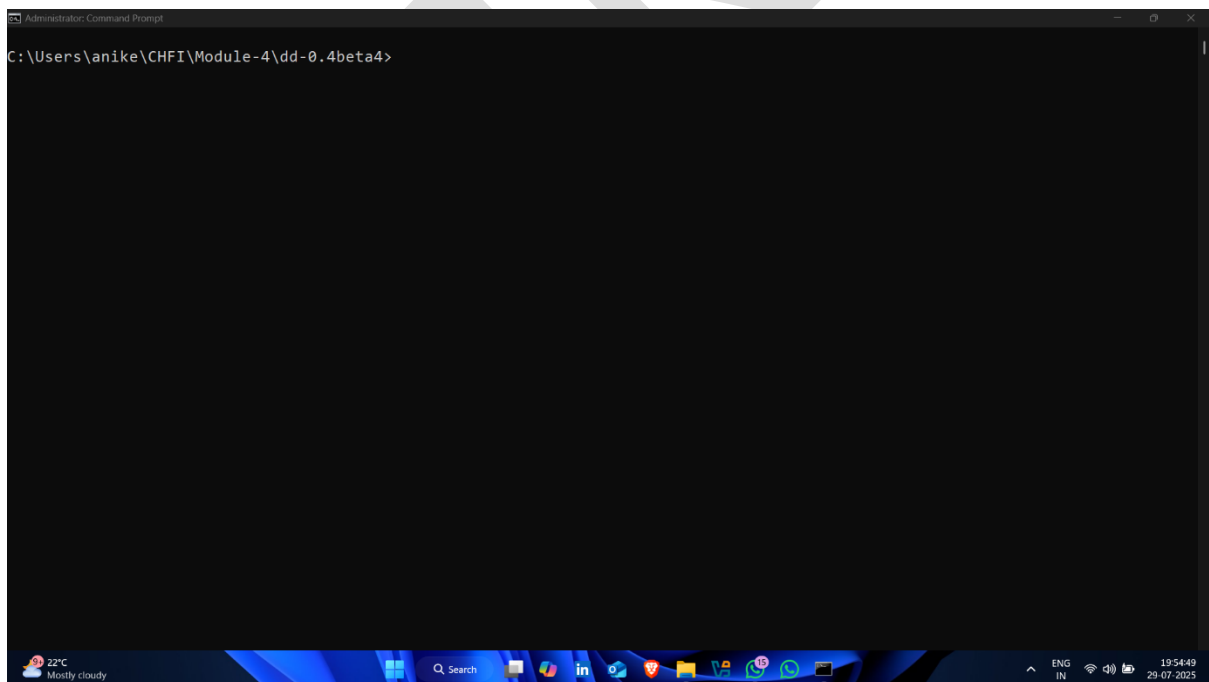
- click on **Run as administrator**



- paste DD tool path



A screenshot of a Windows Command Prompt window titled "Administrator: Command Prompt". The window has a black background with white text. The command prompt shows the path `C:\>"C:\Users\anike\CHFI\Module-4\dd-0.4beta4"` with a cursor at the end. The Windows taskbar is visible at the bottom, showing the Start button, search bar, and several application icons. The system tray on the right indicates the temperature is 22°C, weather is mostly cloudy, and the time is 19:54:25 on 29-07-2025.



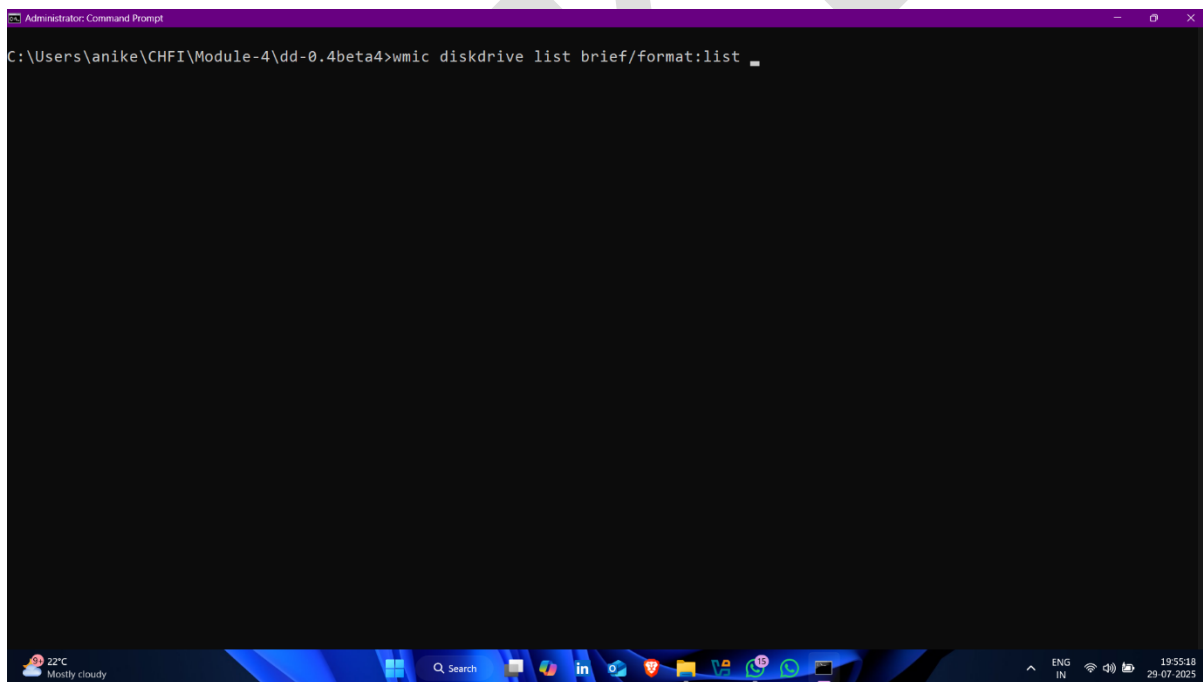
A screenshot of a Windows Command Prompt window titled "Administrator: Command Prompt". The window has a black background with white text. The command prompt shows the path `C:\Users\anike\CHFI\Module-4\dd-0.4beta4>` with a cursor at the end. The Windows taskbar is visible at the bottom, showing the Start button, search bar, and several application icons. The system tray on the right indicates the temperature is 22°C, weather is mostly cloudy, and the time is 19:54:49 on 29-07-2025.

- Now , Enter Following command

Command -: wmic diskdrive list brief /format:list

Explanation -:

Part	Meaning
wmic	Stands for Windows Management Instrumentation Command-line
diskdrive	Targets physical disk drives on the system
list brief	Lists disk drive info in a brief format (limited details)
/format:list	Overrides the display format to show results as a detailed list



The screenshot shows a Windows Command Prompt window titled "Administrator: Command Prompt". The command prompt is open at the directory "C:\Users\anike\CHFI\Module-4\dd-0.4beta4". The command entered is "wmic diskdrive list brief /format:list". The command prompt is running on a Windows 10 desktop with a taskbar at the bottom showing various application icons and system tray information including temperature (22°C), weather (Mostly cloudy), and time (19:55:18 on 29-07-2025).

- **Result – we have two physical drive**

```

Select Administrator: Command Prompt
C:\Users\anike\CHFI\Module-4\dd-0.4beta4>wmic diskdrive list brief/format:list

Caption=JetFlash Transcend 4GB USB Device
DeviceID=\\.\PHYSICALDRIVE1
Model=JetFlash Transcend 4GB USB Device
Partitions=1
Size=4013936640

Caption=WDC PC SN530 SDBPNPZ-512G-1006
DeviceID=\\.\PHYSICALDRIVE0
Model=WDC PC SN530 SDBPNPZ-512G-1006
Partitions=5
Size=512105932800

C:\Users\anike\CHFI\Module-4\dd-0.4beta4>
  
```

- Now enter next command

Command -: `.\dd.exe if=\\.\PHYSICALDRIVE0
of=C:\Users\anike\Downloads\DD-Image\pendrive.dd bs=512k
--size --progress`

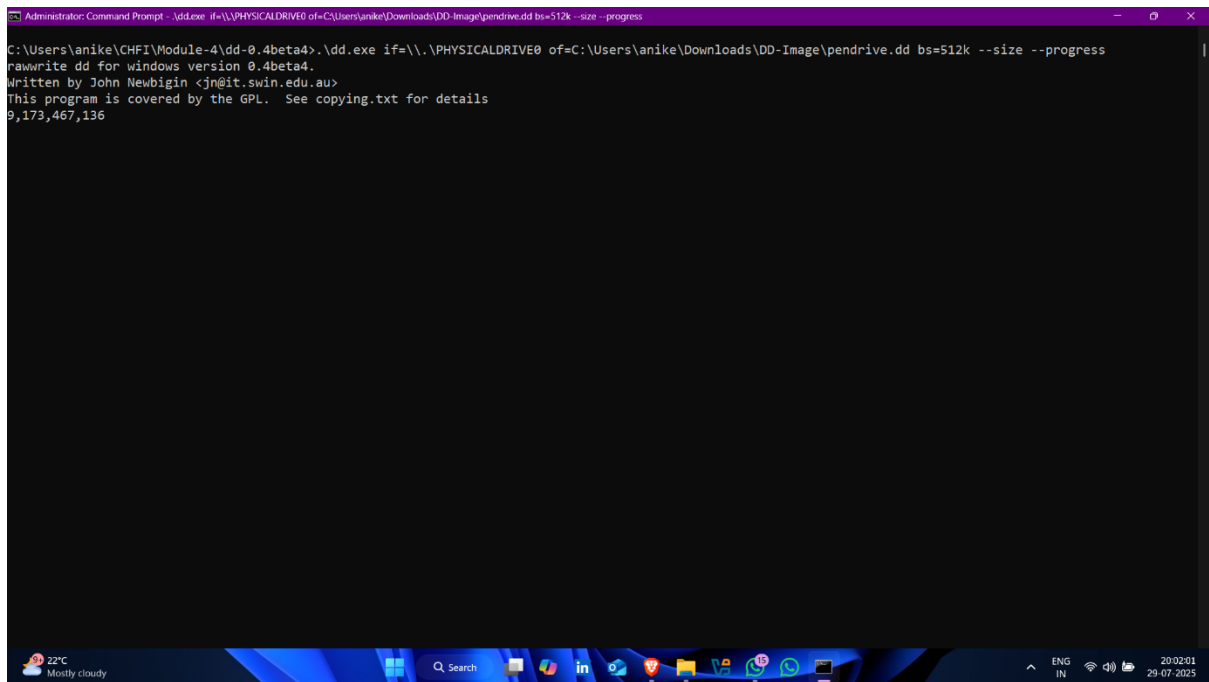
Explanation -:

Part	Meaning
<code>.\dd.exe</code>	Executes the dd.exe program from the current folder (probably in Module-4\dd-0.4beta4).
<code>if=\\.\PHYSICALDRIVE0</code>	Input file: raw access to your main internal SSD (PhysicalDrive0). ⚠ Be cautious — this is your system drive.
<code>of=C:\Users\anike\Downloads\DD-Image\pendrive.dd</code>	Output file: the image will be saved as pendrive.dd in

Part	Meaning
	the specified folder. (But the name is misleading if it's not really a pendrive!)
bs=512k	Block size: reads/writes 512 kilobytes at a time. Increases performance.
--size	Displays total size of the source drive before starting imaging.
--progress	Shows live progress: how much data has been copied, transfer rate, etc.

The screenshot shows a Windows Command Prompt window titled "Administrator: Command Prompt". The command entered is: `C:\Users\anike\CHFI\Module-4\dd-0.4beta4>.dd.exe if=\\.\PHYSICALDRIVE0 of=C:\Users\anike\Downloads\DD-Image\pendrive.dd bs=512k --size --progress`. The command prompt is currently blank, indicating the command has been executed but its output is not visible in the screenshot. The Windows taskbar at the bottom shows the date and time as 20:02:13 on 29-07-2025, along with system icons for network, volume, and battery.

- Image creating started

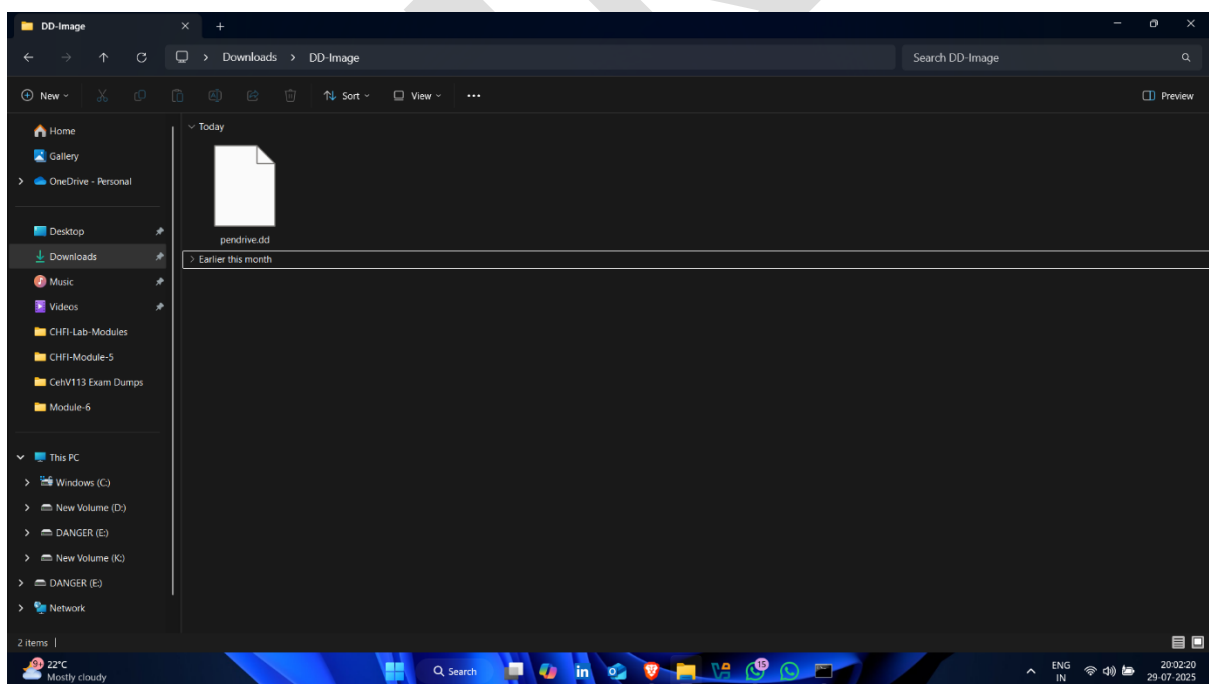


Administrator: Command Prompt - dd.exe if=\\.\PHYSICALDRIVE0 of=C:\Users\anike\Downloads\DD-Image\pendrive.dd bs=512k --size --progress

```
C:\Users\anike\CHFI\Module-4\dd-0.4beta4>.\dd.exe if=\\.\PHYSICALDRIVE0 of=C:\Users\anike\Downloads\DD-Image\pendrive.dd bs=512k --size --progress
rawwrite dd for windows version 0.4beta4.
Written by John Newbigin <jn@it.swin.edu.au>
This program is covered by the GPL. See copying.txt for details
9,173,467,136
```

The screenshot shows a Windows taskbar at the bottom with the system clock displaying 20:02:01 on 29-07-2025. The taskbar includes icons for weather (22°C, Mostly cloudy), search, and various application icons.

- Image Created



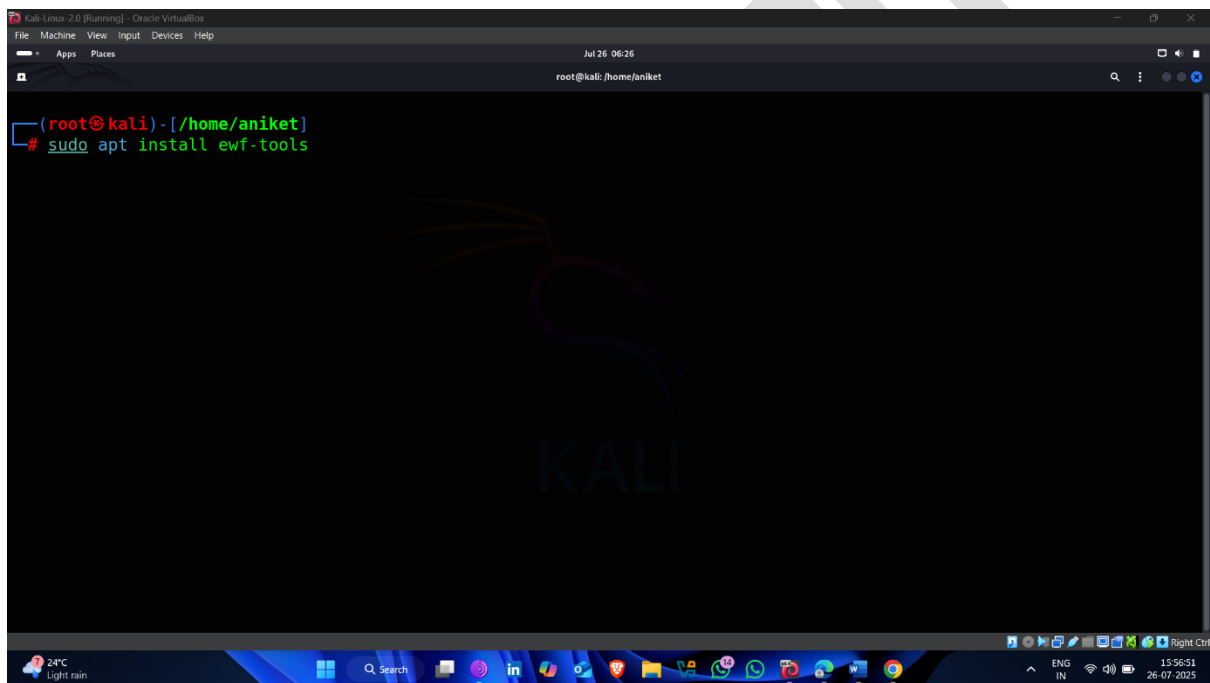
Lab-2 Convert Image File From DD format to E01 Format Using Ewfacquire

ewfacquire is a **forensic tool** used to acquire and store disk images in the **EWf (Expert Witness Format)** — also known as .E01 files.

It is part of the **libewf** toolkit, which is commonly used in **digital forensics**.

How to use it -:

- Open kali linux terminal and install **Ewf-tools**



- now , convert this .dd format image into E01 format

```

Kali Linux 2.0 [running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jul 26 06:29
root@kali:/home/aniket/DD-Image

(root@kali) - [/home/aniket]
# cd DD-Image

(root@kali) - [/home/aniket/DD-Image]
# ls
TestRawImage.dd

(root@kali) - [/home/aniket/DD-Image]
#

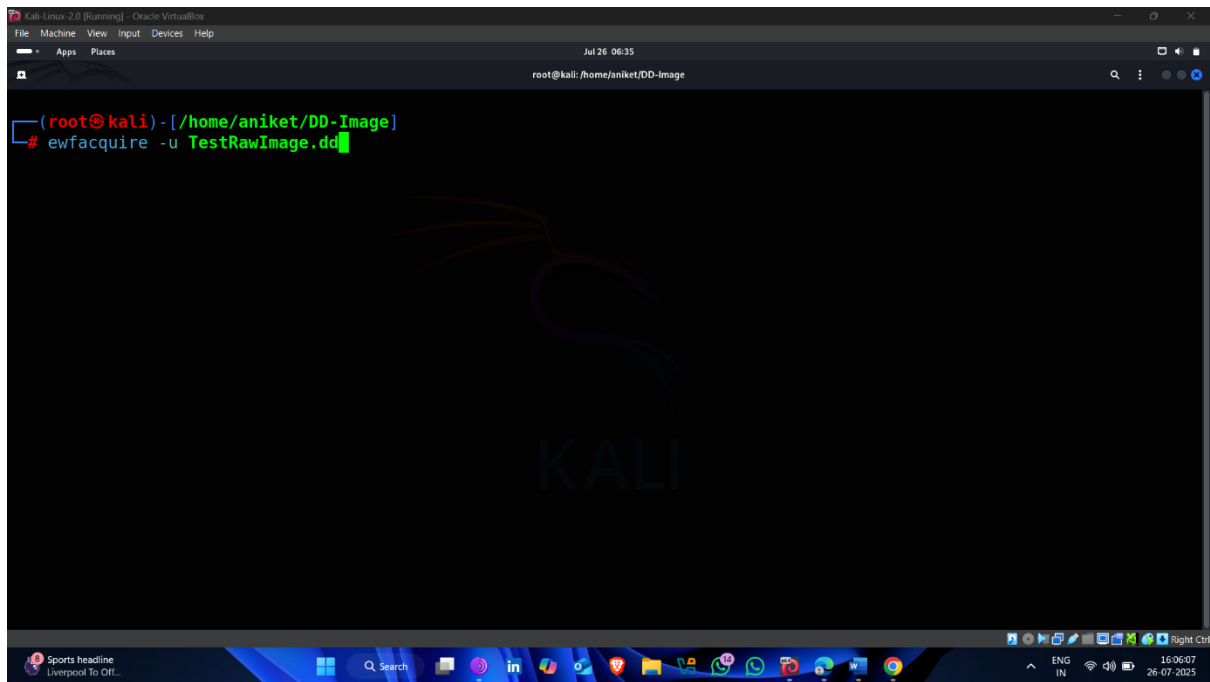
```

- Enter following command

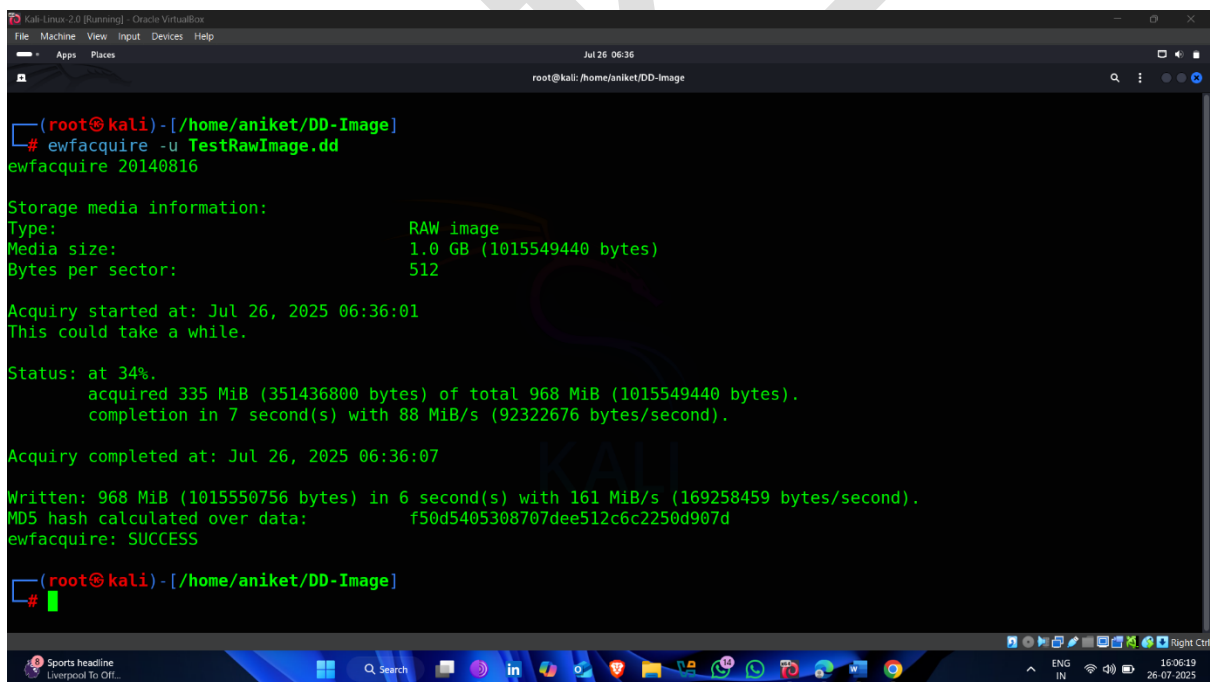
Command -: ewfacquire -u TestRawImage.dd

Explanation -:

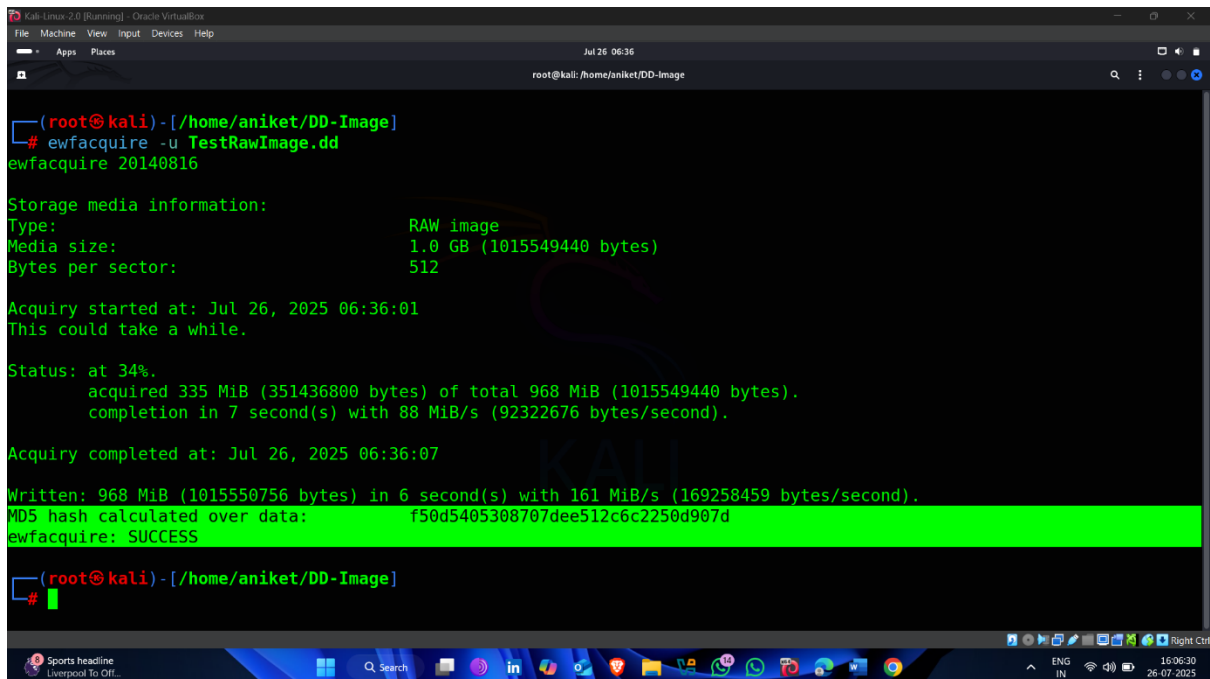
Part	Description
ewfacquire	The tool used to acquire (create) a forensic image in EWF (.E01) format.
-u	This option tells ewfacquire to create a raw image instead of EWF. In this mode, the image will be saved as a single raw file (like .dd) instead of .E01.
TestRawImage.dd	This is the output file name for the raw image you're creating. It will contain the full bit-for-bit copy of the target disk/device.



- **E01 Image creation Successfully** 🙌 ✅



- **MD5 Hash**



```
(root@kali)-[/home/aniket/DD-Image]
# ewf_acquire -u TestRawImage.dd
ewf_acquire 20140816

Storage media information:
Type: RAW image
Media size: 1.0 GB (1015549440 bytes)
Bytes per sector: 512

Acquiry started at: Jul 26, 2025 06:36:01
This could take a while.

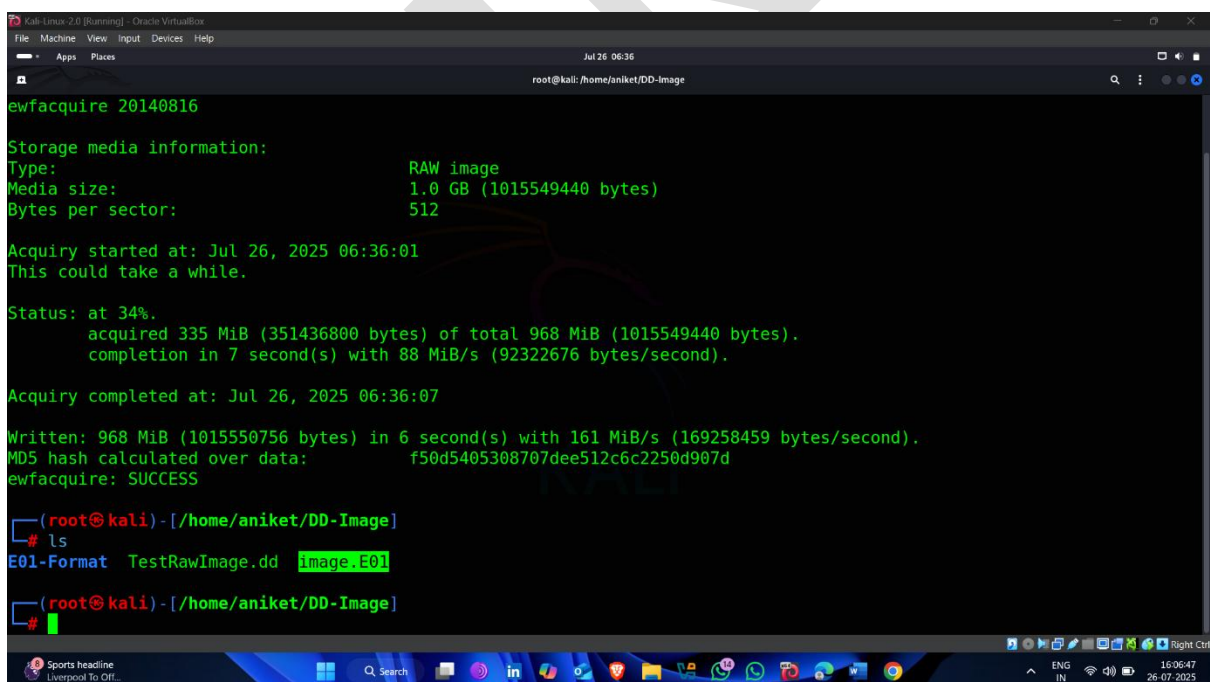
Status: at 34%.
        acquired 335 MiB (351436800 bytes) of total 968 MiB (1015549440 bytes).
        completion in 7 second(s) with 88 MiB/s (92322676 bytes/second).

Acquiry completed at: Jul 26, 2025 06:36:07

Written: 968 MiB (1015550756 bytes) in 6 second(s) with 161 MiB/s (169258459 bytes/second).
MD5 hash calculated over data: f50d5405308707dee512c6c2250d907d
ewf_acquire: SUCCESS

(root@kali)-[/home/aniket/DD-Image]
#
```

- **Image file**



```
ewf_acquire 20140816

Storage media information:
Type: RAW image
Media size: 1.0 GB (1015549440 bytes)
Bytes per sector: 512

Acquiry started at: Jul 26, 2025 06:36:01
This could take a while.

Status: at 34%.
        acquired 335 MiB (351436800 bytes) of total 968 MiB (1015549440 bytes).
        completion in 7 second(s) with 88 MiB/s (92322676 bytes/second).

Acquiry completed at: Jul 26, 2025 06:36:07

Written: 968 MiB (1015550756 bytes) in 6 second(s) with 161 MiB/s (169258459 bytes/second).
MD5 hash calculated over data: f50d5405308707dee512c6c2250d907d
ewf_acquire: SUCCESS

(root@kali)-[/home/aniket/DD-Image]
# ls
E01-Format TestRawImage.dd image.E01

(root@kali)-[/home/aniket/DD-Image]
#
```

Convert Image File From format to DD Format Using Xmount

xmount is a **forensic tool** used to **convert and mount disk image formats as virtual block devices** in Linux. It's especially useful for **digital forensics** and **incident response** when you want to analyze a disk image (like .E01, .dd, .vmdk, etc.) **without altering the original file**.

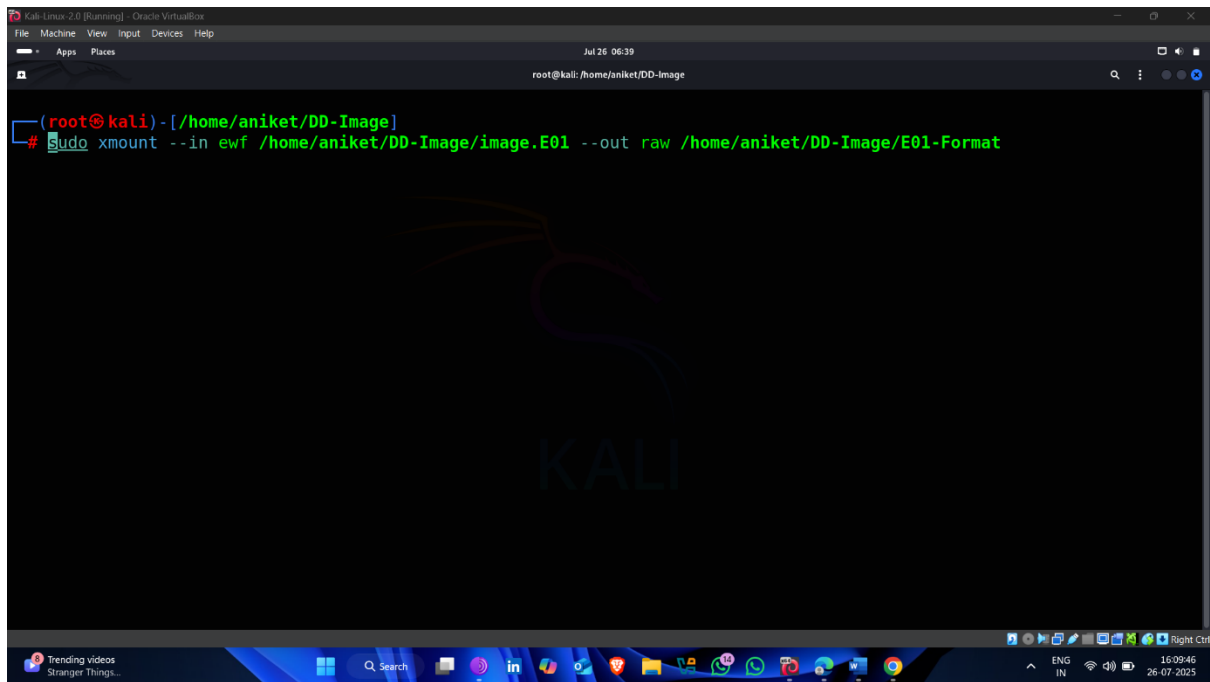
How to use it -:

- Open kali linux terminal and enter following command

Command -: `sudo xmount --in ewf /home/aniket/DD-Image/image.E01 --out raw "/home/aniket/DD-Image/E01 Format"`

Explanation -:

Part	Description
sudo	Required because mounting virtual devices needs root privileges
xmount	Main command to create a virtual block device
--in ewf	Input format is EWF (.E01 file)
/home/aniket/DD-Image/image.E01	Path to your .E01 forensic image
--out raw	Output will be in raw format (block device)
"/home/aniket/DD-Image/E01 Format"	Mount point directory (will contain virtual files like image.dd)

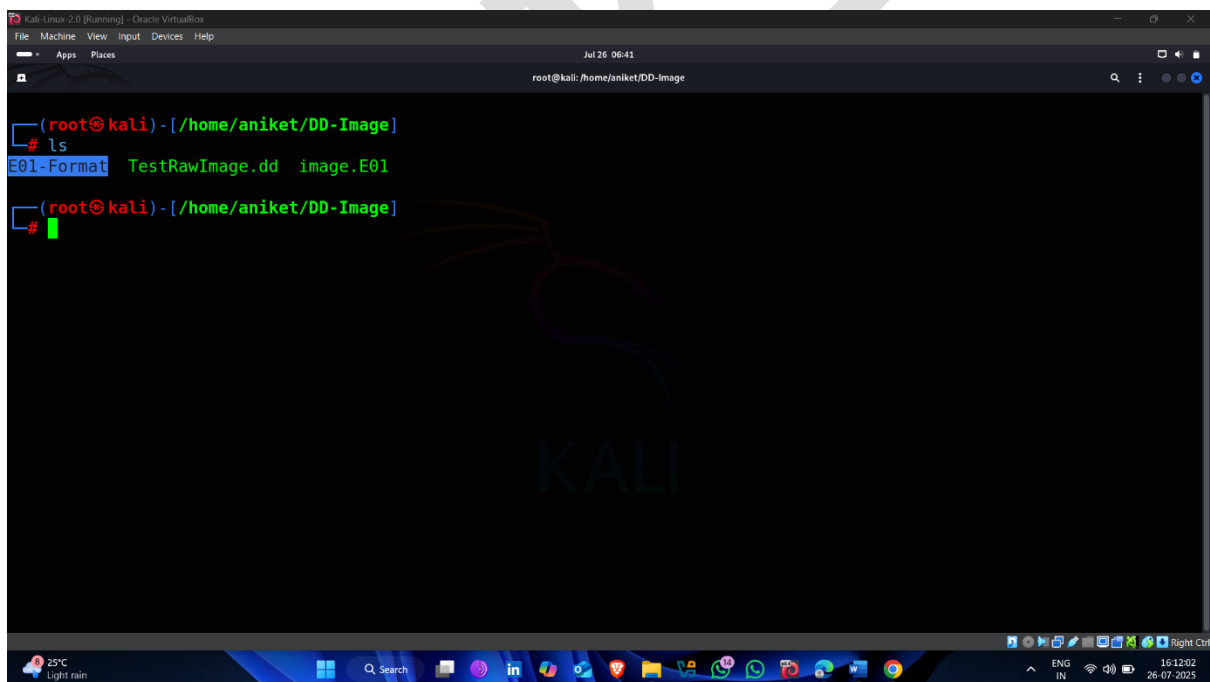


The screenshot shows a Kali Linux terminal window with the following content:

```
(root@kali) - [/home/aniket/DD-Image]
# sudo xmount --in ewf /home/aniket/DD-Image/image.E01 --out raw /home/aniket/DD-Image/E01-Format
```

The terminal window has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. The title bar indicates 'Kali Linux 2.0 [running] - Oracle VM VirtualBox'. The status bar at the bottom shows the date 'Jul 26 06:39', the user 'root@kali', and the path '/home/aniket/DD-Image'. A Windows taskbar is visible at the bottom of the screen.

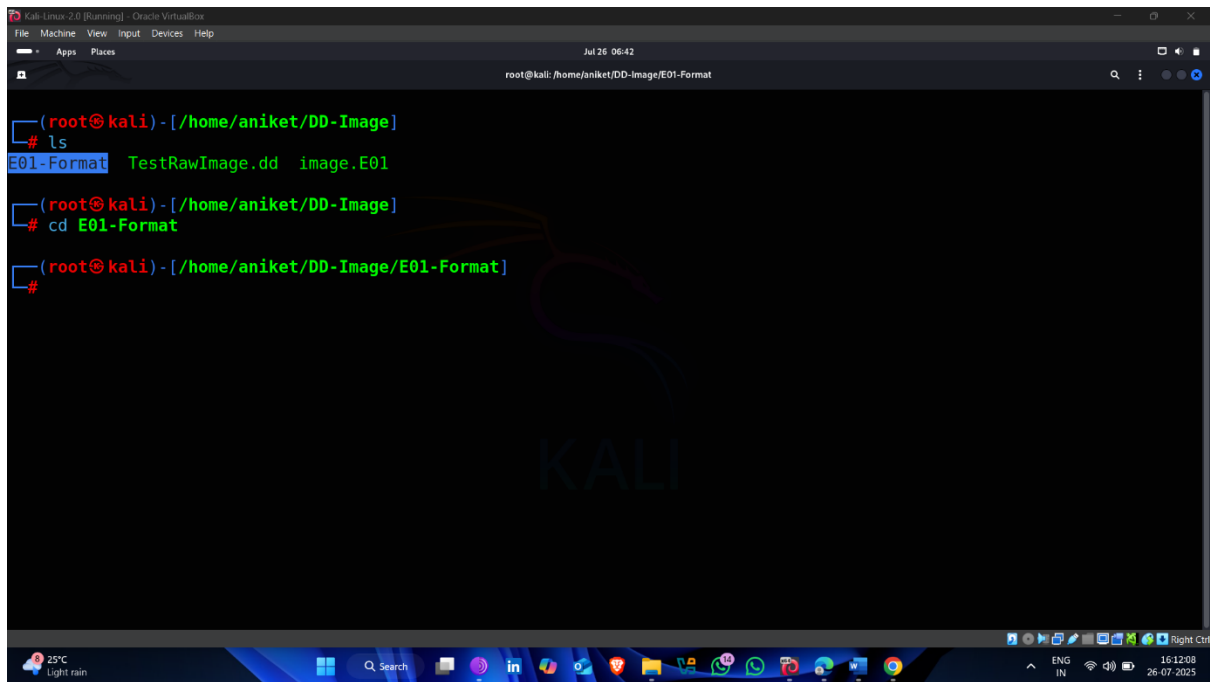
- Now open Directory that you give the output path



The screenshot shows a Kali Linux terminal window with the following content:

```
(root@kali) - [/home/aniket/DD-Image]
# ls
E01-Format  TestRawImage.dd  image.E01
(root@kali) - [/home/aniket/DD-Image]
#
```

The terminal window has a menu bar with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. The title bar indicates 'Kali Linux 2.0 [running] - Oracle VM VirtualBox'. The status bar at the bottom shows the date 'Jul 26 06:41', the user 'root@kali', and the path '/home/aniket/DD-Image'. A Windows taskbar is visible at the bottom of the screen.



A screenshot of a Kali Linux terminal window. The terminal shows the following commands and output:

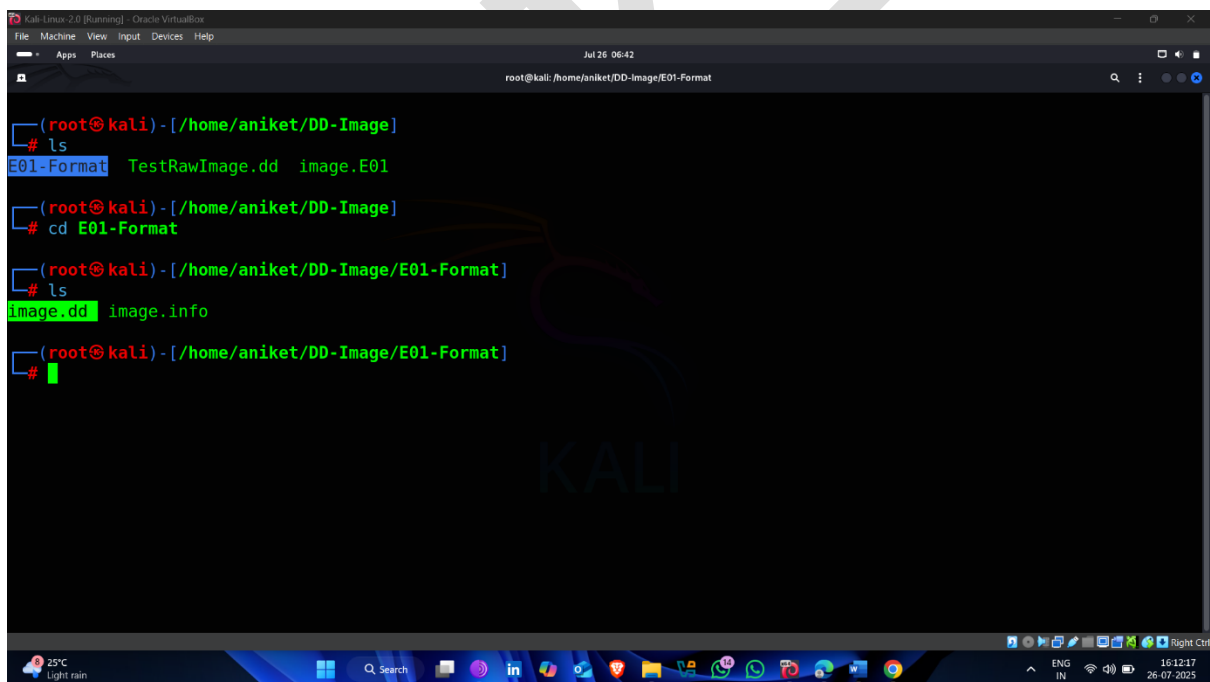
```
(root@kali) - [/home/aniket/DD-Image]
# ls
E01-Format  TestRawImage.dd  image.E01

(root@kali) - [/home/aniket/DD-Image]
# cd E01-Format

(root@kali) - [/home/aniket/DD-Image/E01-Format]
#
```

The terminal window has a title bar that says "Kali Linux 2.0 [Running] - Oracle VM VirtualBox". The system tray at the bottom shows the date and time as "Jul 26 06:42" and "16:12:08 26-07-2025".

- Here **.dd format** image Created



A screenshot of a Kali Linux terminal window. The terminal shows the following commands and output:

```
(root@kali) - [/home/aniket/DD-Image]
# ls
E01-Format  TestRawImage.dd  image.E01

(root@kali) - [/home/aniket/DD-Image]
# cd E01-Format

(root@kali) - [/home/aniket/DD-Image/E01-Format]
# ls
image.dd  image.info

(root@kali) - [/home/aniket/DD-Image/E01-Format]
#
```

The terminal window has a title bar that says "Kali Linux 2.0 [Running] - Oracle VM VirtualBox". The system tray at the bottom shows the date and time as "Jul 26 06:42" and "16:12:17 26-07-2025".

Lab-3:- Mount Image On A Linux Forensics Workstation.

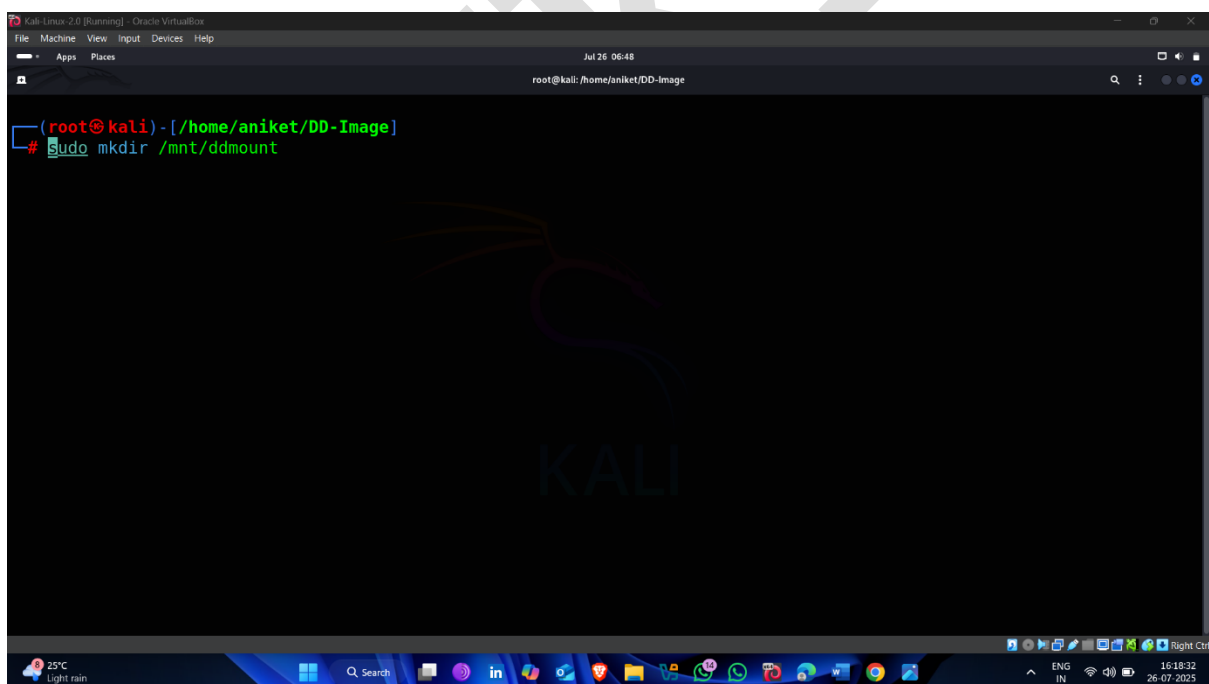
Mounting an image means making the **contents of a disk image file** (like .iso, .dd, .img, .E01, etc.) **accessible as if it were a physical drive** attached to your system.

A) Mount Image On A Linux Forensics Workstation Using Mount command

the mount command is used to **attach a file system** (like a disk image) to a directory so you can access its contents. Here's a detailed explanation specific to **mounting forensic image files** like .dd, .img, .iso, or from tools like ewfacquire.

How to use it -:

- Enter Kali linux terminal and make following Directory



- Directory Created without any error

```

Kali Linux 2.0 [running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Jul 26 06:48
root@kali: /home/aniket/DD-Image

(root@kali) - [/home/aniket/DD-Image]
# sudo mkdir /mnt/ddmount

(root@kali) - [/home/aniket/DD-Image]
#
  
```

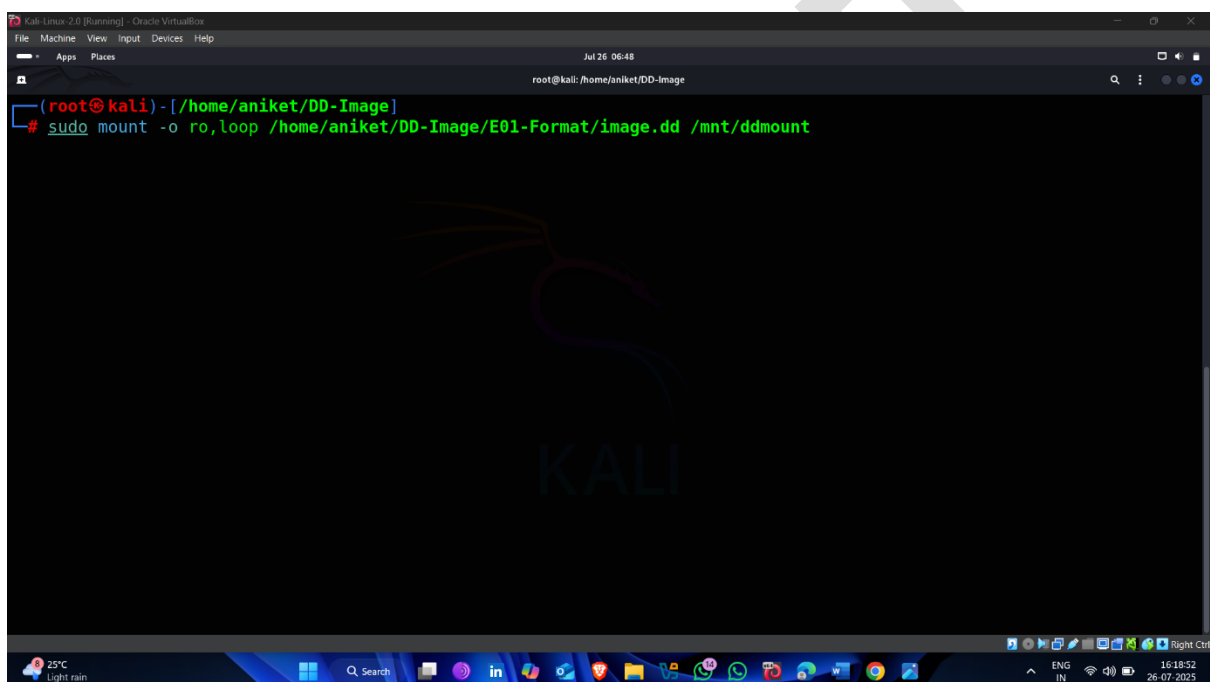
- Now enter following command

Command -: `mount -o ro,loop /home/aniket/DD-image/E01-Format/Image.dd /mnt/ddformat`

Explanation -:

Part	Description
sudo	Runs the command with root privileges (required for mounting).
mount	Main Linux command to attach a filesystem to a mount point.
-o	Option flag to pass additional mount parameters.
loop	Tells Linux to treat the .dd file as a loopback device (i.e., a virtual block device).
ro	Mount the image in read-only mode (safe for forensics).

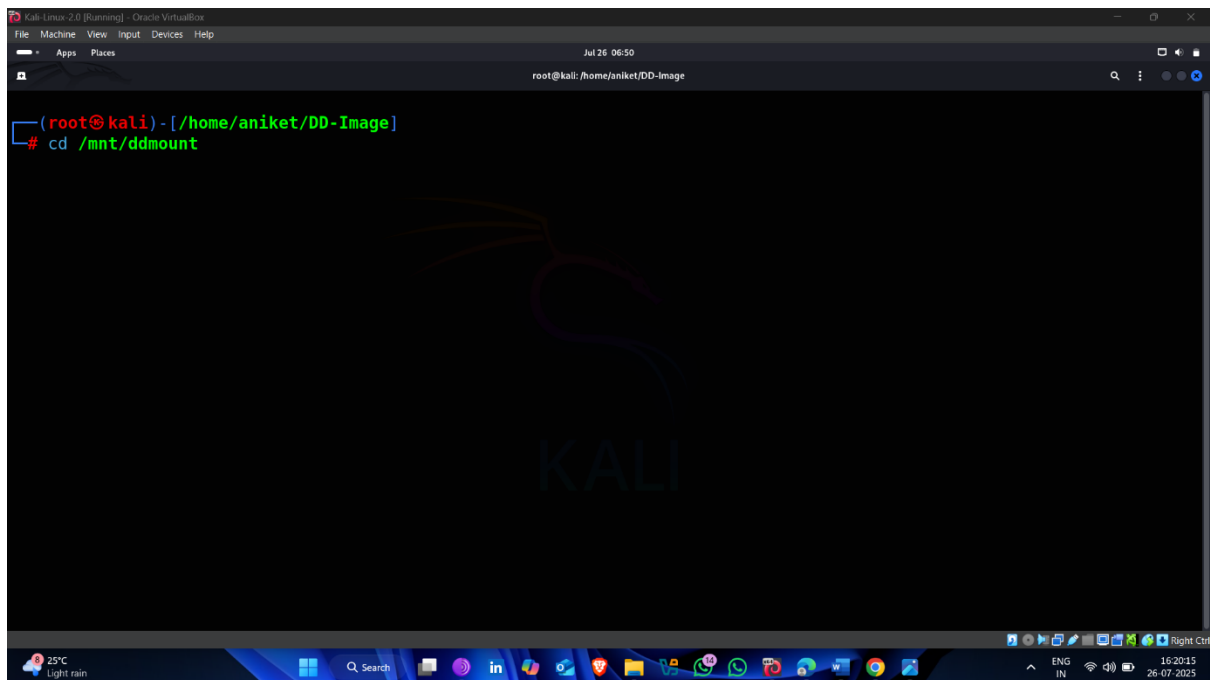
Part	Description
/home/aniket/DD-image/E01-Format/Image.dd	Path to the raw disk image you want to mount.
/mnt/ddformat	Directory where the image will be mounted and accessed like a normal file system.



The screenshot shows a terminal window titled 'Kali Linux 2.0 [running] - Oracle VM VirtualBox'. The prompt is 'root@kali: /home/aniket/DD-Image'. The command entered is `sudo mount -o ro,loop /home/aniket/DD-Image/E01-Format/image.dd /mnt/ddmount`. The terminal background features a large, faint 'KALI' logo. The window's taskbar at the bottom shows various application icons and system status information including '25°C Light rain', '16:18:52', and '26-07-2025'.

```
(root@kali) - [/home/aniket/DD-Image]
# sudo mount -o ro,loop /home/aniket/DD-Image/E01-Format/image.dd /mnt/ddmount
```

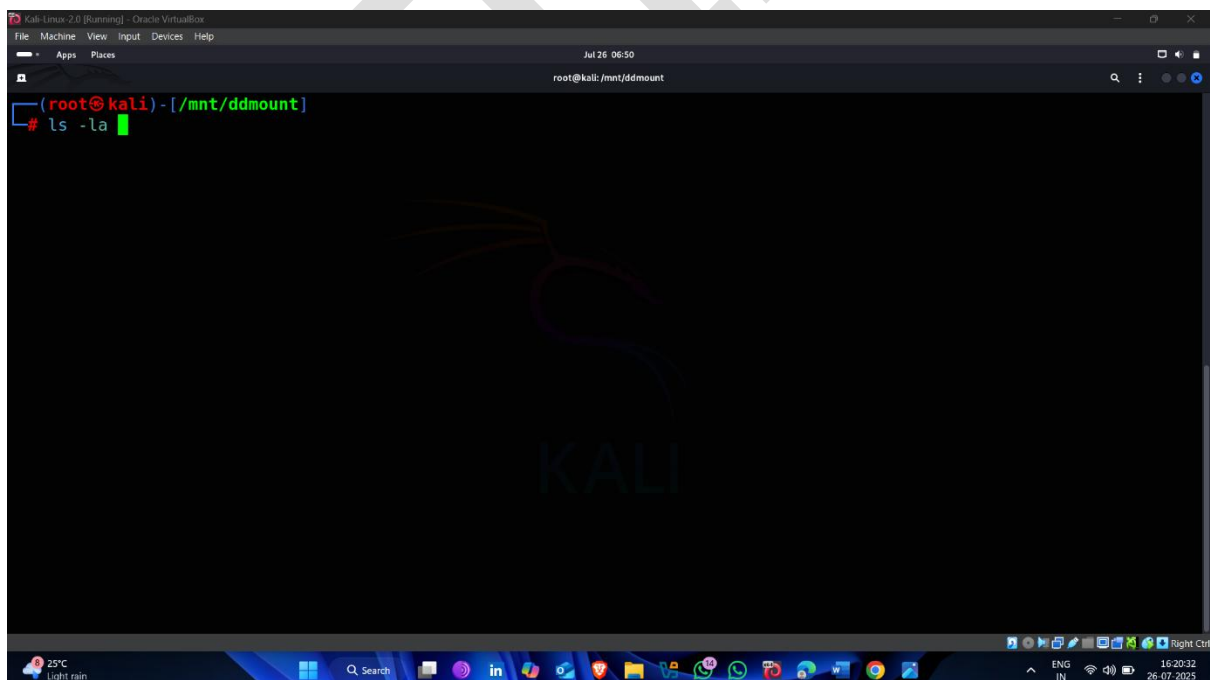
- After successfully run above command
- Go to /mnt/ddmount directory



A screenshot of a Kali Linux terminal window. The window title is "Kali Linux 2.0 [Running] - Oracle VM VirtualBox". The menu bar includes "File", "Machine", "View", "Input", "Devices", and "Help". The address bar shows "root@kali: /home/aniket/DD-Image". The terminal prompt is "(root@kali) - [/home/aniket/DD-Image]". The command "# cd /mnt/ddmount" has been entered and executed. The background of the terminal is black with a faint "KALI" logo. The bottom of the window shows a Windows taskbar with various icons and a system tray displaying "25°C Light rain" and the date "16:20:15 26-07-2025".

```
(root@kali) - [/home/aniket/DD-Image]
# cd /mnt/ddmount
```

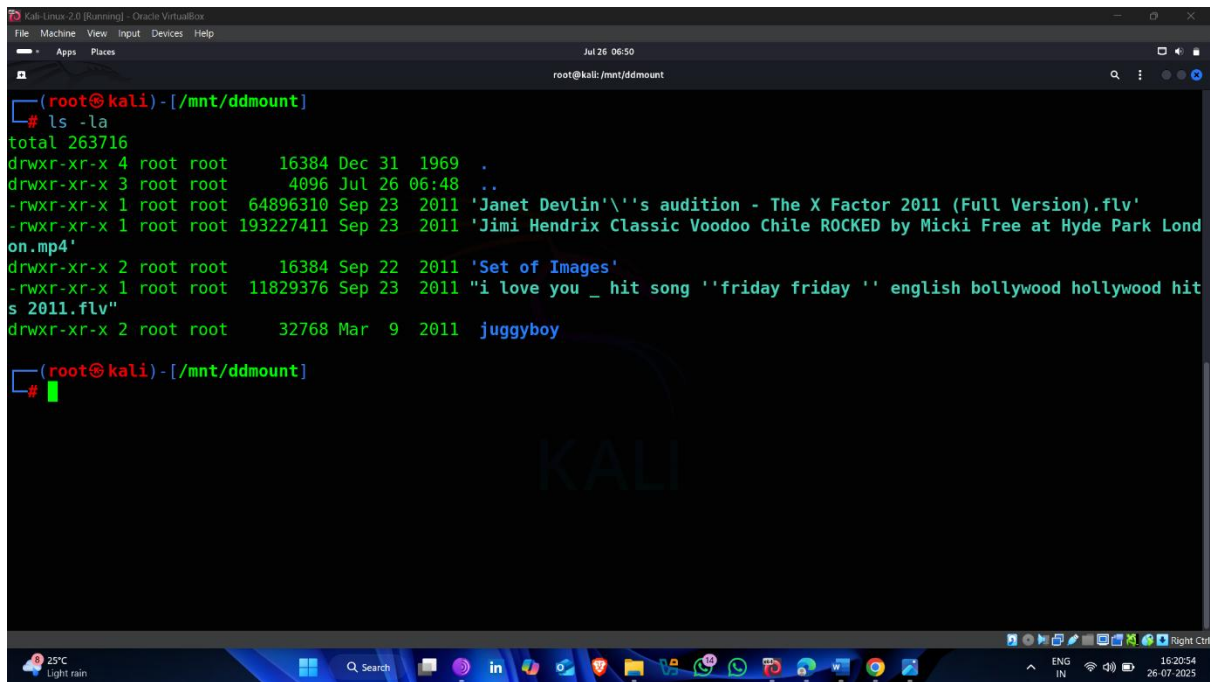
- Enter ls command



A screenshot of a Kali Linux terminal window. The window title is "Kali Linux 2.0 [Running] - Oracle VM VirtualBox". The menu bar includes "File", "Machine", "View", "Input", "Devices", and "Help". The address bar shows "root@kali: /mnt/ddmount". The terminal prompt is "(root@kali) - [/mnt/ddmount]". The command "# ls -la" has been entered and executed. The background of the terminal is black with a faint "KALI" logo. The bottom of the window shows a Windows taskbar with various icons and a system tray displaying "25°C Light rain" and the date "16:20:32 26-07-2025".

```
(root@kali) - [/mnt/ddmount]
# ls -la
```

- Here image mounted

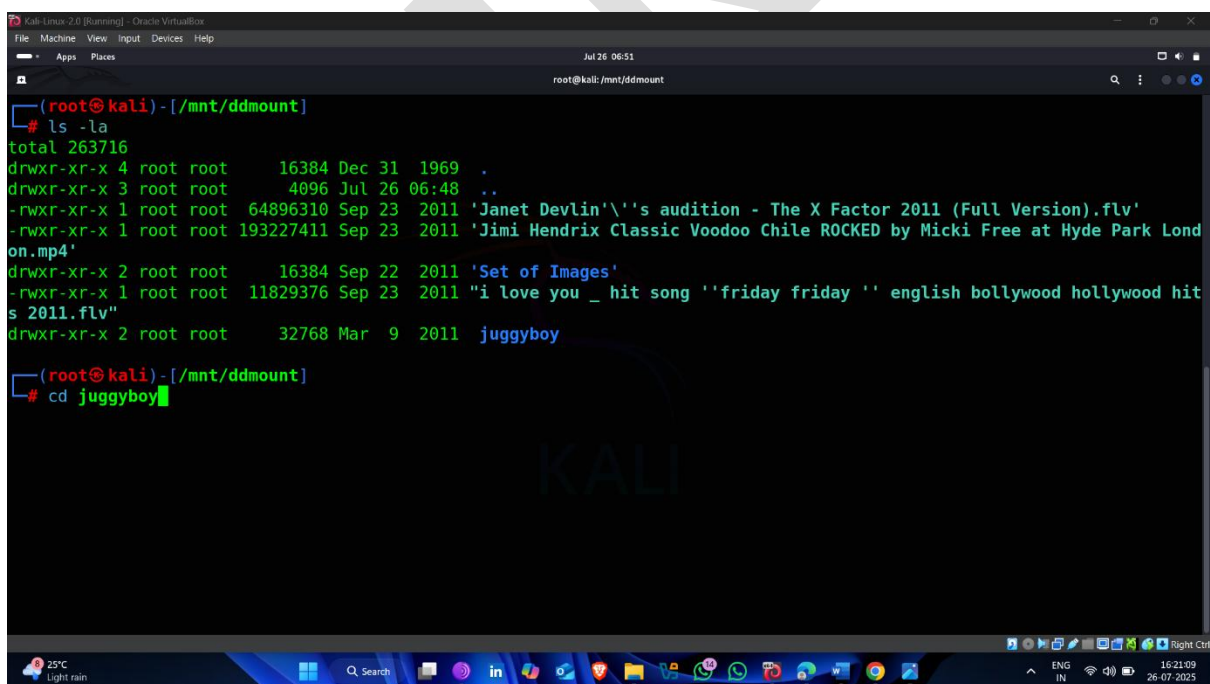


The screenshot shows a Kali Linux terminal window with the title bar 'Kali Linux 2.0 [Running] - Oracle VM VirtualBox'. The terminal prompt is '(root@kali)-[/mnt/ddmount]'. The user has entered the command 'ls -la', and the output is as follows:

```
total 263716
drwxr-xr-x 4 root root    16384 Dec 31 1969  .
drwxr-xr-x 3 root root     4096 Jul 26 06:48  ..
-rwxr-xr-x 1 root root 64896310 Sep 23 2011 'Janet Devlin\'\'s audition - The X Factor 2011 (Full Version).flv'
-rwxr-xr-x 1 root root 193227411 Sep 23 2011 'Jimi Hendrix Classic Voodoo Chile ROCKED by Micki Free at Hyde Park London.mp4'
drwxr-xr-x 2 root root    16384 Sep 22 2011 'Set of Images'
-rwxr-xr-x 1 root root 11829376 Sep 23 2011 "i love you _ hit song 'friday friday ' english bollywood hollywood hits 2011.flv"
drwxr-xr-x 2 root root    32768 Mar  9 2011 juggyboy
```

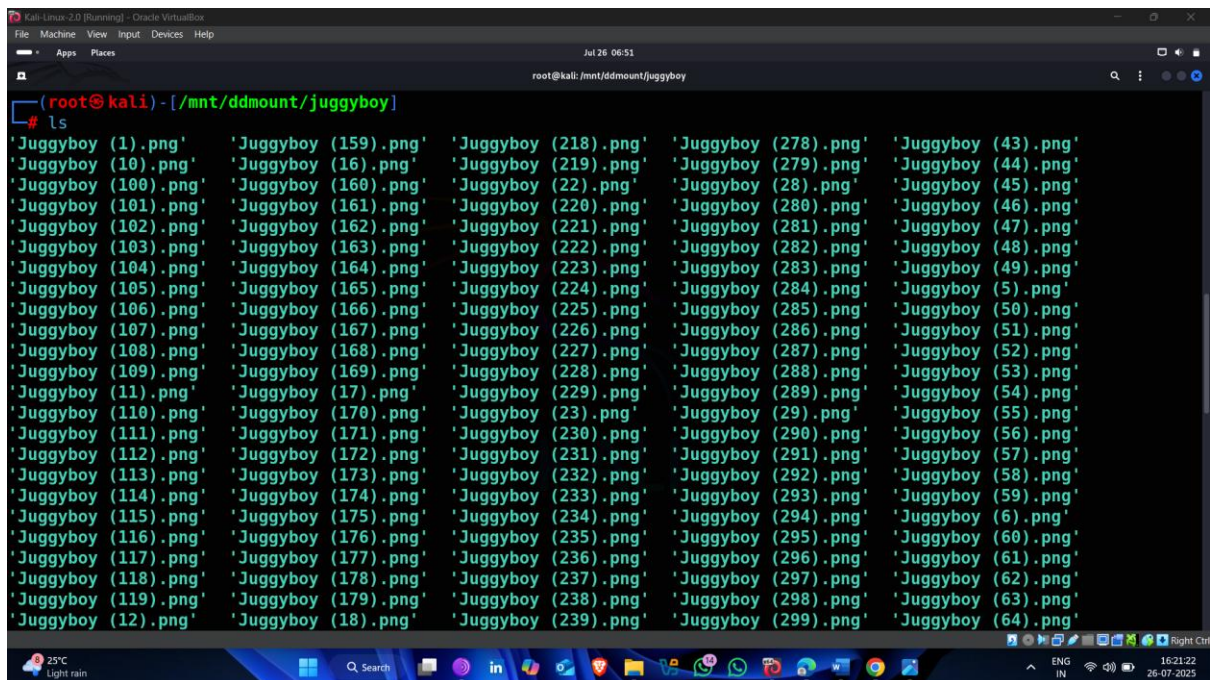
The terminal window also shows a taskbar at the bottom with various application icons and system status information like '25°C Light rain' and '16:20:54 26-07-2025'.

- Now go to directory for access files



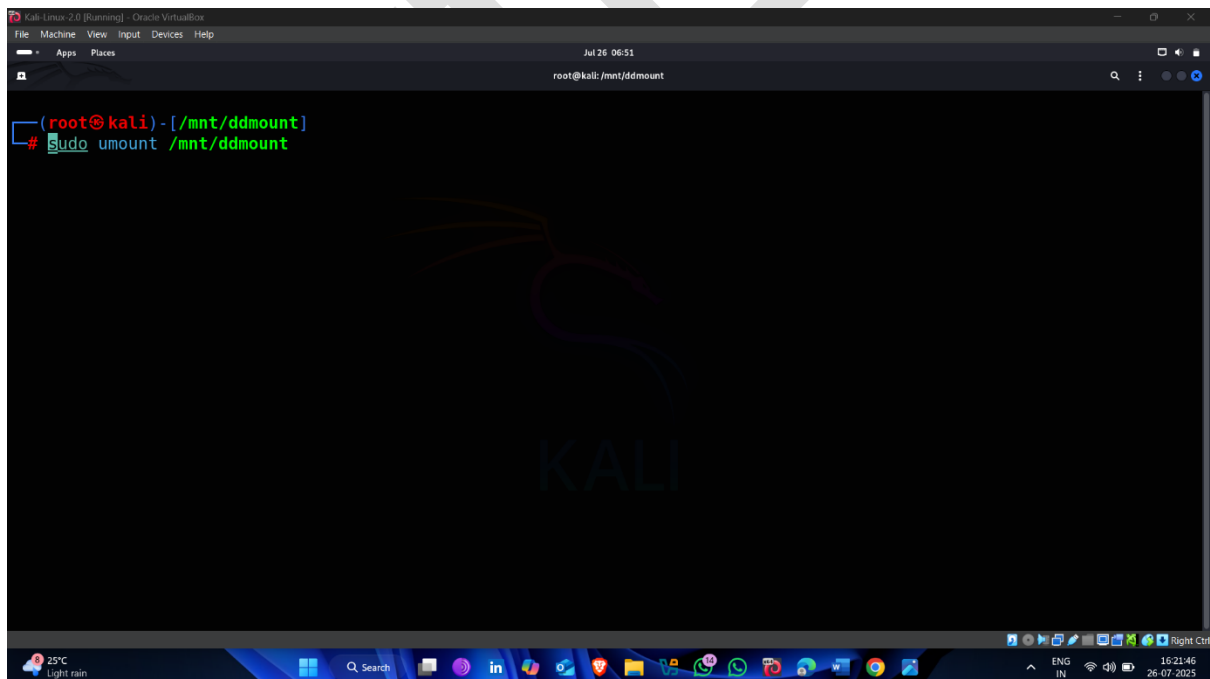
The screenshot shows the same Kali Linux terminal window. The user has entered the command 'cd juggyboy', and the terminal prompt has changed to '(root@kali)-[/mnt/ddmount/juggyboy]'. The output of the 'ls -la' command is the same as in the previous screenshot.

- Here all the data visible ✓👉



```
(root@kali) - [/mnt/ddmount/juggyboy]
# ls
'Juggyboy (1).png'  'Juggyboy (159).png'  'Juggyboy (218).png'  'Juggyboy (278).png'  'Juggyboy (43).png'
'Juggyboy (10).png'  'Juggyboy (16).png'  'Juggyboy (219).png'  'Juggyboy (279).png'  'Juggyboy (44).png'
'Juggyboy (100).png'  'Juggyboy (160).png'  'Juggyboy (22).png'  'Juggyboy (28).png'  'Juggyboy (45).png'
'Juggyboy (101).png'  'Juggyboy (161).png'  'Juggyboy (220).png'  'Juggyboy (280).png'  'Juggyboy (46).png'
'Juggyboy (102).png'  'Juggyboy (162).png'  'Juggyboy (221).png'  'Juggyboy (281).png'  'Juggyboy (47).png'
'Juggyboy (103).png'  'Juggyboy (163).png'  'Juggyboy (222).png'  'Juggyboy (282).png'  'Juggyboy (48).png'
'Juggyboy (104).png'  'Juggyboy (164).png'  'Juggyboy (223).png'  'Juggyboy (283).png'  'Juggyboy (49).png'
'Juggyboy (105).png'  'Juggyboy (165).png'  'Juggyboy (224).png'  'Juggyboy (284).png'  'Juggyboy (5).png'
'Juggyboy (106).png'  'Juggyboy (166).png'  'Juggyboy (225).png'  'Juggyboy (285).png'  'Juggyboy (50).png'
'Juggyboy (107).png'  'Juggyboy (167).png'  'Juggyboy (226).png'  'Juggyboy (286).png'  'Juggyboy (51).png'
'Juggyboy (108).png'  'Juggyboy (168).png'  'Juggyboy (227).png'  'Juggyboy (287).png'  'Juggyboy (52).png'
'Juggyboy (109).png'  'Juggyboy (169).png'  'Juggyboy (228).png'  'Juggyboy (288).png'  'Juggyboy (53).png'
'Juggyboy (11).png'  'Juggyboy (17).png'  'Juggyboy (229).png'  'Juggyboy (289).png'  'Juggyboy (54).png'
'Juggyboy (110).png'  'Juggyboy (170).png'  'Juggyboy (23).png'  'Juggyboy (29).png'  'Juggyboy (55).png'
'Juggyboy (111).png'  'Juggyboy (171).png'  'Juggyboy (230).png'  'Juggyboy (290).png'  'Juggyboy (56).png'
'Juggyboy (112).png'  'Juggyboy (172).png'  'Juggyboy (231).png'  'Juggyboy (291).png'  'Juggyboy (57).png'
'Juggyboy (113).png'  'Juggyboy (173).png'  'Juggyboy (232).png'  'Juggyboy (292).png'  'Juggyboy (58).png'
'Juggyboy (114).png'  'Juggyboy (174).png'  'Juggyboy (233).png'  'Juggyboy (293).png'  'Juggyboy (59).png'
'Juggyboy (115).png'  'Juggyboy (175).png'  'Juggyboy (234).png'  'Juggyboy (294).png'  'Juggyboy (6).png'
'Juggyboy (116).png'  'Juggyboy (176).png'  'Juggyboy (235).png'  'Juggyboy (295).png'  'Juggyboy (60).png'
'Juggyboy (117).png'  'Juggyboy (177).png'  'Juggyboy (236).png'  'Juggyboy (296).png'  'Juggyboy (61).png'
'Juggyboy (118).png'  'Juggyboy (178).png'  'Juggyboy (237).png'  'Juggyboy (297).png'  'Juggyboy (62).png'
'Juggyboy (119).png'  'Juggyboy (179).png'  'Juggyboy (238).png'  'Juggyboy (298).png'  'Juggyboy (63).png'
'Juggyboy (12).png'  'Juggyboy (18).png'  'Juggyboy (239).png'  'Juggyboy (299).png'  'Juggyboy (64).png'
```

- Now unmount image using following command 👉



```
(root@kali) - [/mnt/ddmount]
# sudo umount /mnt/ddmount
```

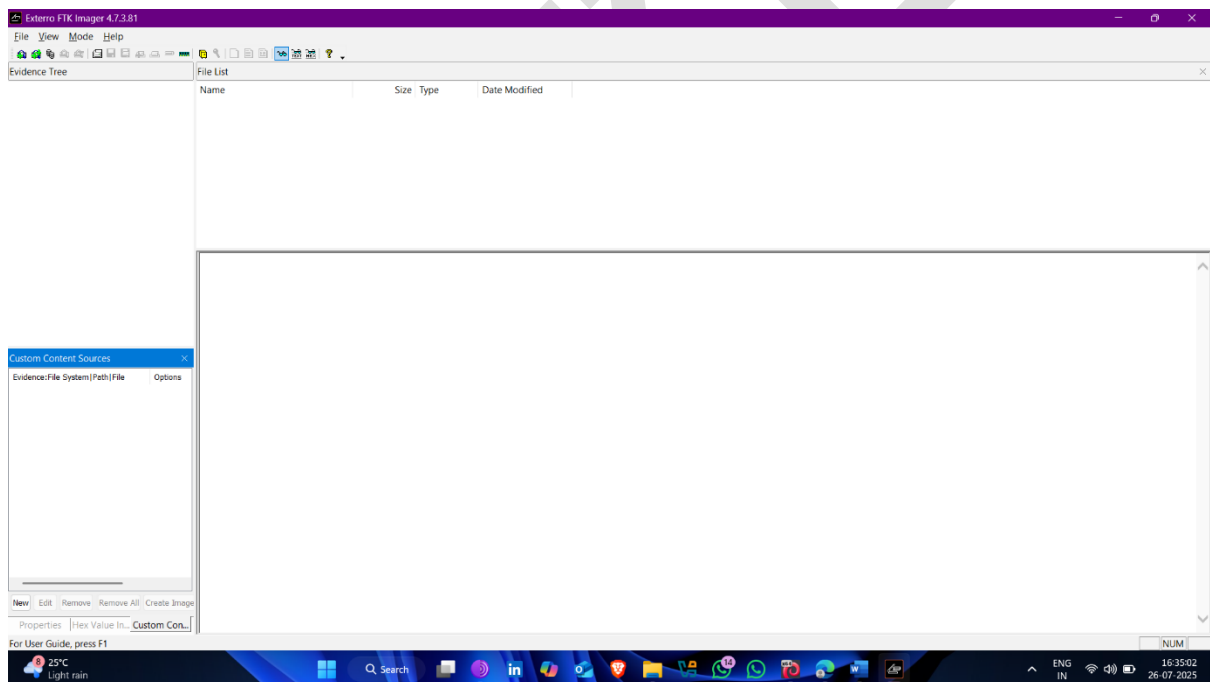
A) Mount Image On A Windows Forensics Workstation Using FTK Imager

FTK Imager (Forensic Toolkit Imager) is a **forensic imaging and analysis tool** developed by **AccessData (Exterro)**. It's widely used in **digital forensics** to:

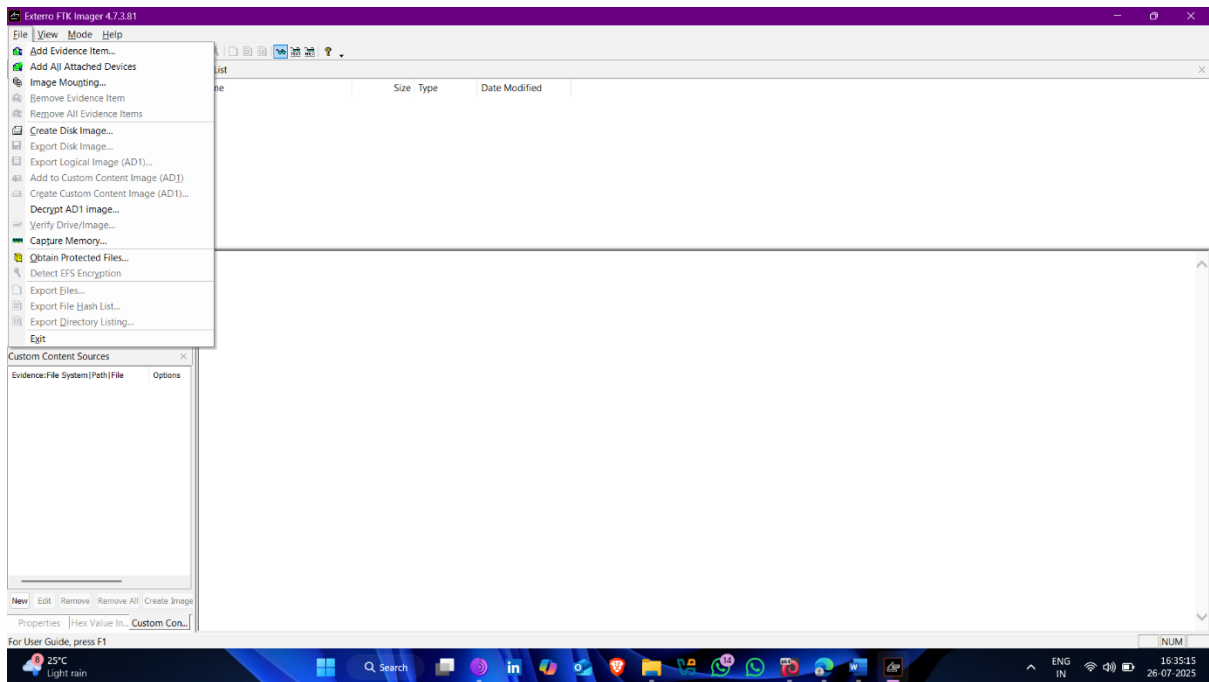
- Acquire (create) forensic disk images.
- **Mount and analyze disk images (like .E01, .dd, .img, etc.).**
- Export files, folders, and registry keys.
- Create forensic hashes (MD5, SHA1).

How to use it -:

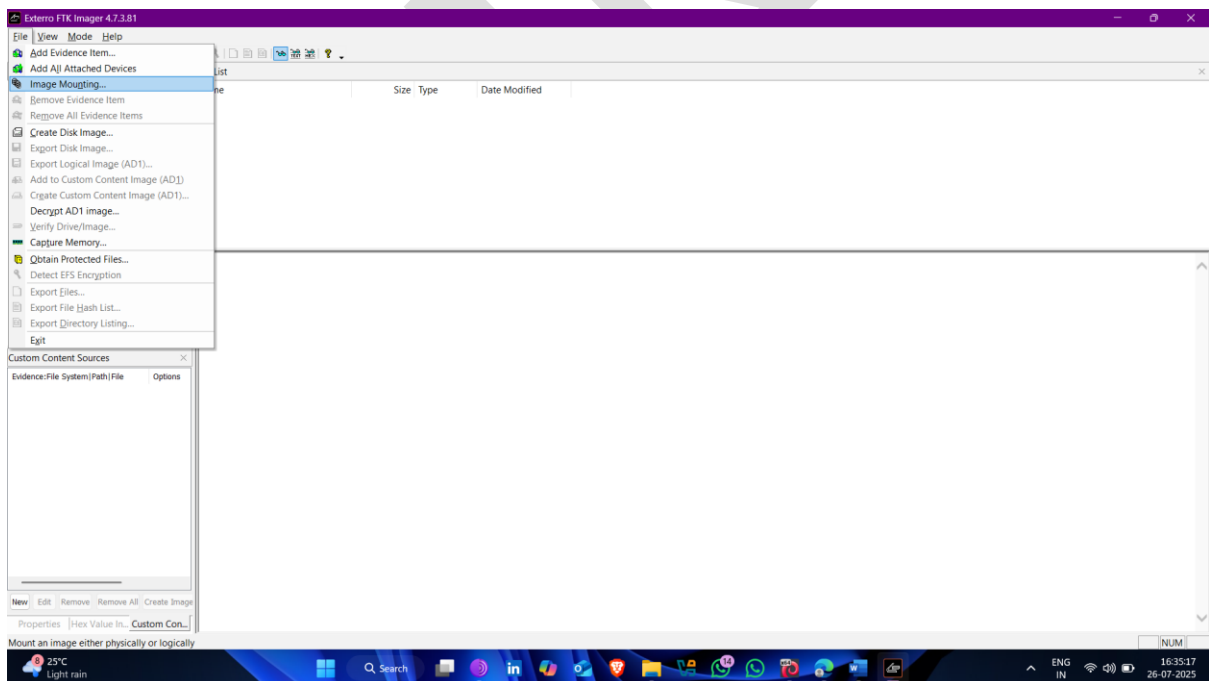
- Open FTK imager



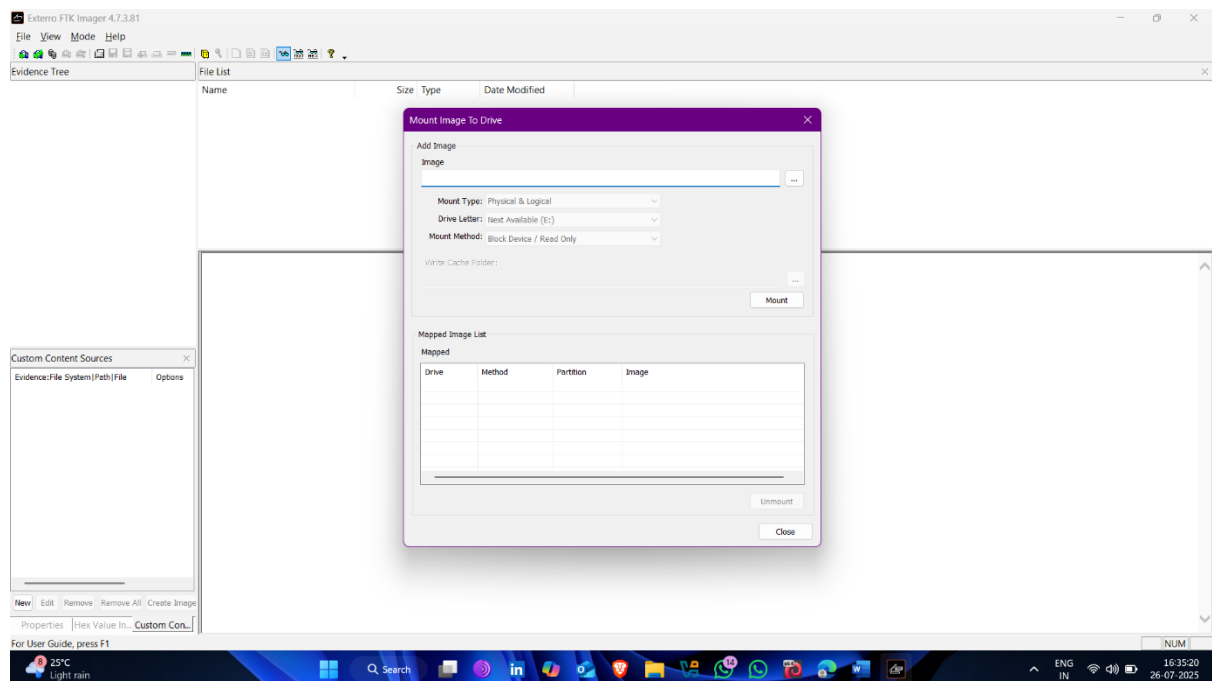
- Click on File option



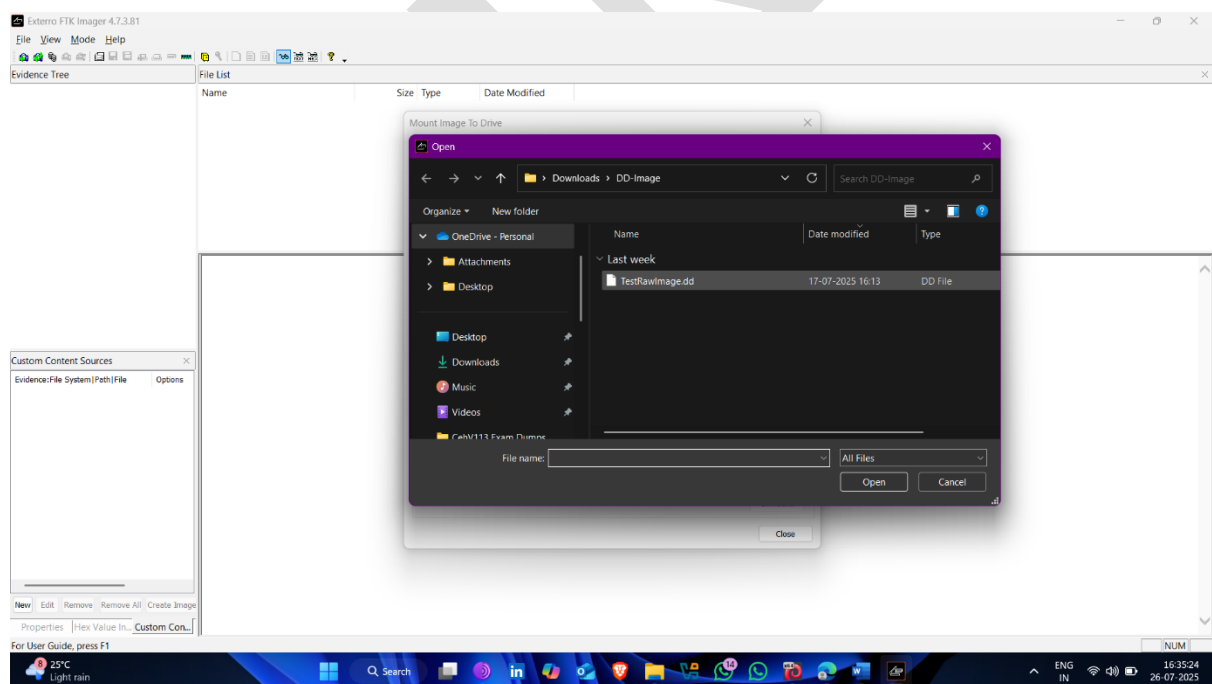
- Now click on **image mounting**



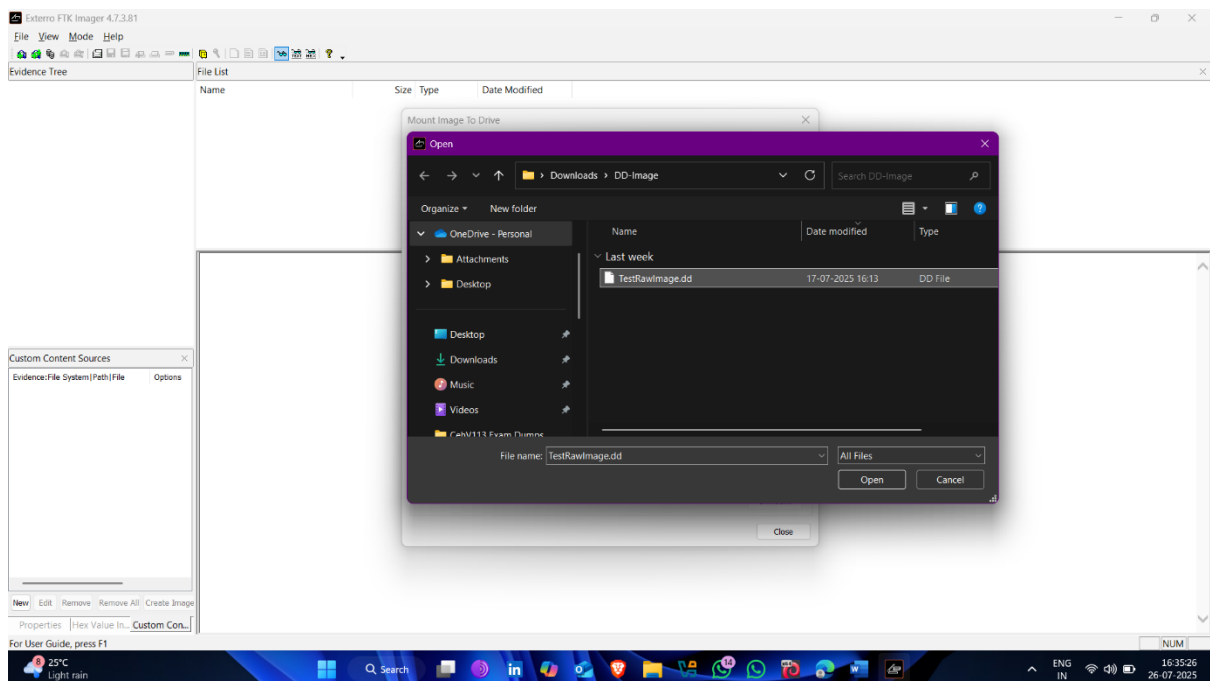
- Add Image file



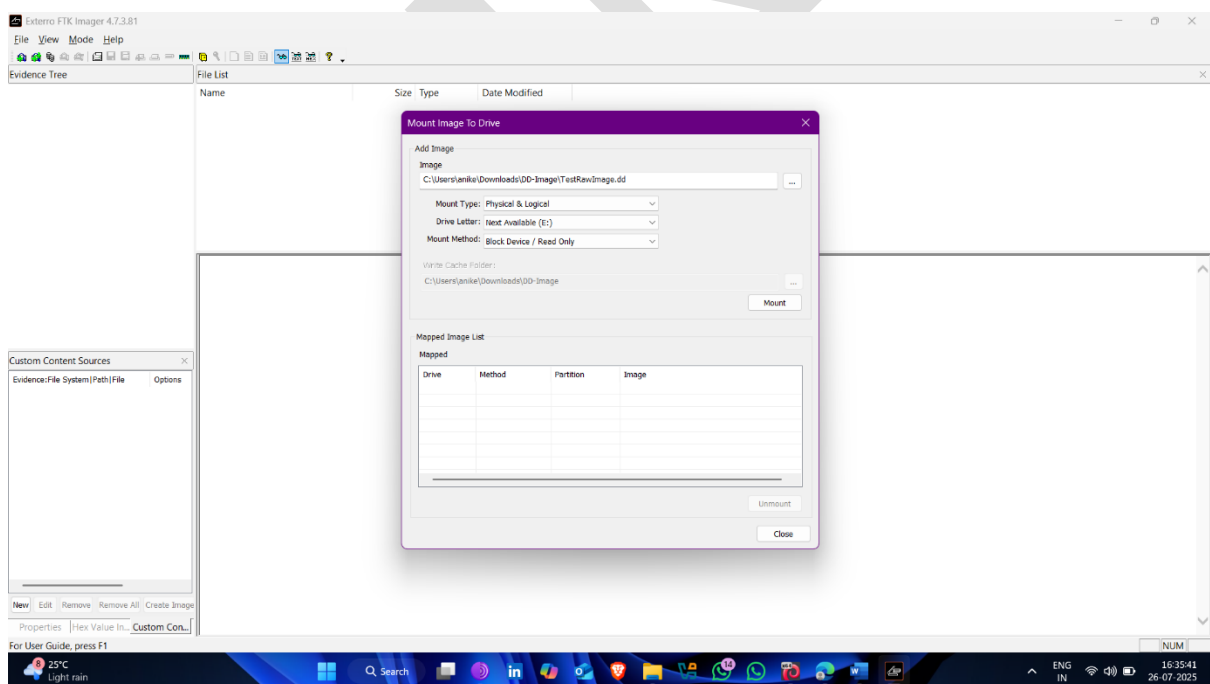
- Select Image



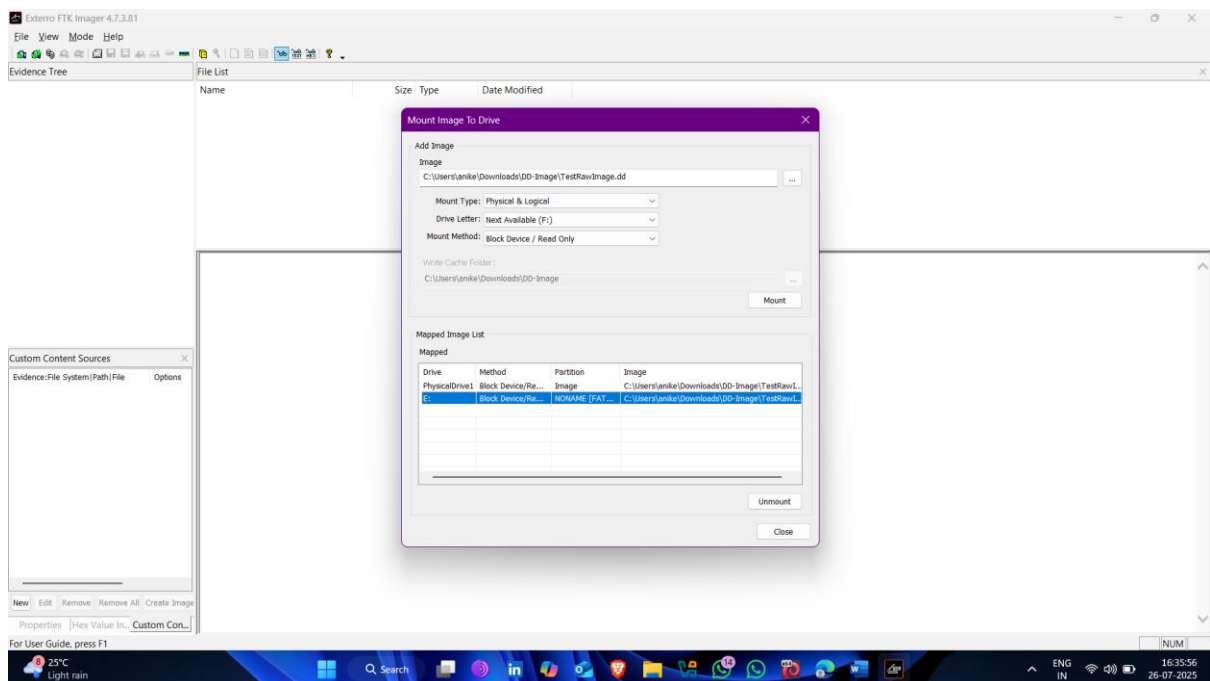
- Click on Open



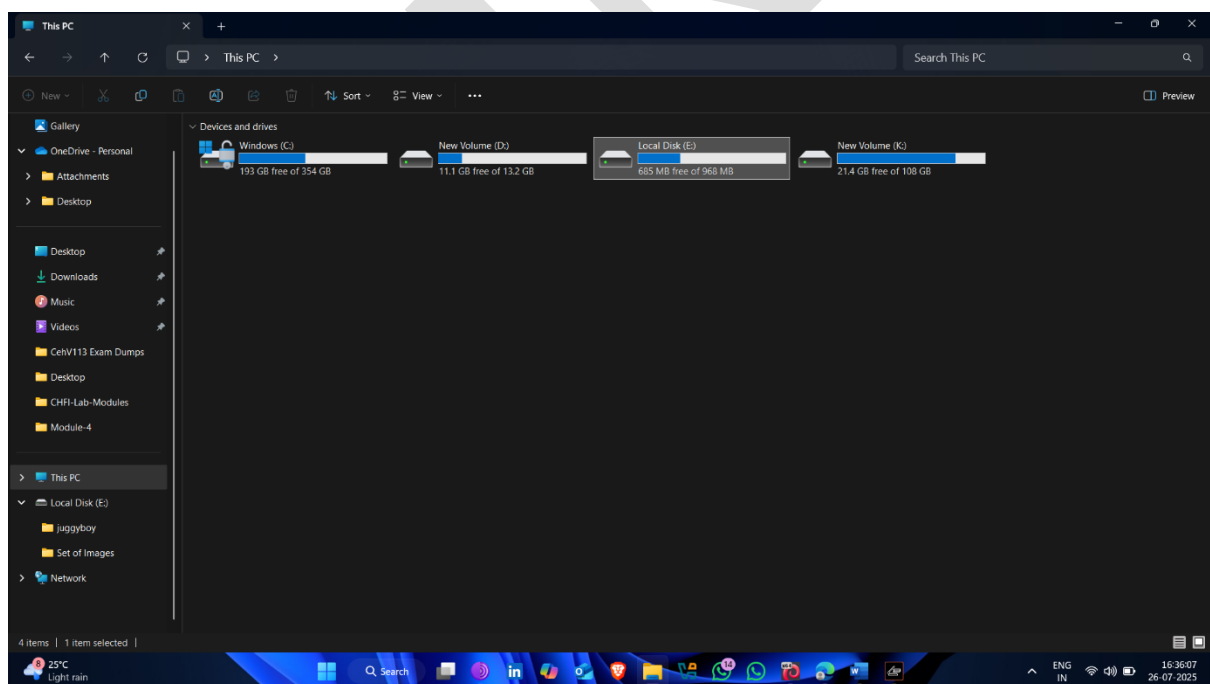
- Click on **Mount**



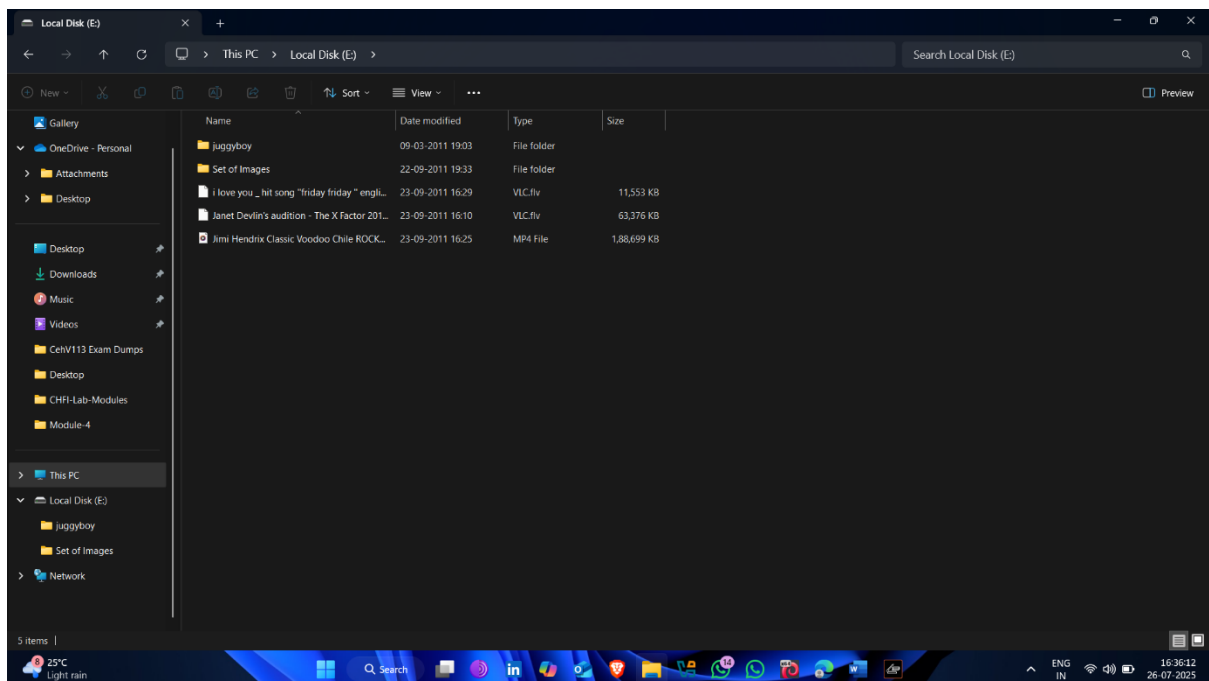
- Mounted Successfully



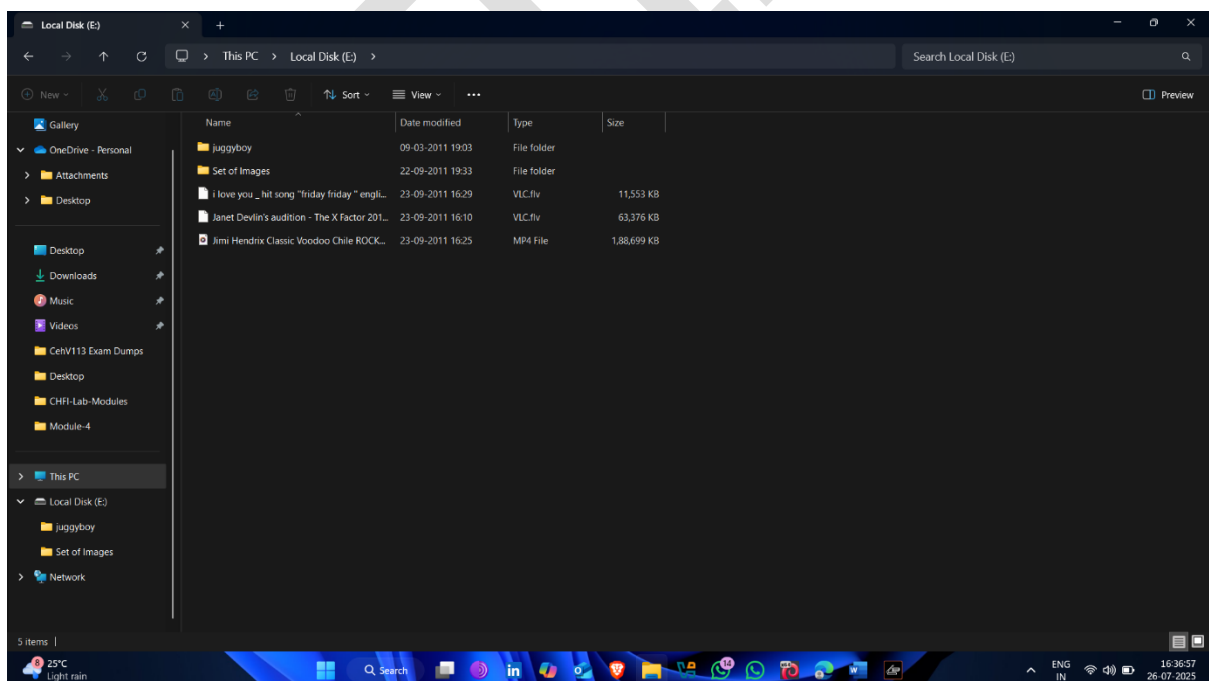
- New disk partition Created



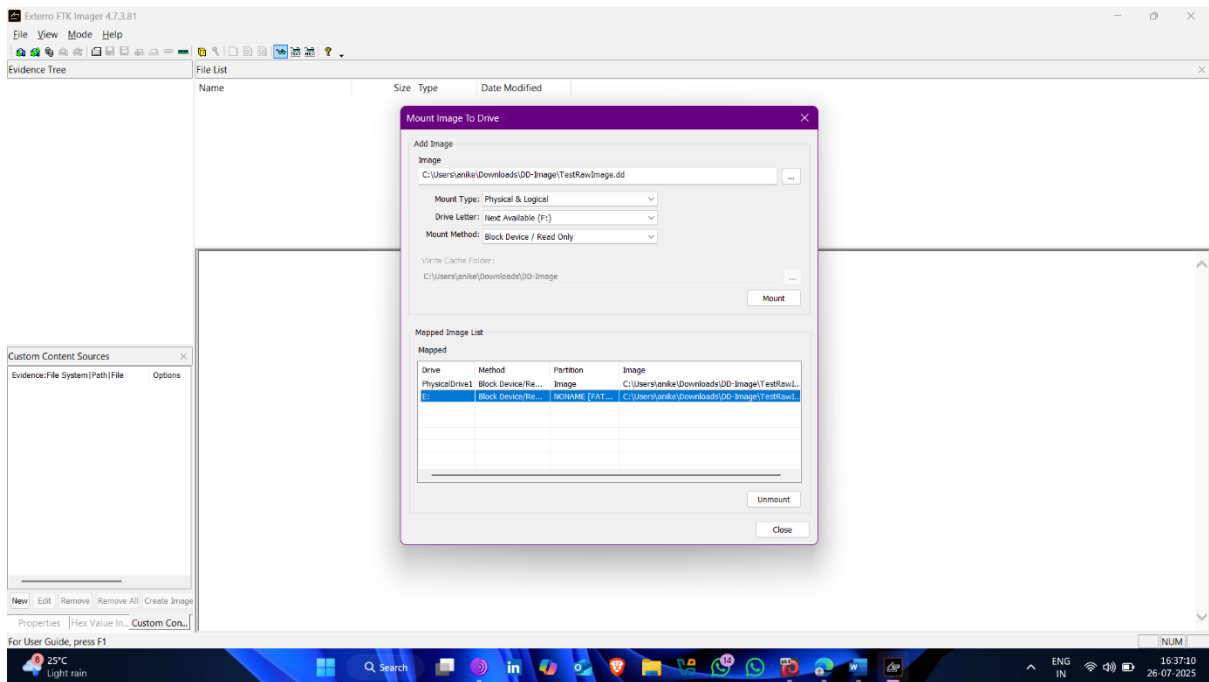
- Data inside the mount image



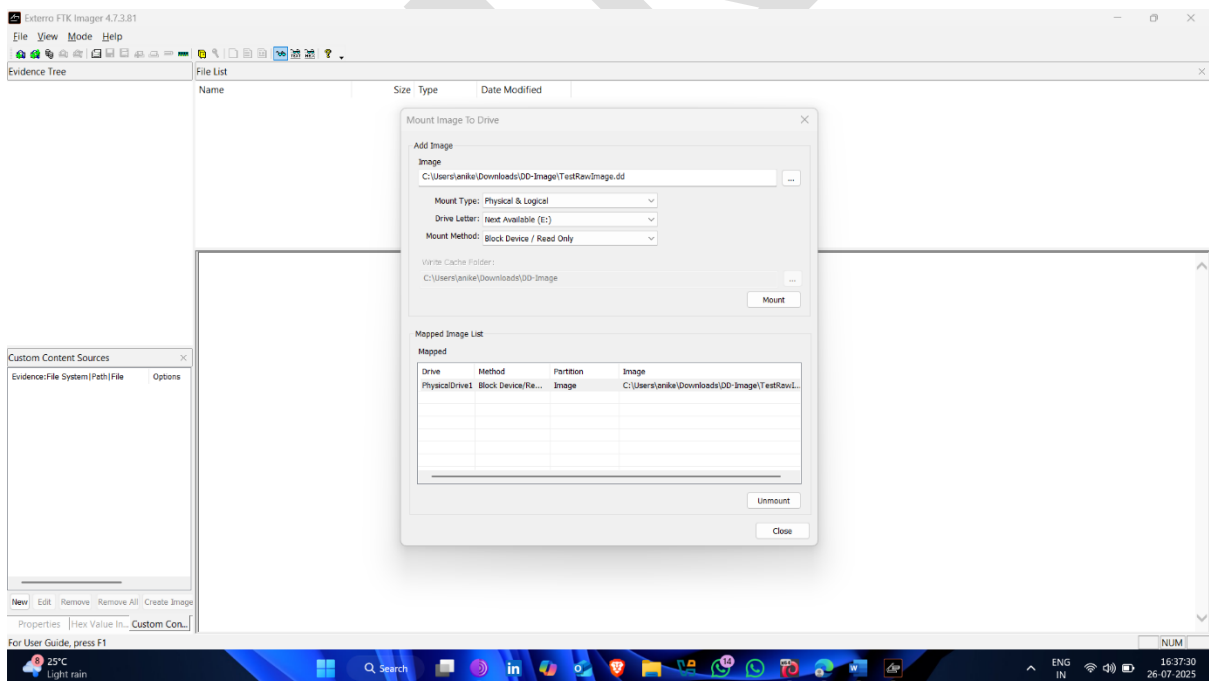
- If you want to unmount this image just simply back to FTK Imager



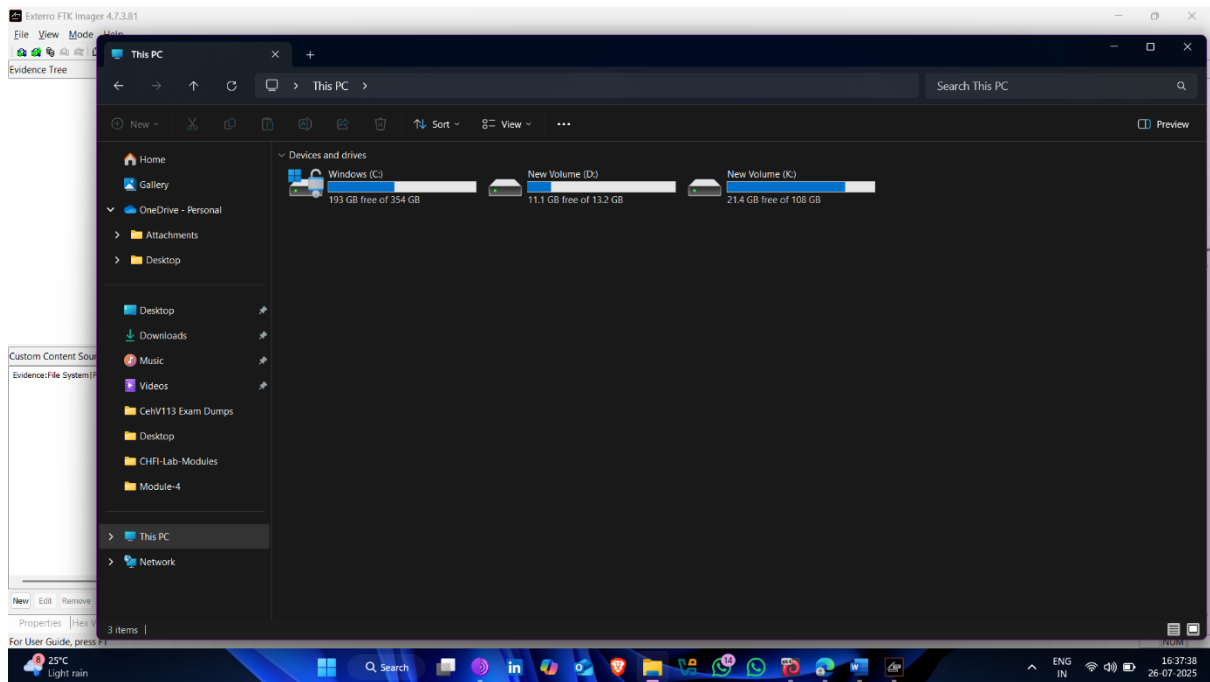
- Select **Mount** file and then click on **Unmount** option



- Image is unmounted



- Also removed in disk partition



Lab-4 -: Acquire RAM From Windows And Linux Workstation

RAM Acquisition (also called **Live Memory Acquisition**) is the process of **capturing the contents of a system's RAM (Random Access Memory)** for digital forensic investigation.

Why is RAM Acquisition Important?

RAM holds **volatile data** that disappears once the computer is turned off. It may include:

Type of Data	Example
 Passwords	Login credentials in plaintext
 Chat history	IM, Discord, or Messenger messages
 Running processes	Malware, keyloggers, reverse shells
 Open files	Unsaved Word docs, decrypted data
 Network connections	IPs, ports, active sessions
 Encryption keys	Used by tools like VeraCrypt, BitLocker

A) Acquire RAM From Windows Workstation Using Belkasoft Ram Capturer

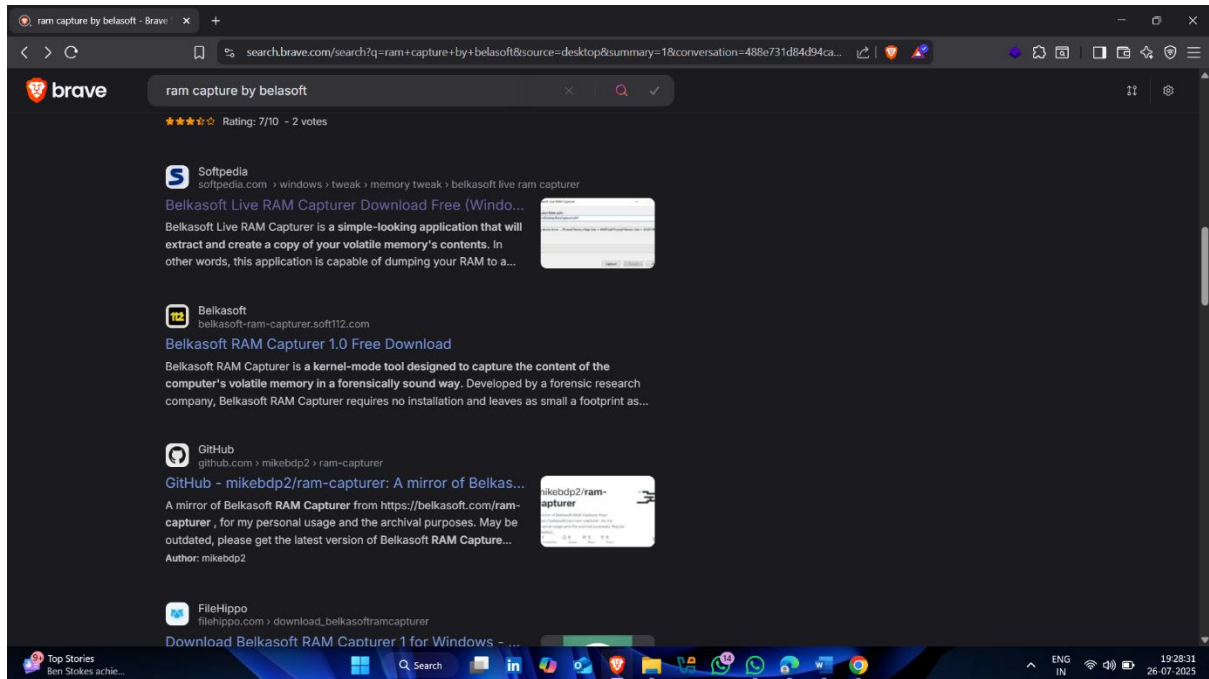
Belkasoft RAM Capturer is a **free Windows forensic tool** designed to **acquire the contents of volatile memory (RAM)** from a running computer. It is commonly used in digital forensics and incident response investigations.

Download Link :-

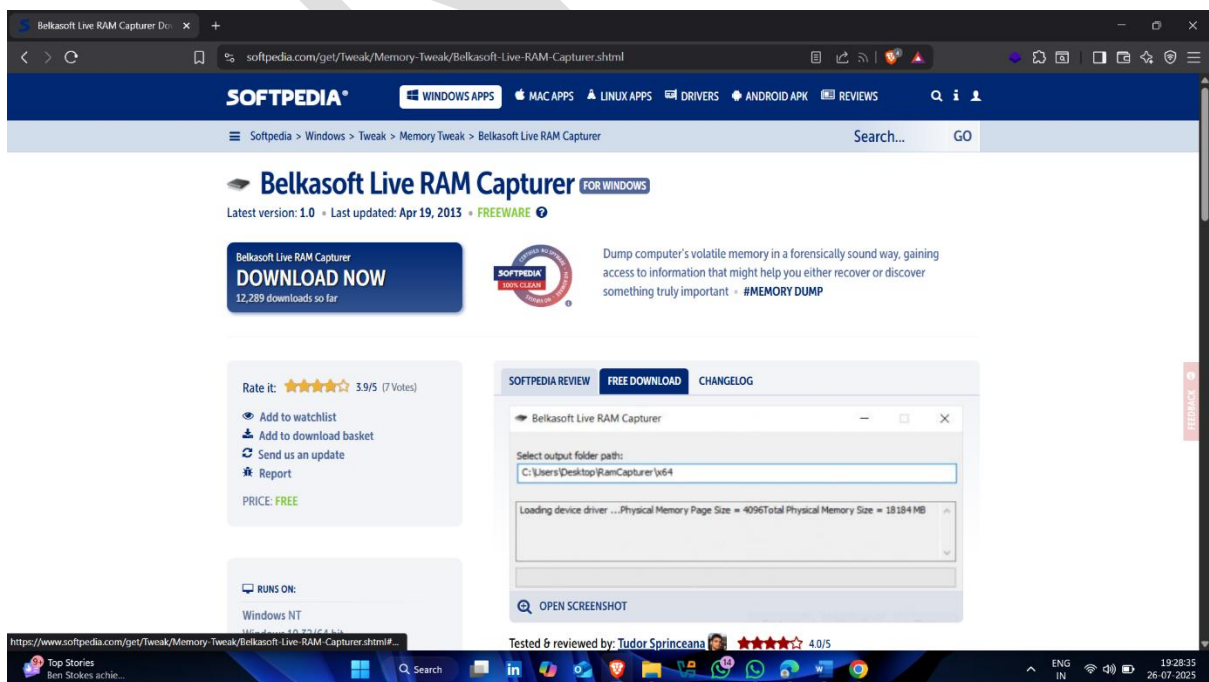
<https://www.softpedia.com/get/Tweak/Memory-Tweak/Belkasoft-Live-RAM-Capturer.shtml>

How to download it :-

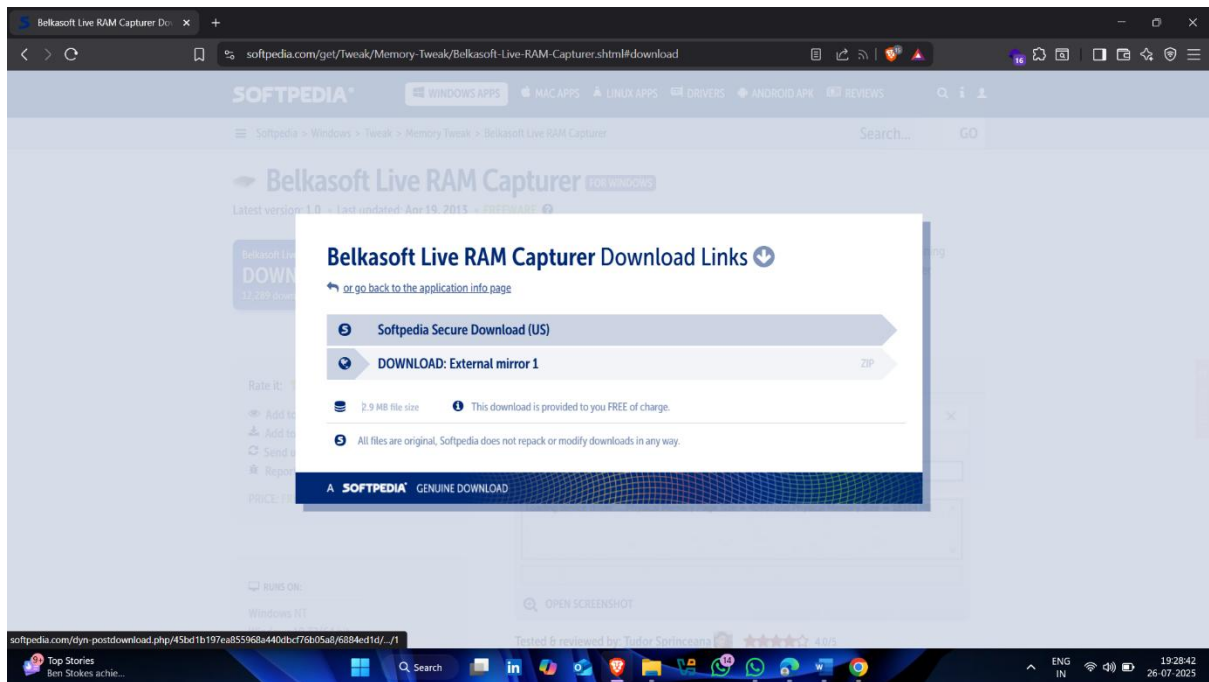
- Open browser and search **Ram Capture by belkasoft**



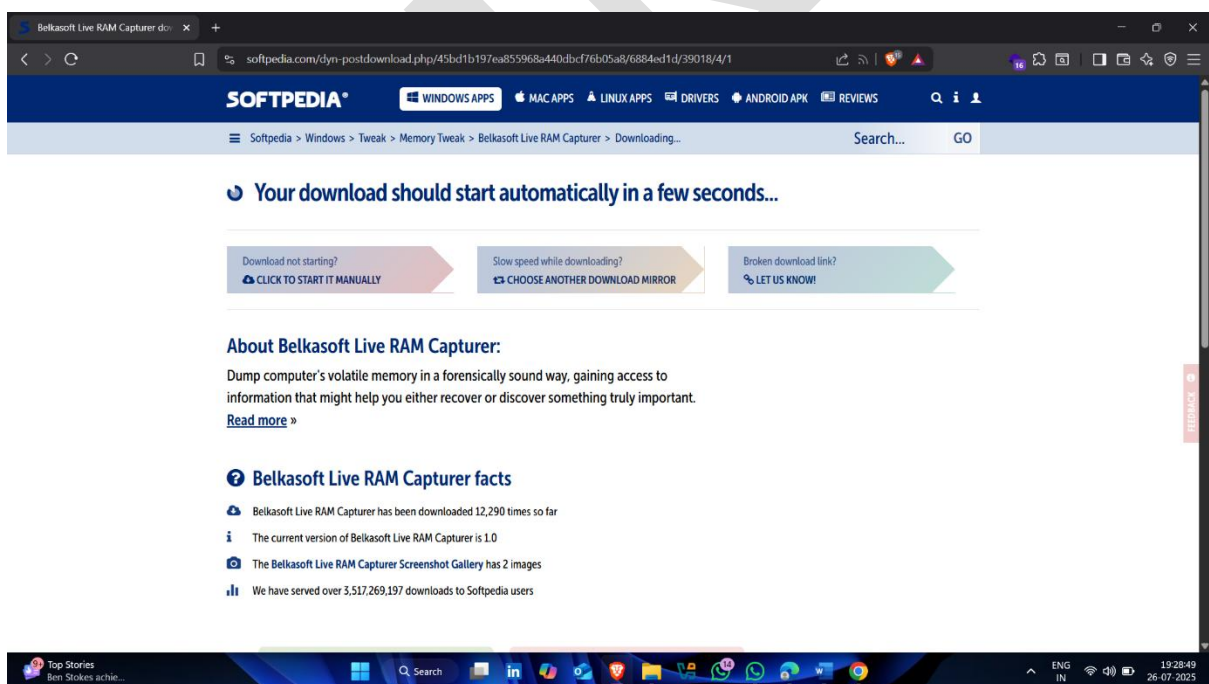
- Click on **download now**



- Click on First option

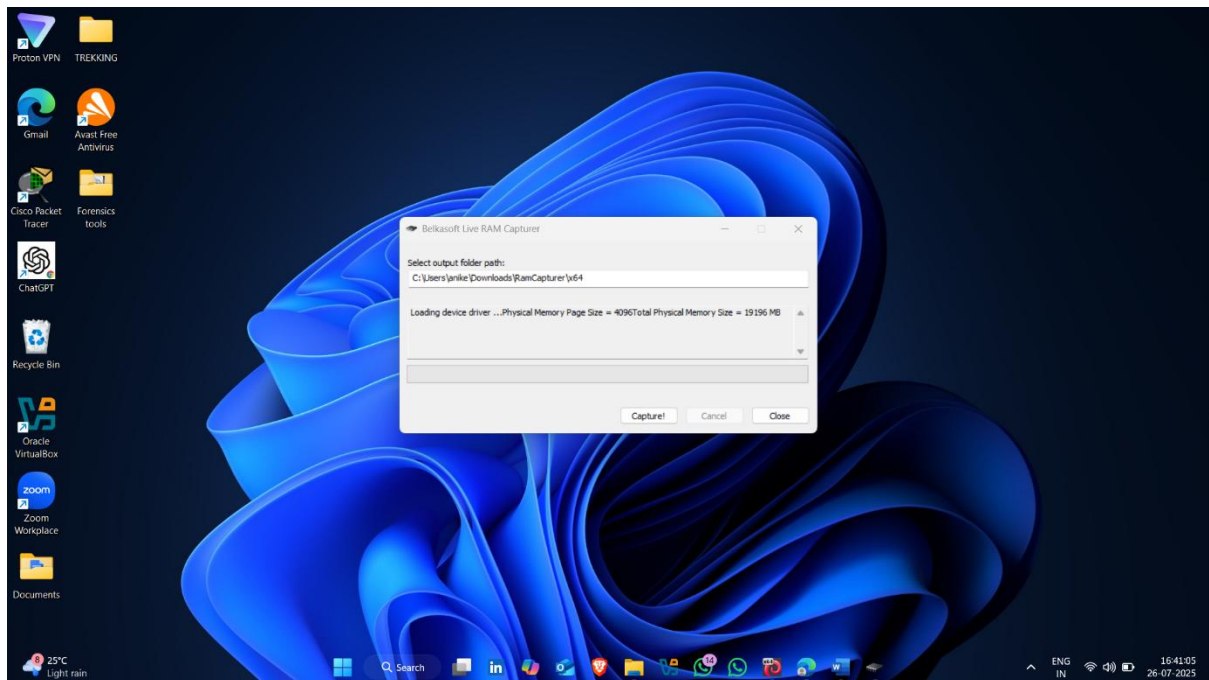


- Wait few seconds , download will automatic started

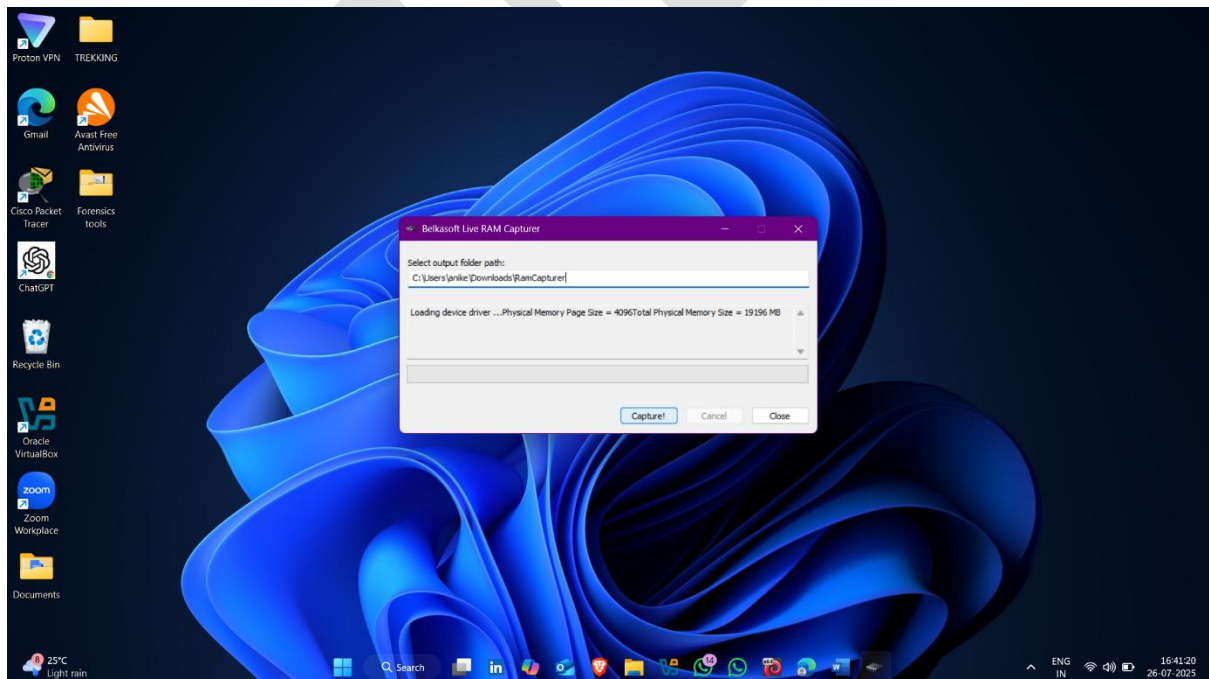


How to use it -:

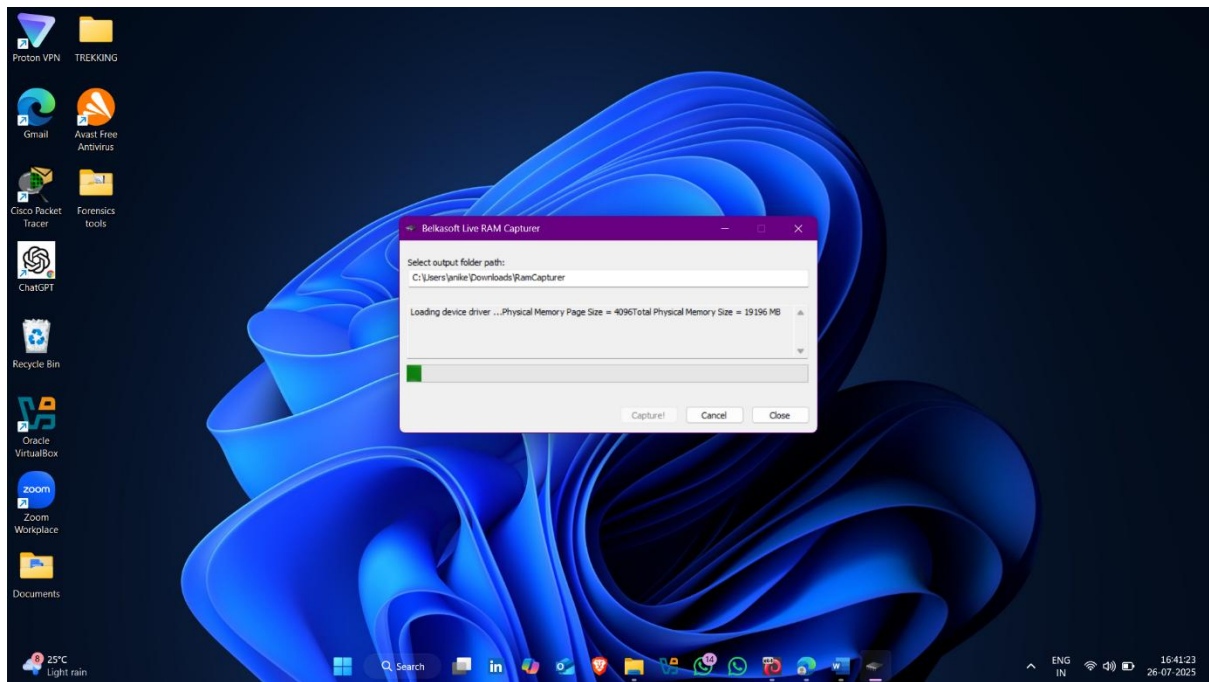
- **After install , open the application**



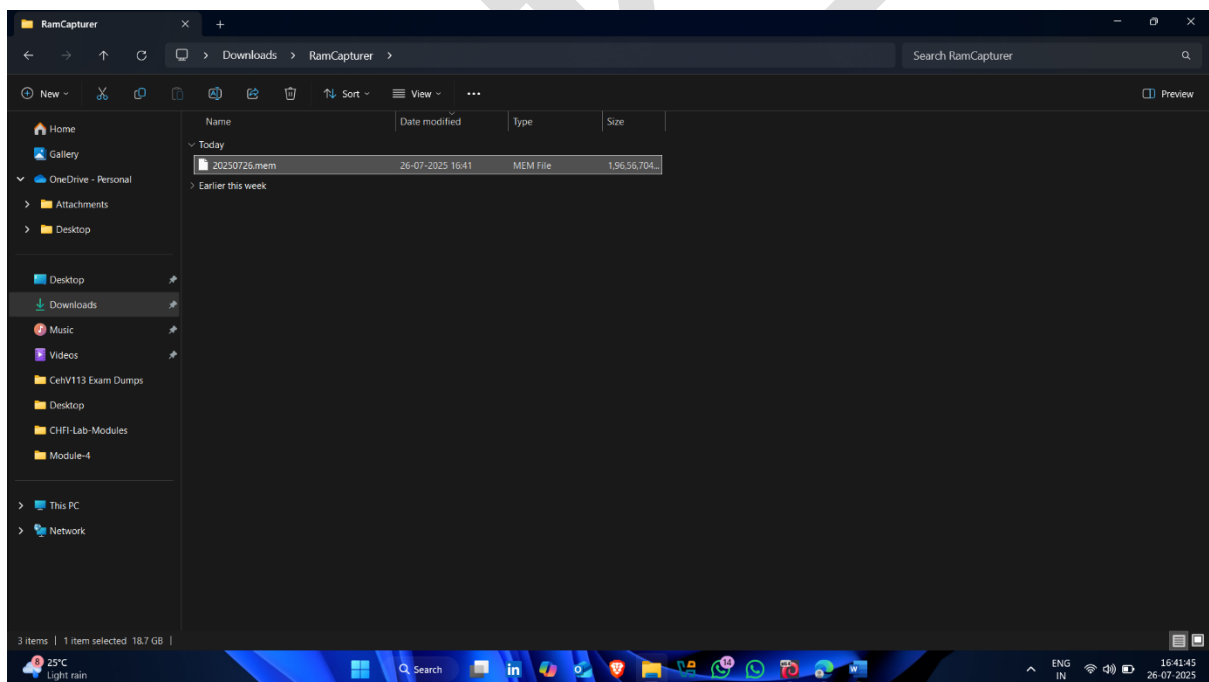
- **Now Browse the output location and then click on **capture****



- **Ram Capture started**



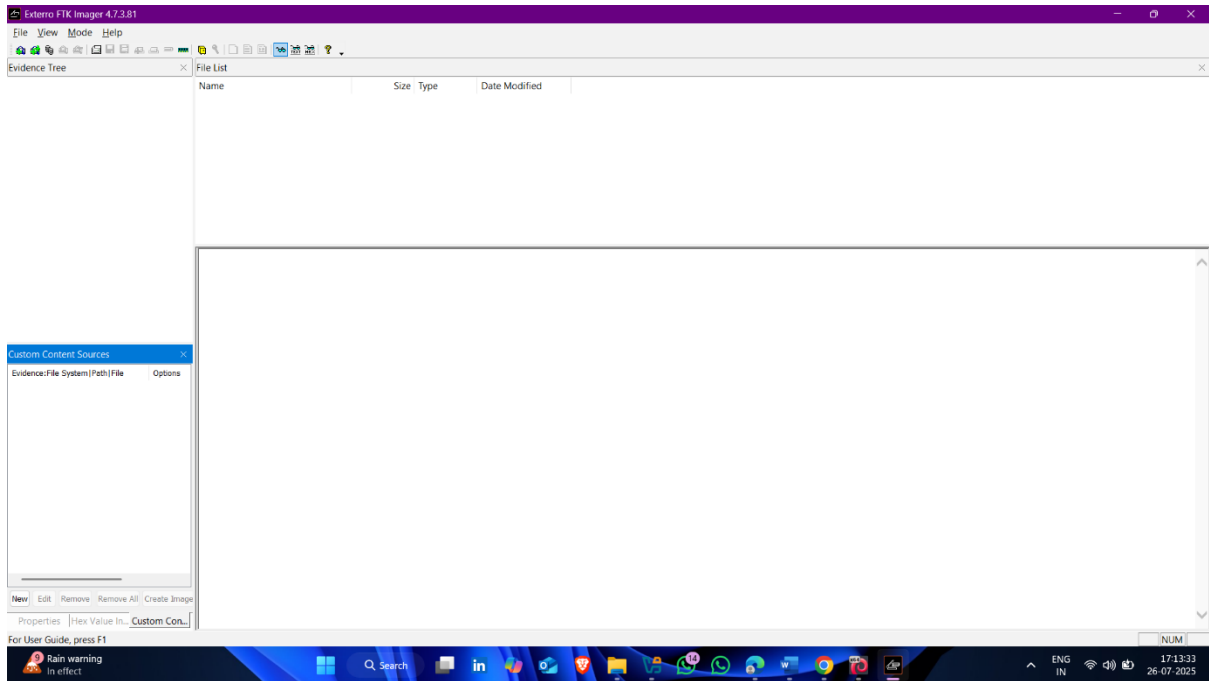
- Volatile memory (RAM) 🖱️✅



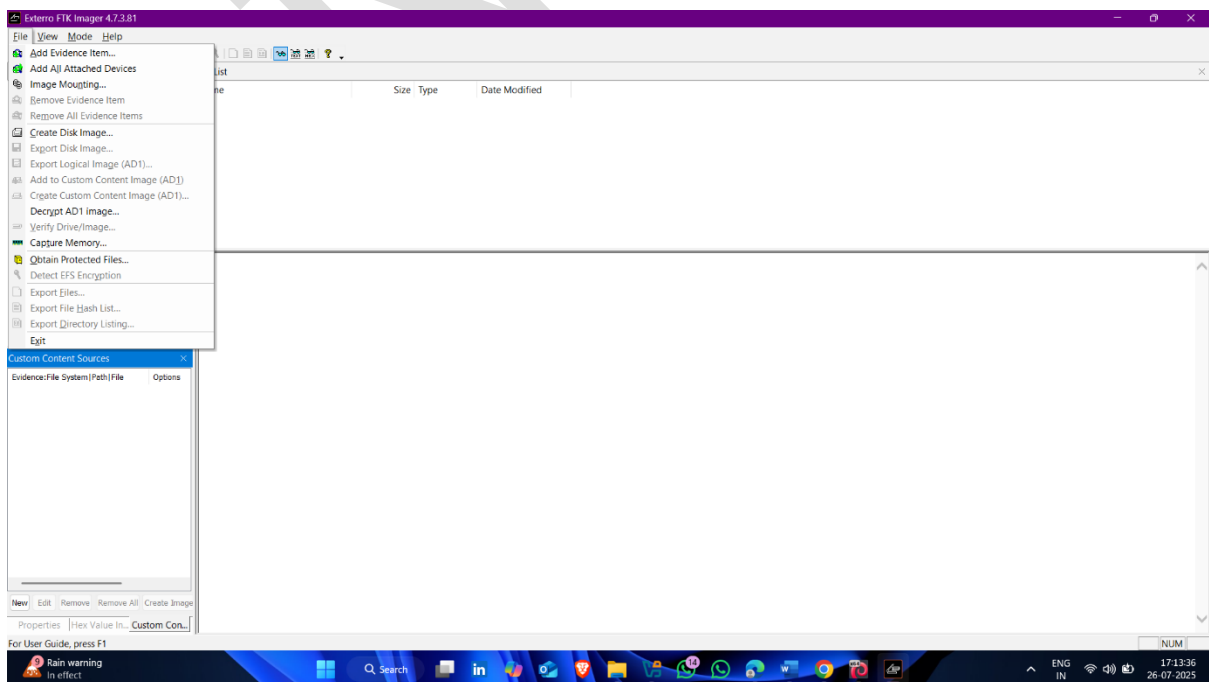
A) Acquire RAM From Windows Workstation Using FTK Imager

How to use it -:

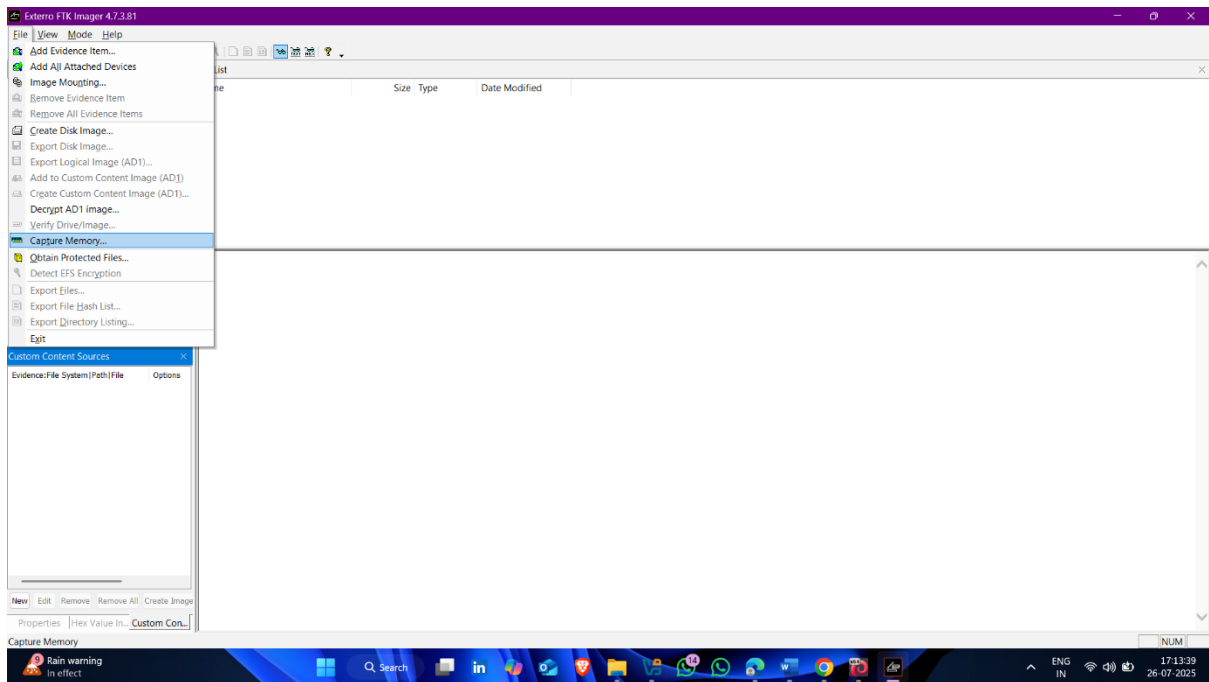
- Open FTK Imager



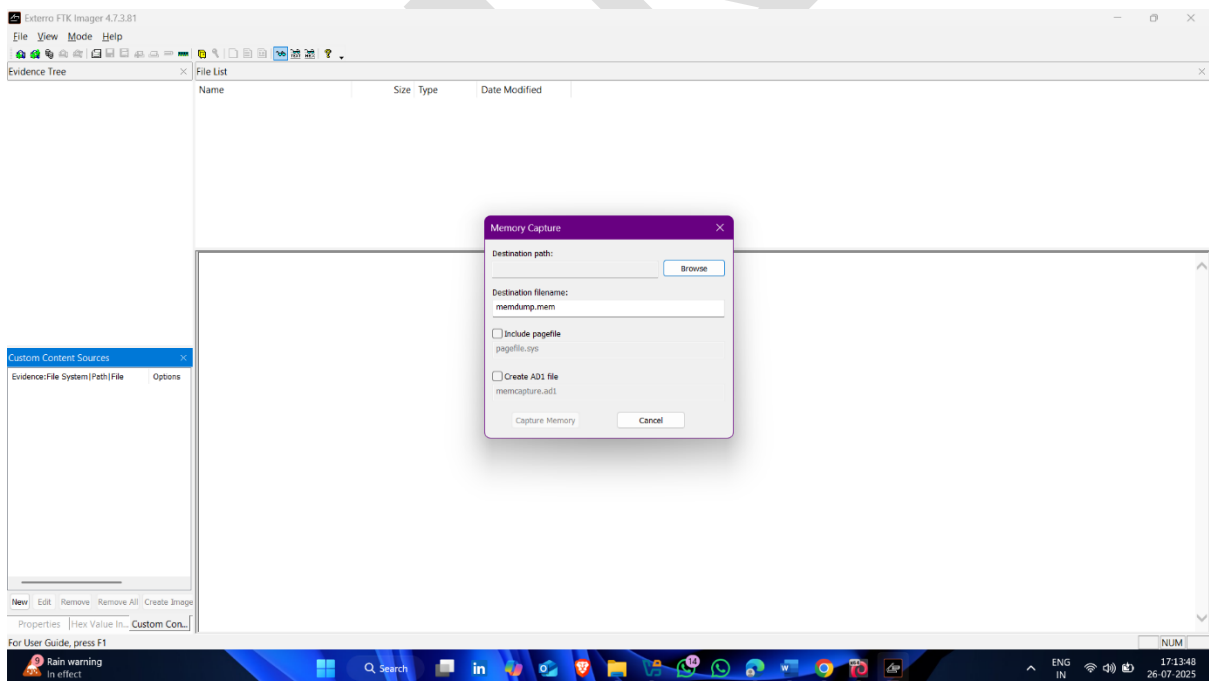
- Click on File Section



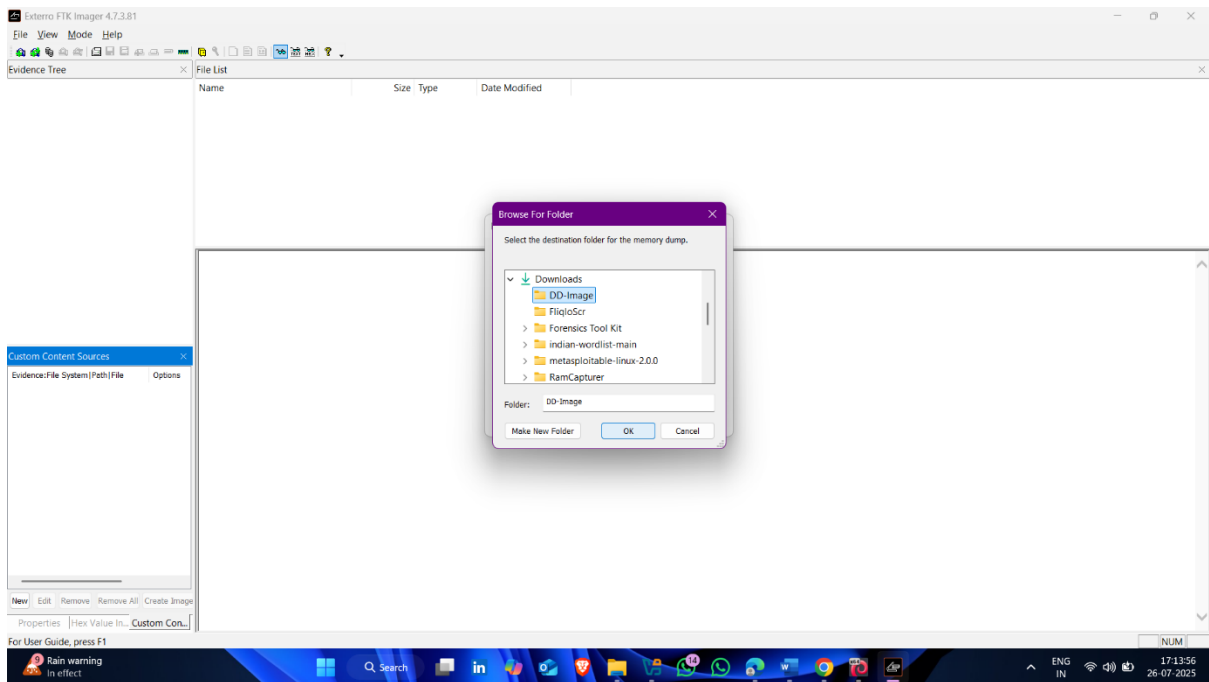
- Click on **Capture Memory**



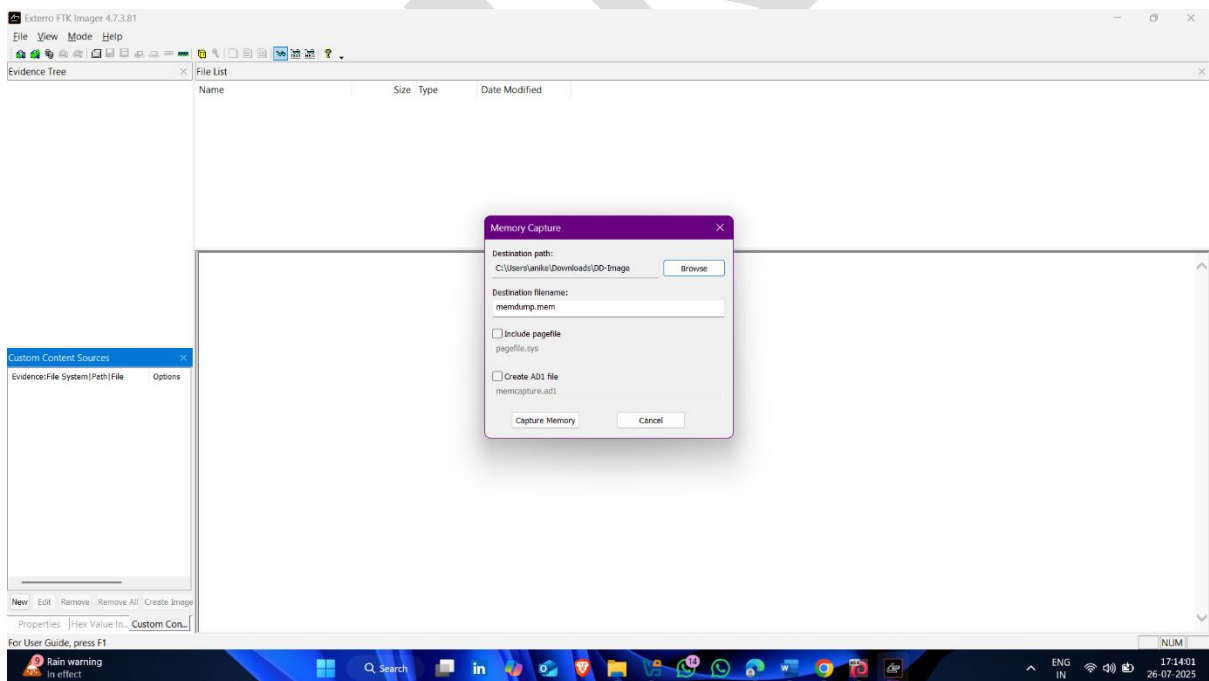
- Click on **Browse** to select destination Path



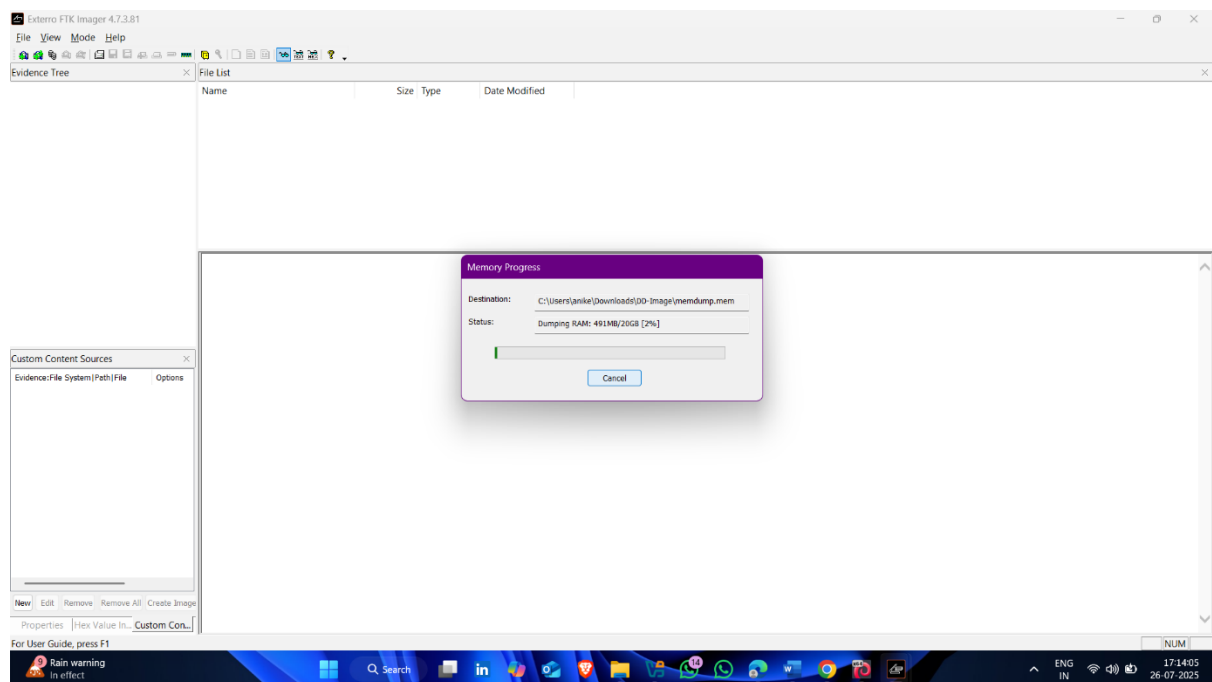
- Click on **ok**



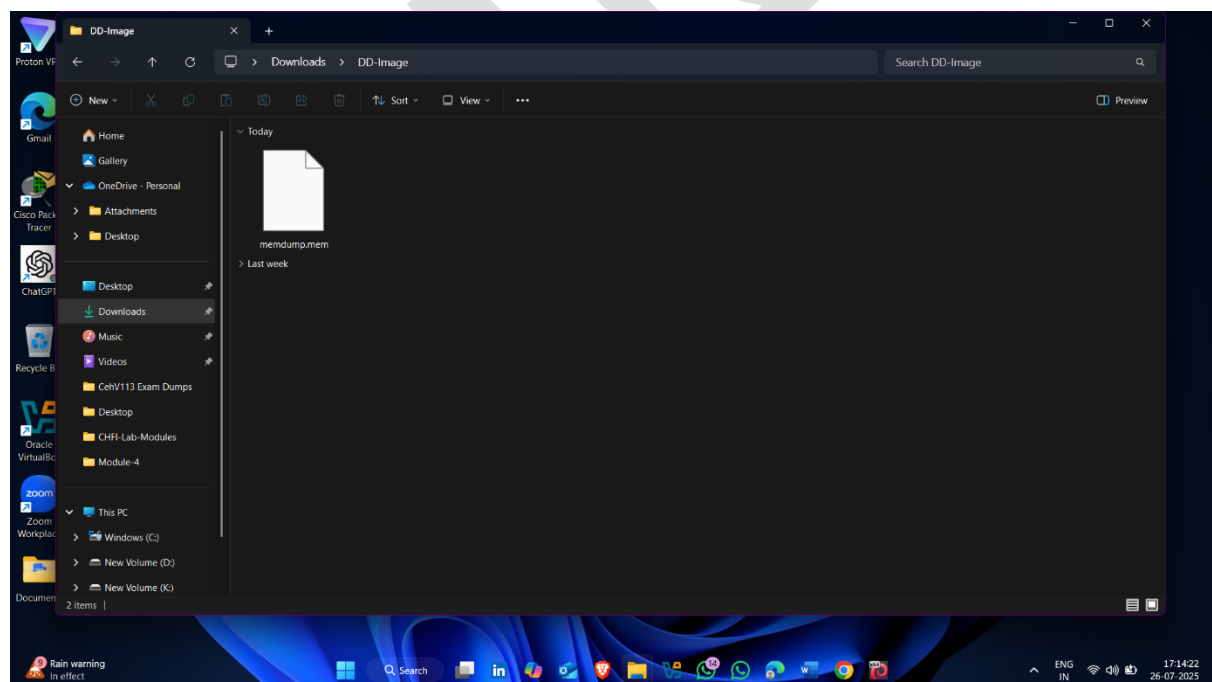
- Click on **capture memory**



- Memory capturing started ✓👉



- Volatile Memory(RAM)



C) Acquire RAM From Linux Workstation Using LiME Tool

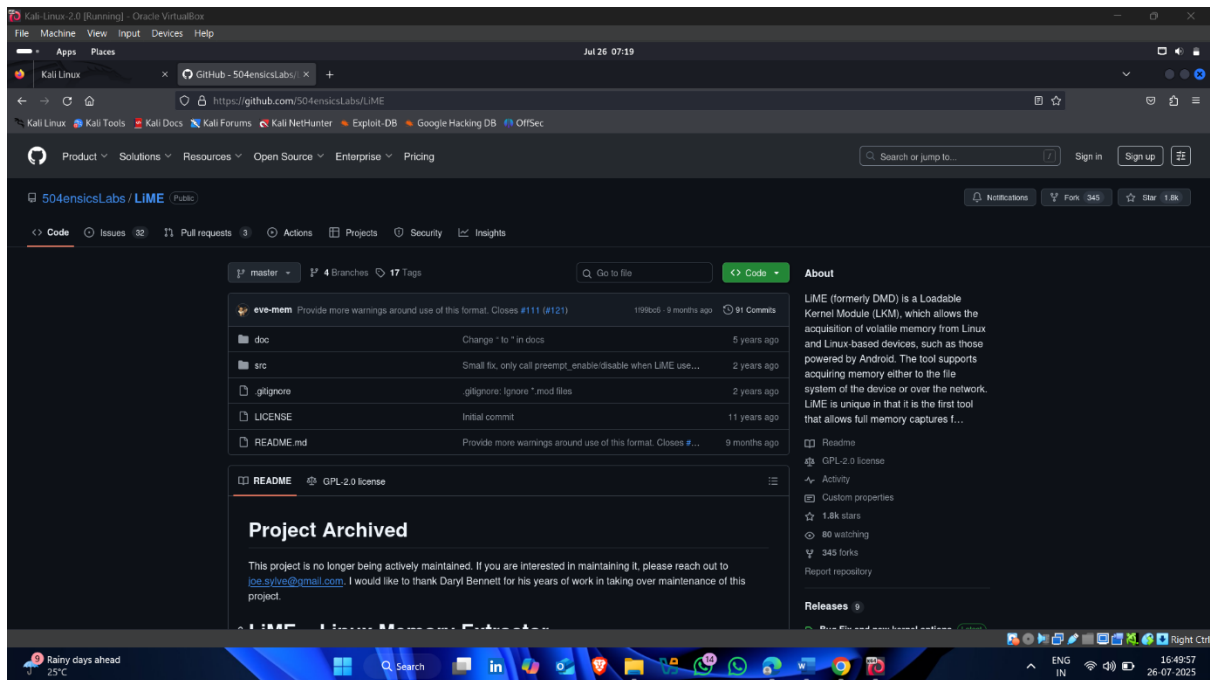
LiME (Linux Memory Extractor) is an open-source **Linux kernel module** used for **capturing volatile memory (RAM)** from Linux systems. It's a popular tool in **digital forensics, incident response, and malware analysis**.

Feature	Description
 Linux Only	Designed to capture RAM on live Linux systems
 Captures full RAM	Dumps physical memory to a file for forensic analysis
 Runs as a kernel module	Operates at a low level to access RAM contents safely
 Supports different outputs	TCP stream, raw format, and padded format for flexibility
 Forensic-friendly	Minimal footprint; aims to preserve evidence integrity
 Compatible with analysis tools	Works well with Volatility, Rekall, and other forensic tools

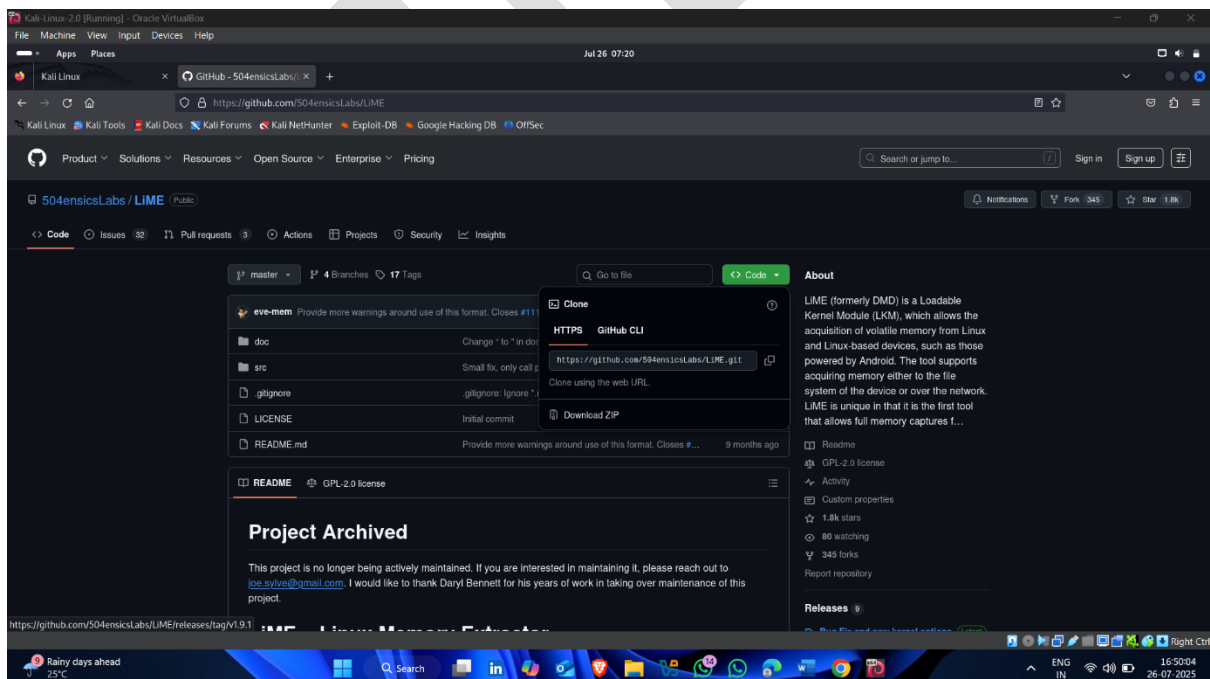
Download Link -: <https://github.com/504ensicsLabs/LiME>

How to use it :-

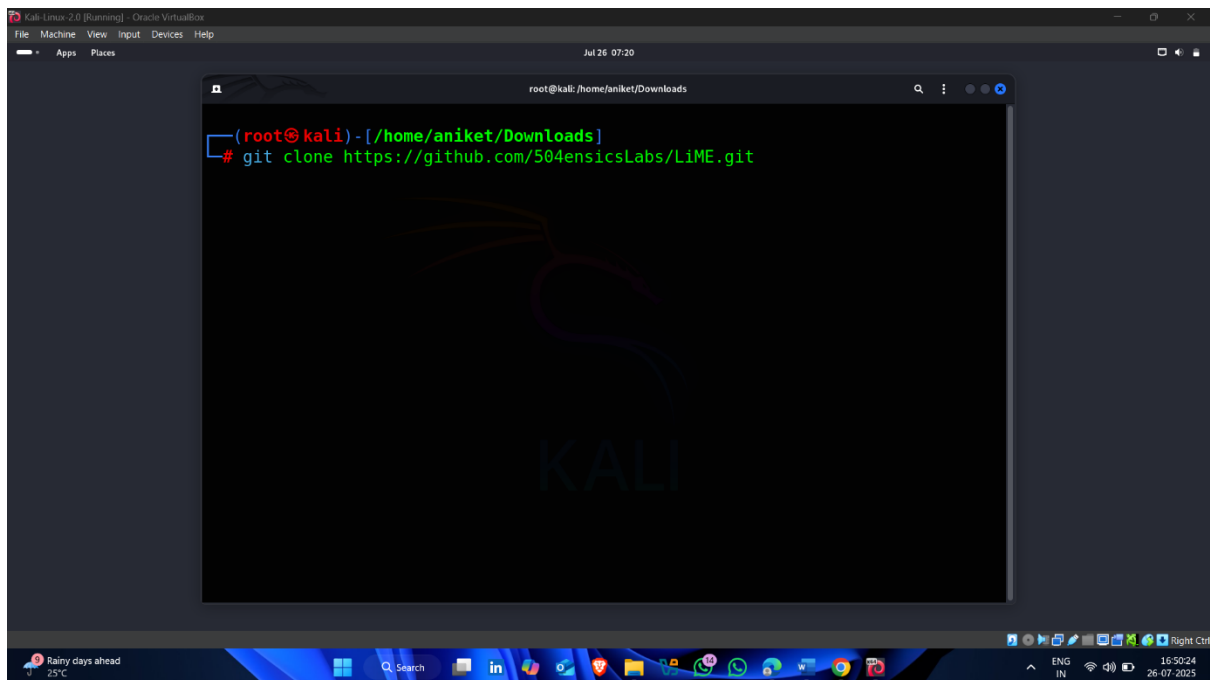
- Click on Green Code Button



- Copy Url 🖱️ ✅



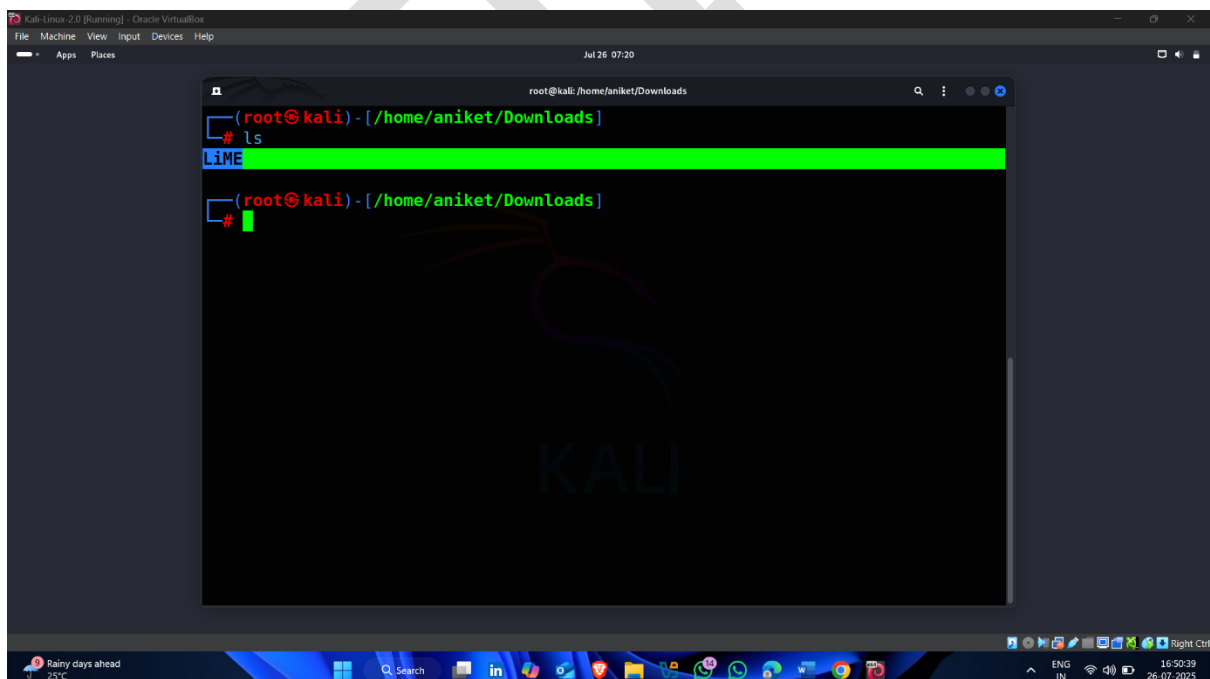
- Open terminal and paste url with git clone ✓👉



A screenshot of a Kali Linux terminal window. The terminal title bar shows 'root@kali: /home/aniket/Downloads'. The prompt is '(root@kali) - [/home/aniket/Downloads]'. The command '# git clone https://github.com/504ensicsLabs/LiME.git' has been entered and executed. The terminal background features a large, faint 'KALI' logo. The window is part of a desktop environment with a taskbar at the bottom showing various application icons and system status indicators like 'Rainy days ahead 25°C' and '16:50:24 26-07-2025'.

```
(root@kali) - [/home/aniket/Downloads]
# git clone https://github.com/504ensicsLabs/LiME.git
```

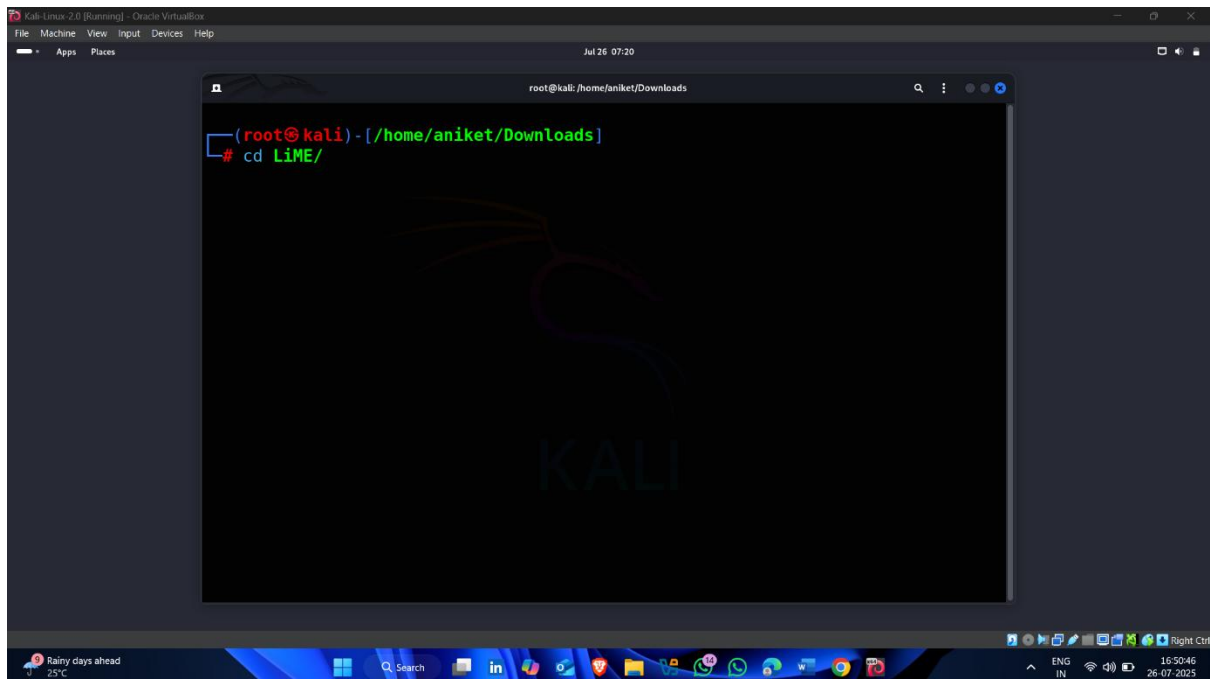
- Install successfully



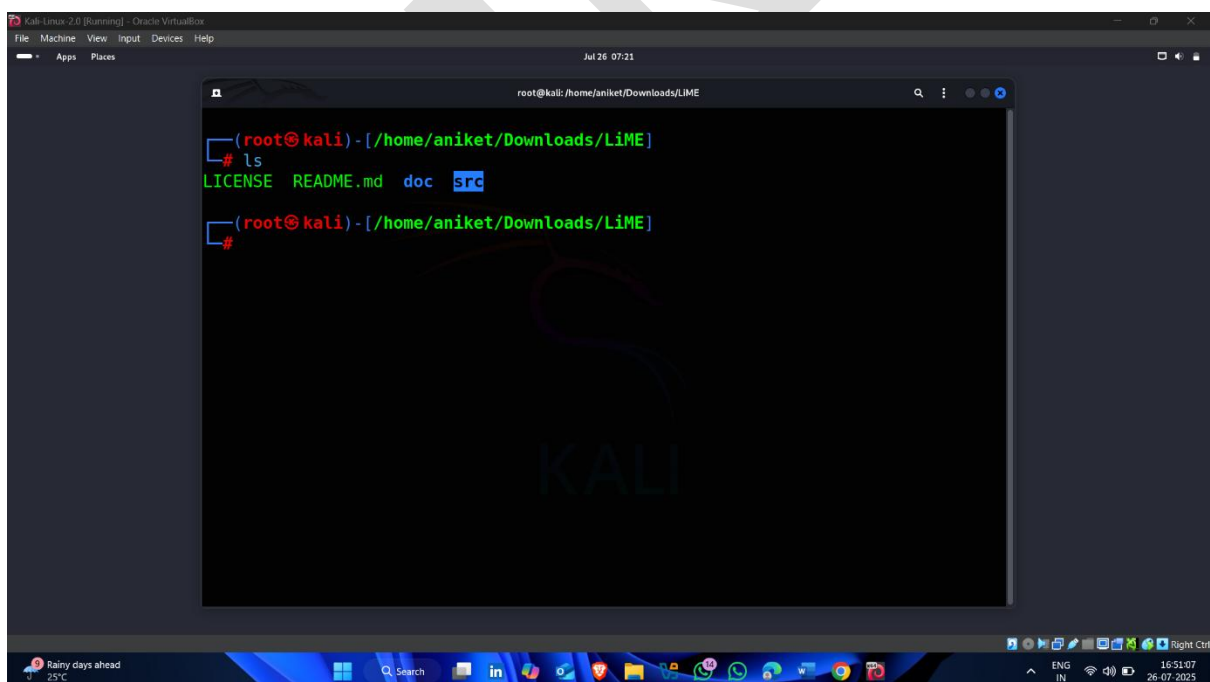
A screenshot of a Kali Linux terminal window. The terminal title bar shows 'root@kali: /home/aniket/Downloads'. The prompt is '(root@kali) - [/home/aniket/Downloads]'. The command '# ls' has been entered and executed, resulting in a green highlight over the word 'LiME'. The terminal background features a large, faint 'KALI' logo. The window is part of a desktop environment with a taskbar at the bottom showing various application icons and system status indicators like 'Rainy days ahead 25°C' and '16:50:39 26-07-2025'.

```
(root@kali) - [/home/aniket/Downloads]
# ls
LiME
(root@kali) - [/home/aniket/Downloads]
#
```

- Go to the LiME Directory



- Now open this Directory

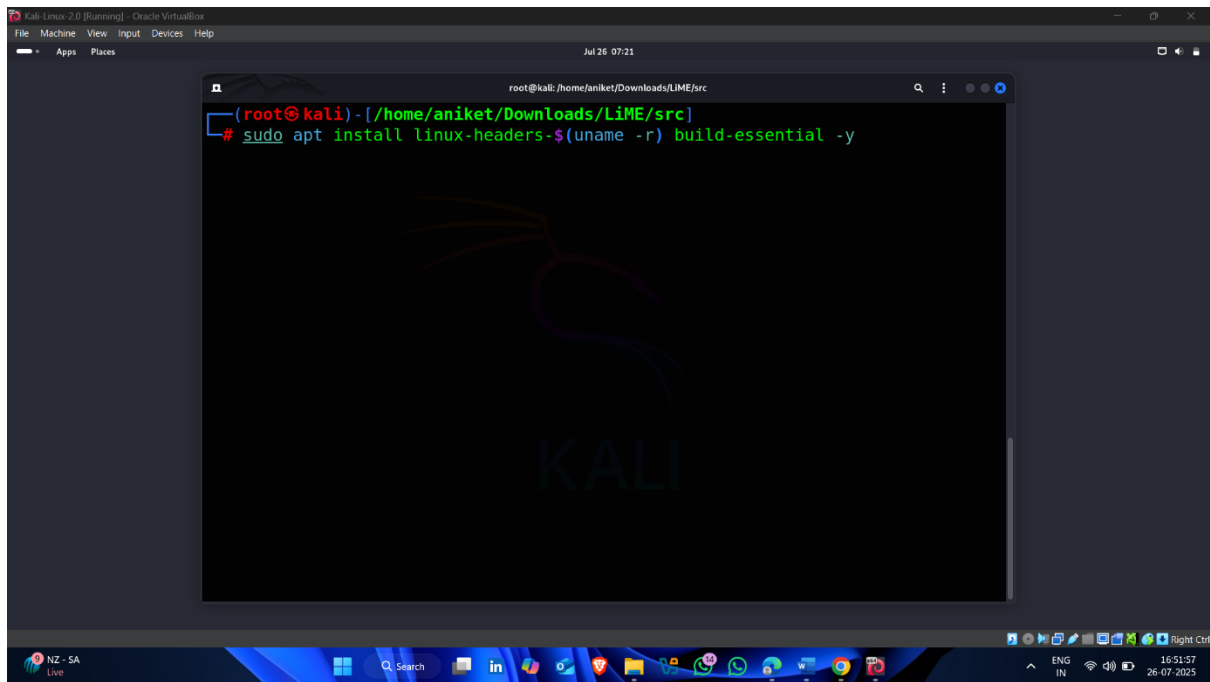


- And the use following command

Command -: `sudo apt install linux-headers-$(uname -r) build-essential -y`

Explanation -:

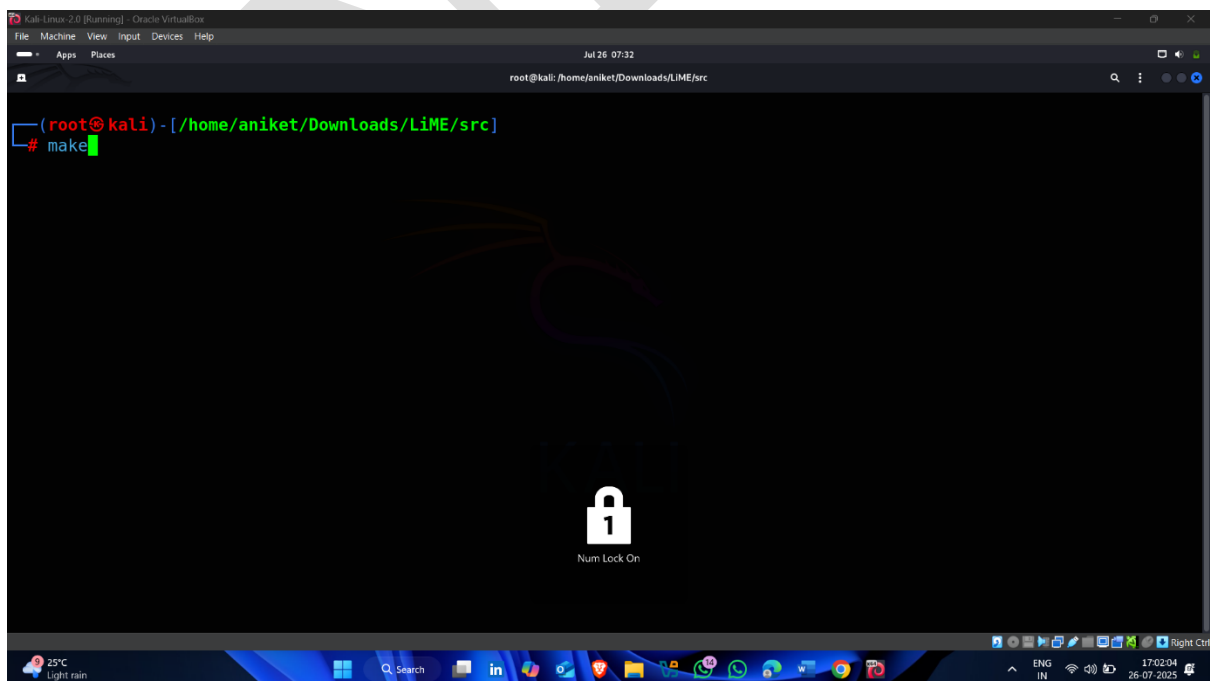
Part	Meaning
sudo	Runs the command with superuser (admin) privileges. Required for system-level changes like installing packages.
apt install	Tells the APT package manager to install one or more packages.
linux-headers-\$(uname -r)	Installs the Linux kernel header files for your currently running kernel version. \$(uname -r) dynamically fetches the kernel version (e.g., 6.8.11-amd64).
build-essential	A meta-package that installs the core development tools : GCC compiler, make, and other libraries needed for compiling software (especially kernel modules like LiME).
-y	Automatically answers "yes" to all installation prompts. Useful for scripting or automating installs.



- Now type next Command

Command -: make

Explanation -: The make command is a build automation tool used primarily to compile and build software from source code—especially in Linux and Unix-like systems.



- Now run this file 🖱️ ✅

```

(root@kali) - [/home/aniket/Downloads/LiME/src]
# ls
Makefile      disk.c        lime.h        main.c
Makefile.sample disk.o        lime.mod      main.o
Module.symvers hash.c        lime.mod.c    modules.order
deflate.c     hash.o        lime.mod.o    tcp.c
deflate.o     lime-6.12.33+kali-amd64.ko lime.o        tcp.o
  
```

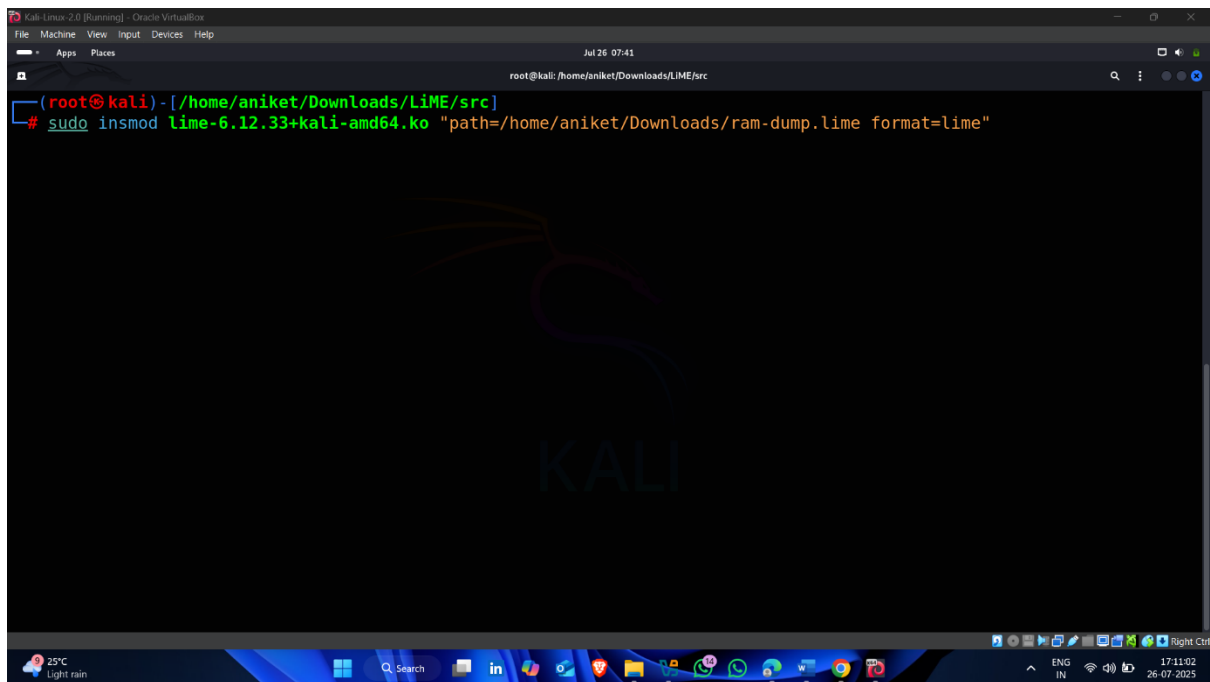
- Enter next command

Command -: `sudo insmod lime-6.12.33+kali-amd64.ko`
 “path=/home/aniket/Downloads/ram-dump.lime format=lime”

Explanation -:

Part	Meaning
sudo	Runs the command as root/superuser , required for kernel-level operations.
insmod	Inserts a kernel module into the currently running Linux kernel.
lime-6.12.33+kali-amd64.ko	The compiled LiME kernel module (specific to your current kernel version).
"path=... format=lime"	Module parameters passed to LiME for RAM capture: ♦ path=... tells LiME where to store the memory dump file. ♦ format=lime tells LiME to

Part	Meaning
	use the "LiME" memory dump format, which includes metadata.

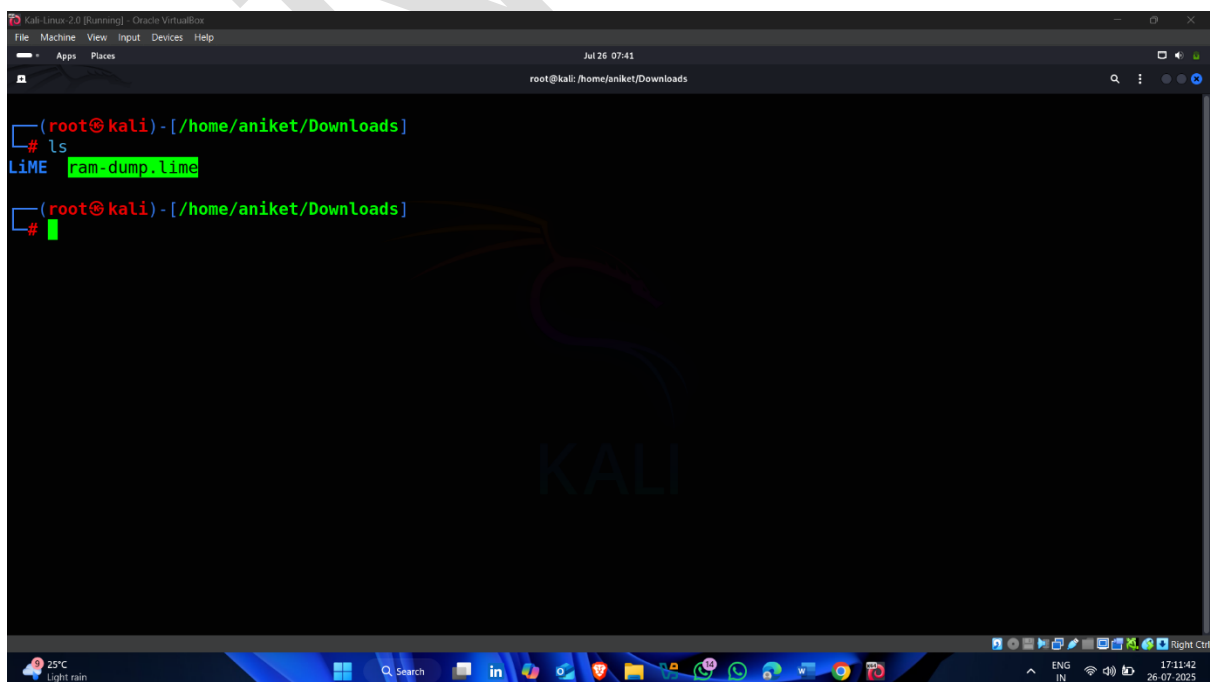


```

(root@kali) - [/home/aniket/Downloads/LiME/src]
# sudo insmod lime-6.12.33+kali-amd64.ko "path=/home/aniket/Downloads/ram-dump.lime format=lime"

```

- Ram captured ✓👉



```

(root@kali) - [/home/aniket/Downloads]
# ls
LiME  ram-dump.lime
(root@kali) - [/home/aniket/Downloads]
#

```

Lab-5 -: Create Customized Images From An Image Containing NTFS File System

The **NTFS (New Technology File System)** is a proprietary file system developed by **Microsoft** for the **Windows NT family of operating systems**.

Key Features of NTFS:

Feature	Description
 Supports large files	Can handle very large files (up to 16 TB on modern systems).
 File permissions	Built-in security with Access Control Lists (ACLs) for controlling user access.
 Journaling	Keeps a log of file system changes to recover from crashes more easily.
 Metadata support	Stores additional information (timestamps, attributes, etc.) with files.
 Compression and Encryption	Files/folders can be compressed or encrypted (EFS) natively.
 File timestamps	Stores multiple time records: created, modified, accessed, and MFT changed.
 Disk quotas	Admins can limit how much disk space each user can use.
 Alternate Data Streams (ADS)	Allows a file to contain hidden metadata or content (used in malware often).

Where NTFS is used:

- **Windows operating systems** (default for internal drives)
- External drives for **Windows compatibility**

- Used in **digital forensics** to analyze Windows systems

A) Create Customized Images From An Image Containing NTFS File System Using DiskExplorer

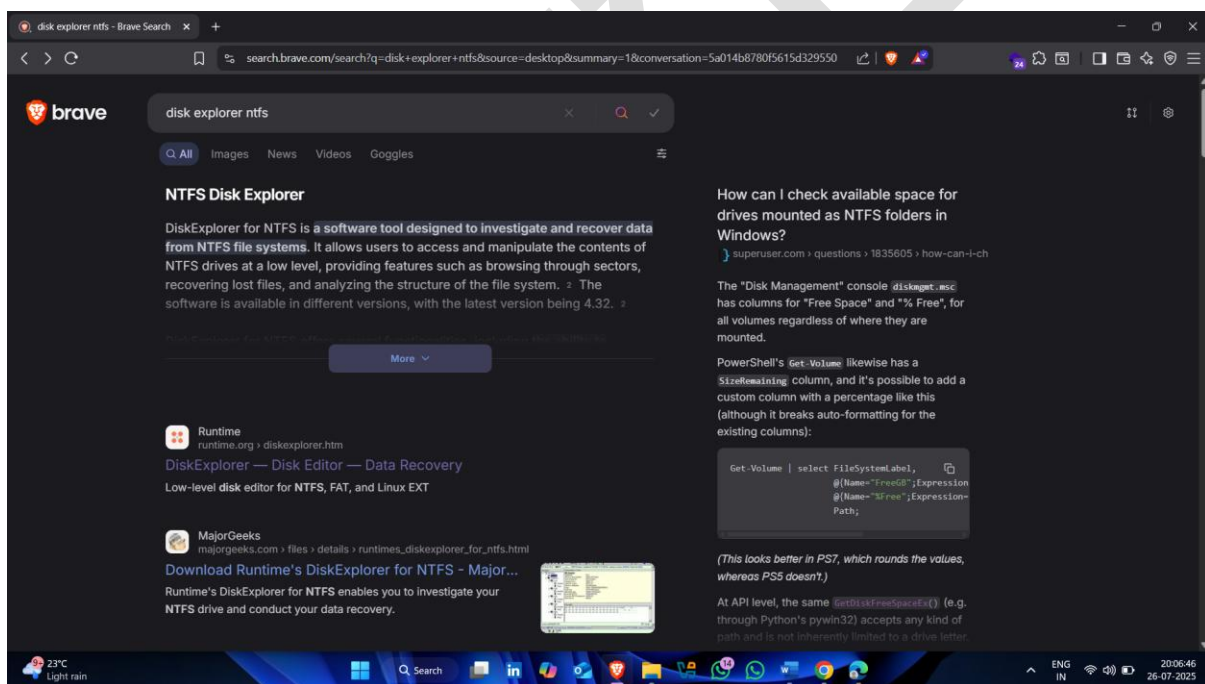
Disk Explorer's role in the NTFS file system is to provide **deep, low-level visibility and control** over NTFS disk structures, metadata, and files — especially useful in digital forensics, incident response, and data recovery.

Download Link :-

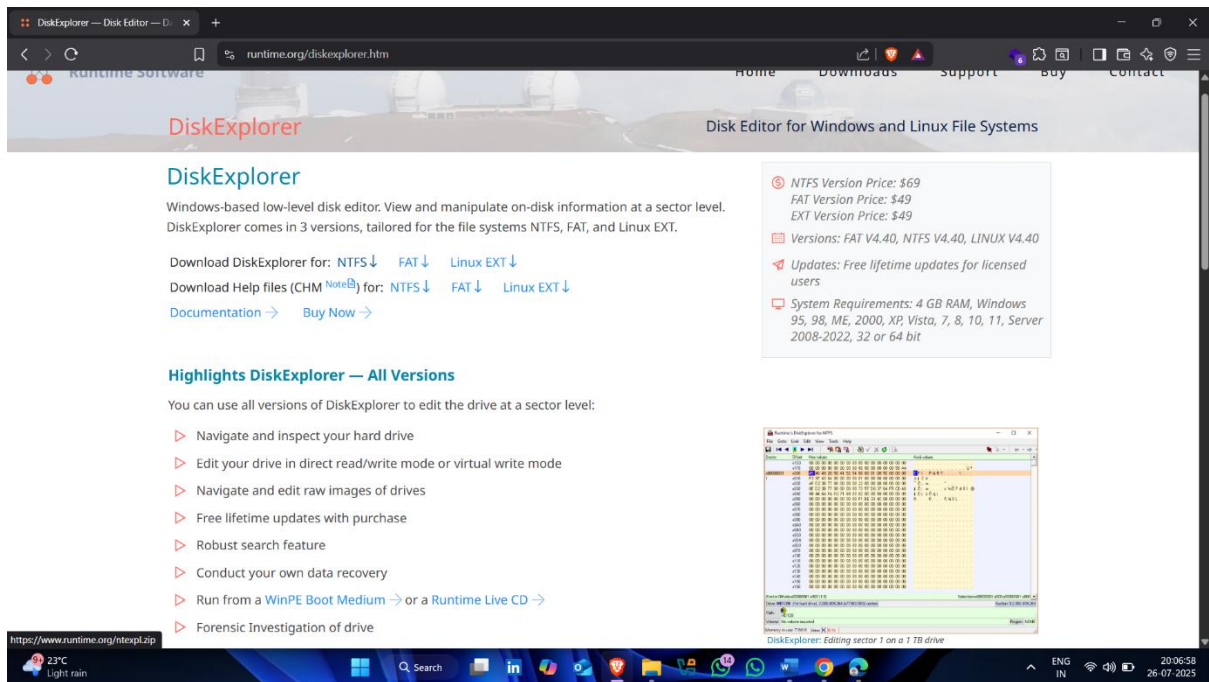
<https://www.runtime.org/diskexplorer.htm>

How to download it :-

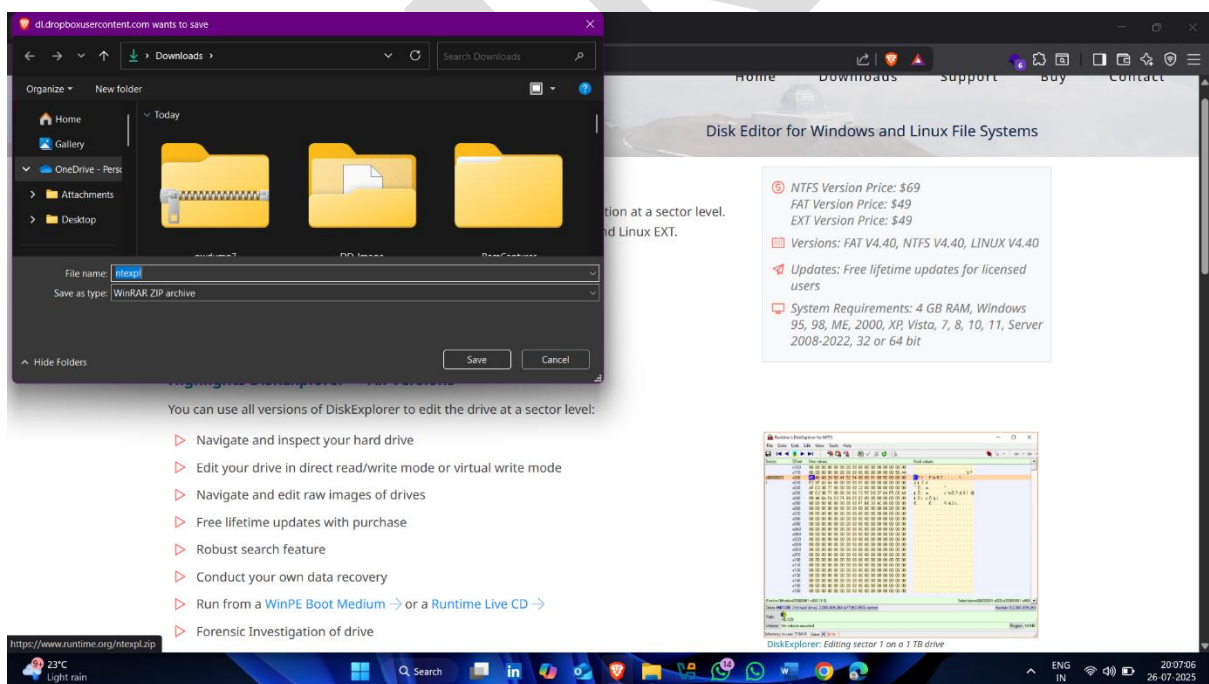
- Open Browser and click on First Website



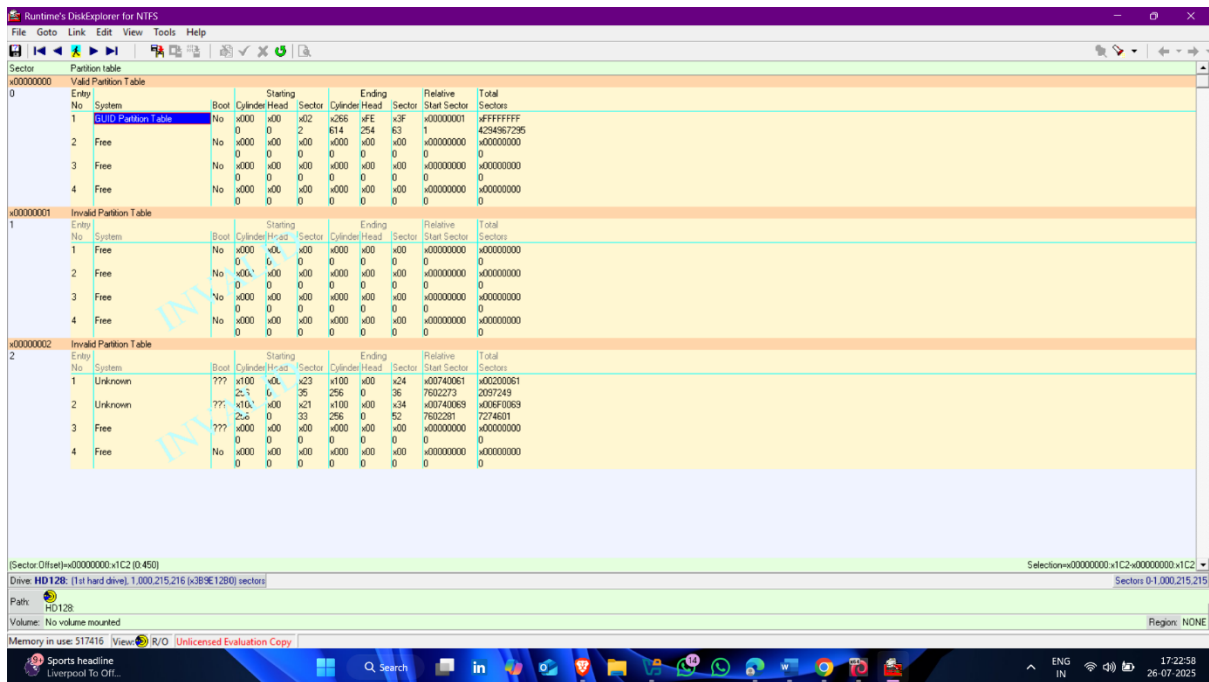
- click on **NTFS**



- Then click on **save** – Download will started



- After installation open the application

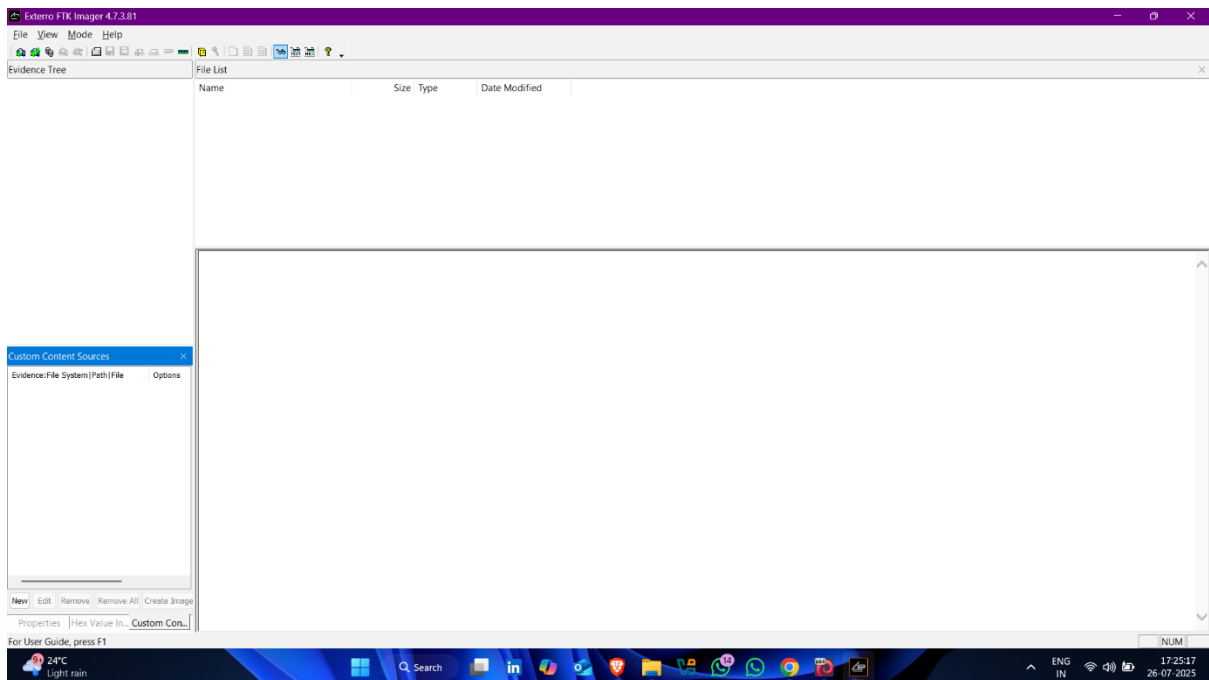


Lab-6-: View Content Of Forensic Image File

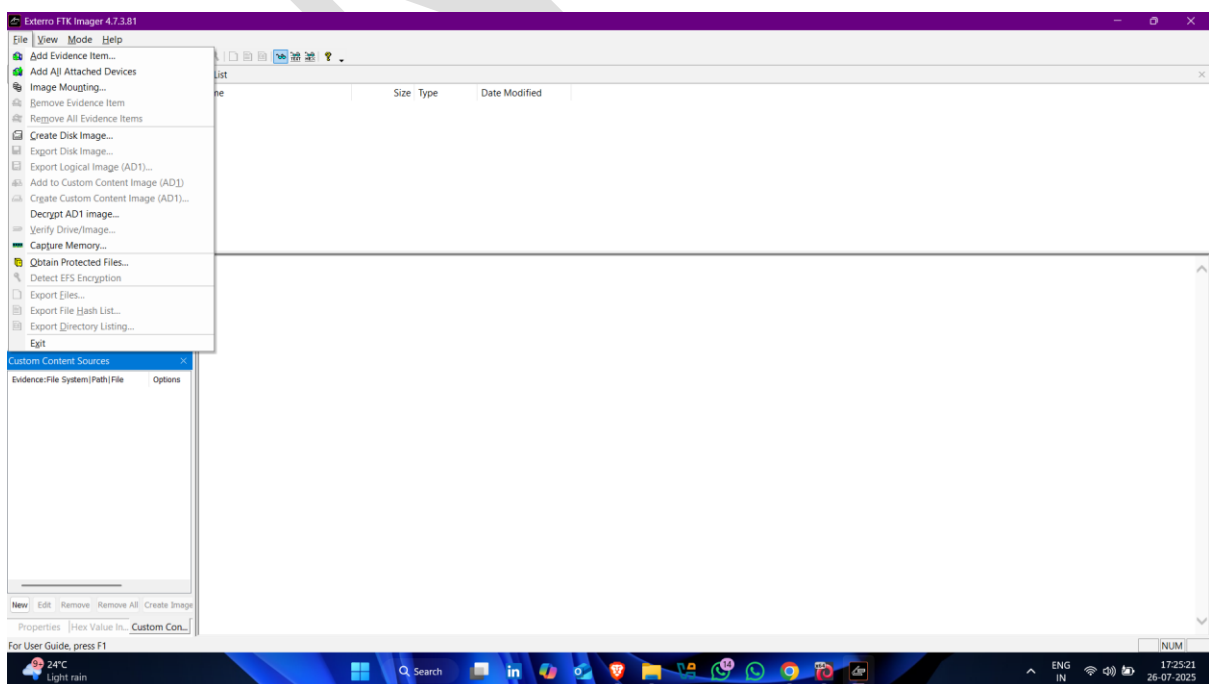
A) View Content Of Forensic Image File Using FTK Imager

How to use it -:

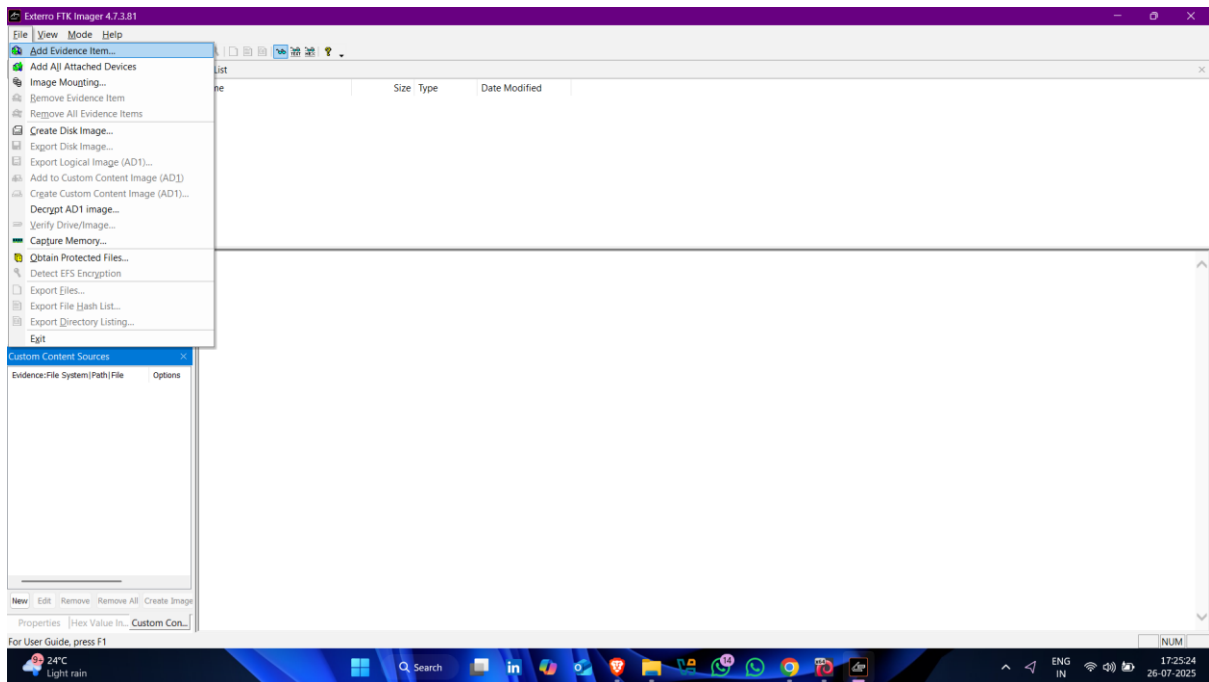
- Open FTK Imager



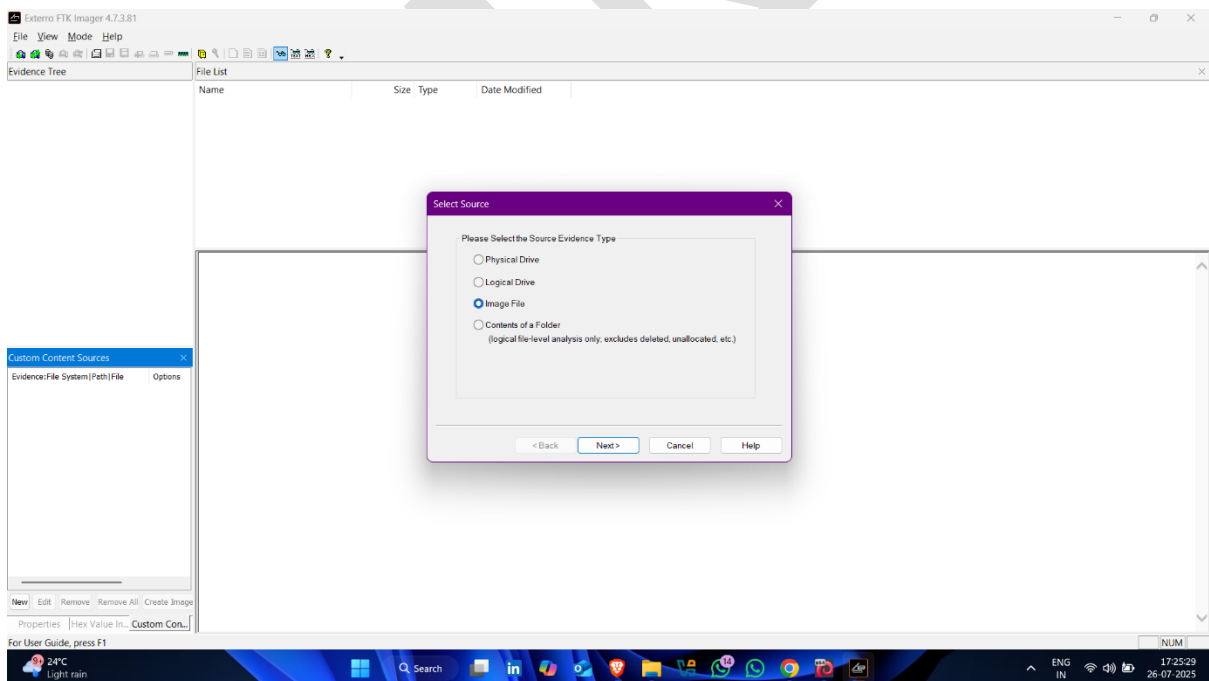
- Click on File Section



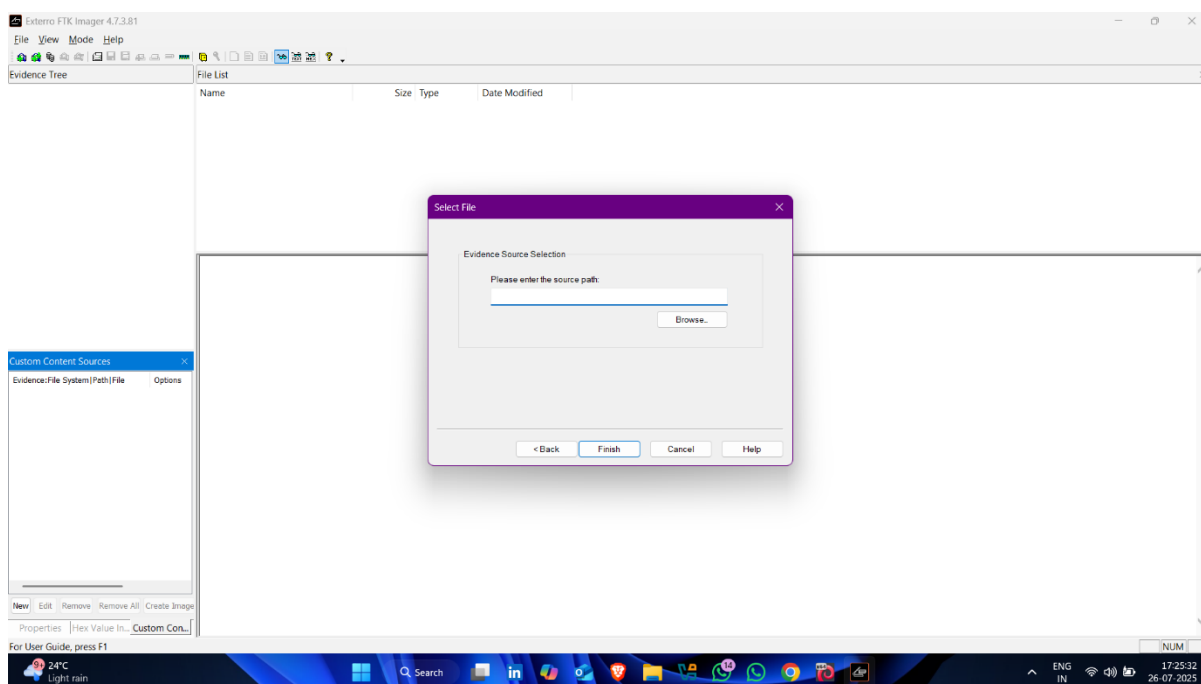
- Click on **Add Evidence Item**



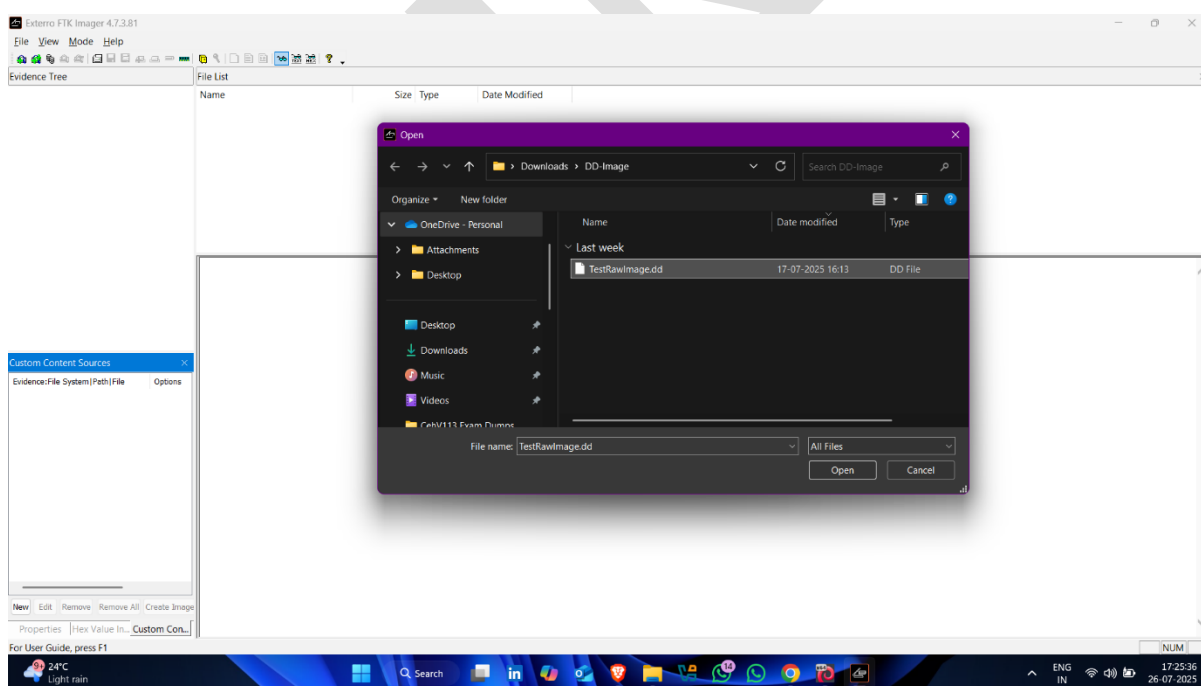
- Click on **Image File**



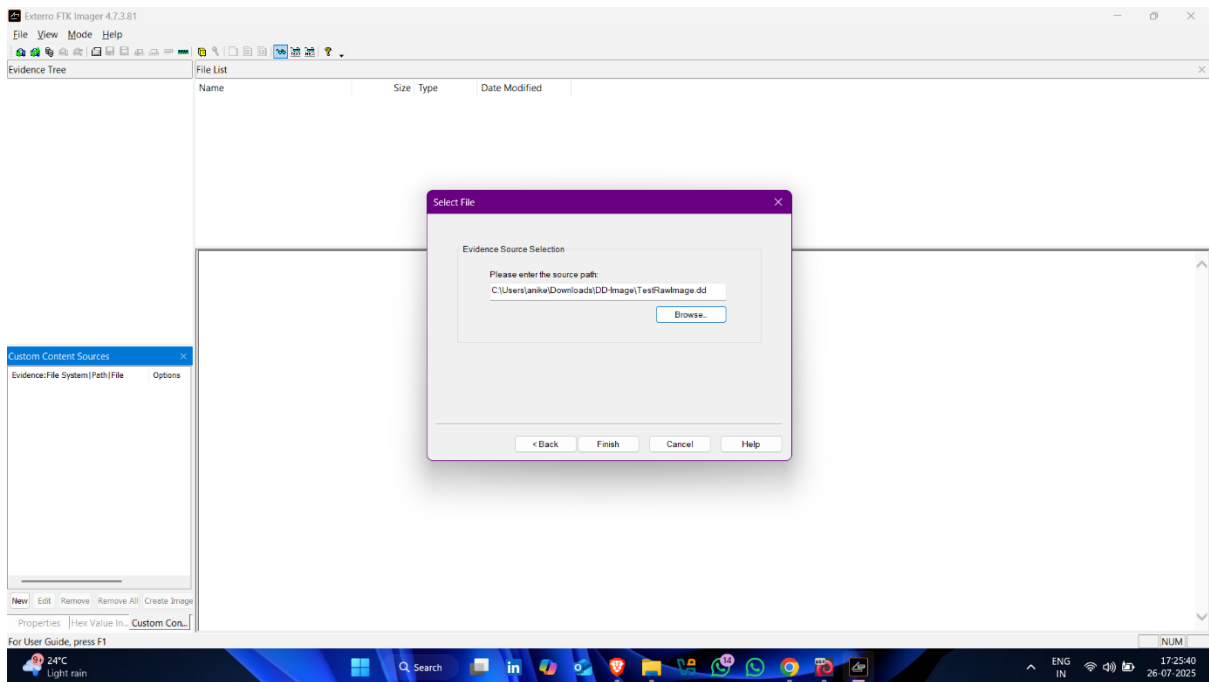
- Click on **Browse** to Select Image File



- Select File and click on **open**



- Click on **Finish**



- Content of image file  

